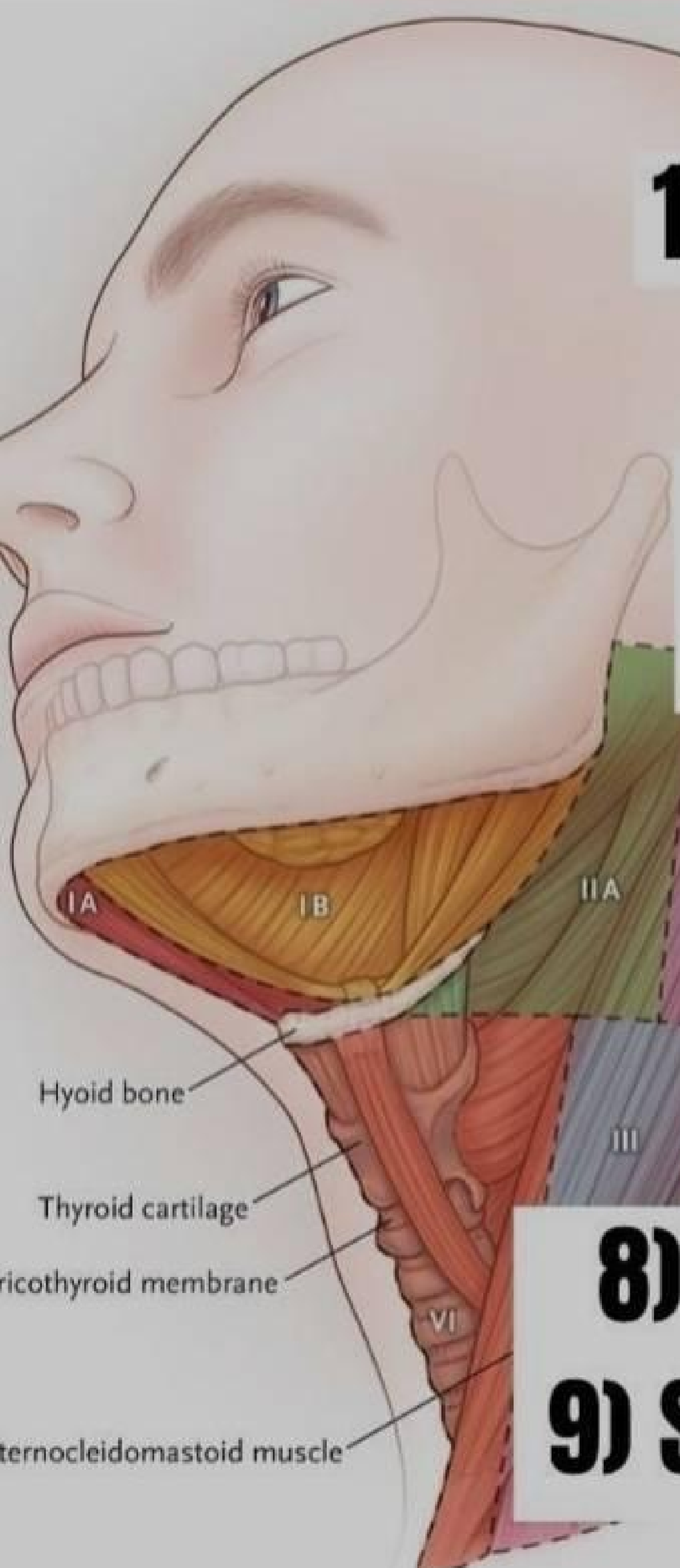
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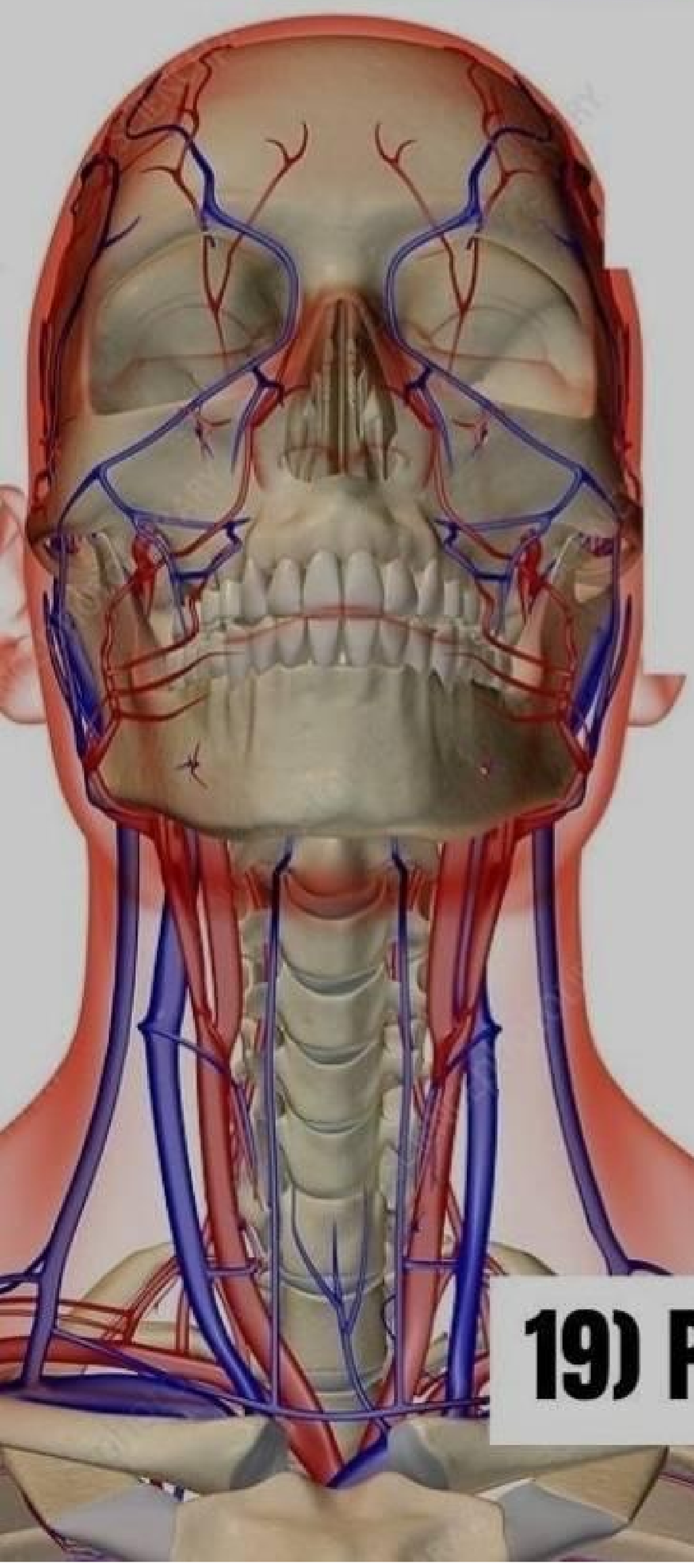
PART-1

1) Scalp | Blood & Nerve supply
2) Lacrimal Apparatus
3) Posterior Triangle of Neck
4) Anterior Triangle of Neck
5) Carodit Triangle
6) Carotid Sheath
7) Ansa Cervicalis
8) Parotid Gland | External feature
9) Secretomotor pathway of parotid
10) Thyroid Gland - 2

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PART-2

11) External Carotid Artery

12) Facial Artery

13) Maxillary Artery

14) Dural Venous Snius

15) Cavernous Sinus

16) Mandibular Nerve

17) Mastication Muscle

18) Soft Palate - 2

19) Palatine Tonsil | External feature

20) Tongue - 3

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PART-3



21) Lateral Wall of Nose - 2

22) Middle Ear - 2

23) Paranasal Air Sinus

24) Tympanic membrane

25) Styloid Process

26) Extraocular Muscle

27) Mastication Muscle

28) Waldeyer lymphatic ring

29) sternocleidomastoid , Digastric ,Orbicularis oculi Muscle

30) Otic,Ciliary,Submandibular Ganglion

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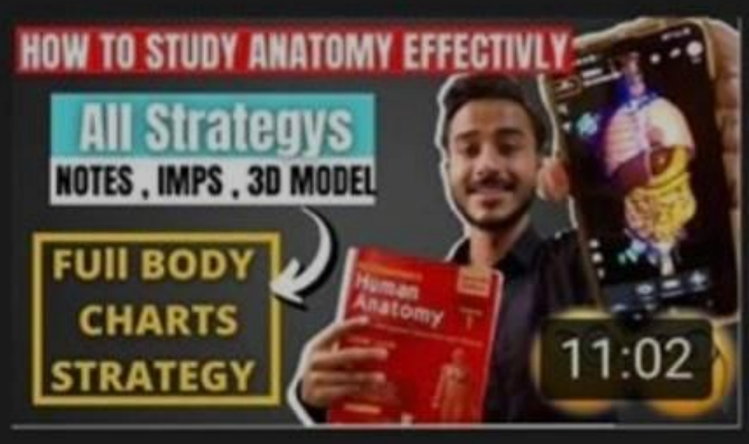
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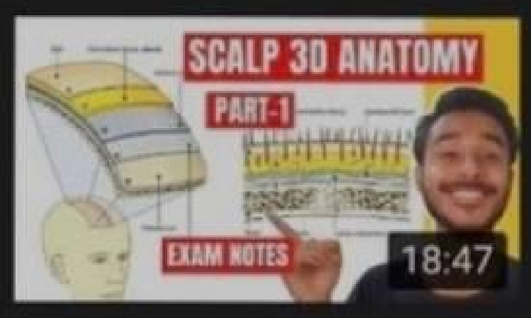
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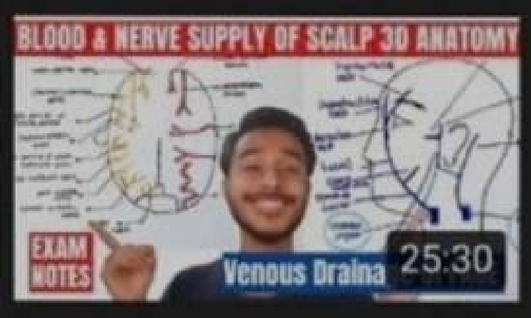
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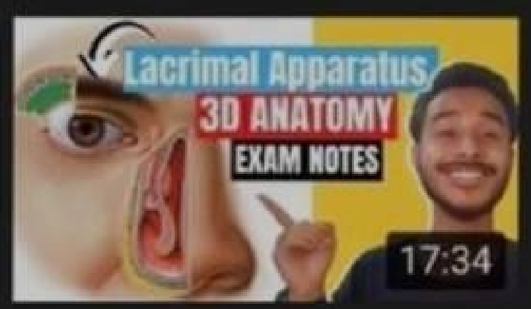
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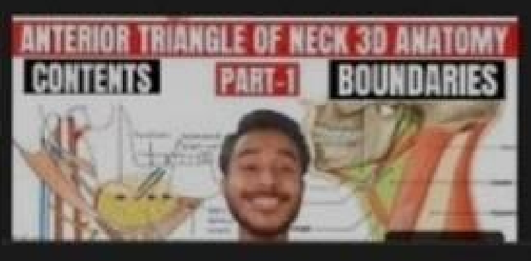
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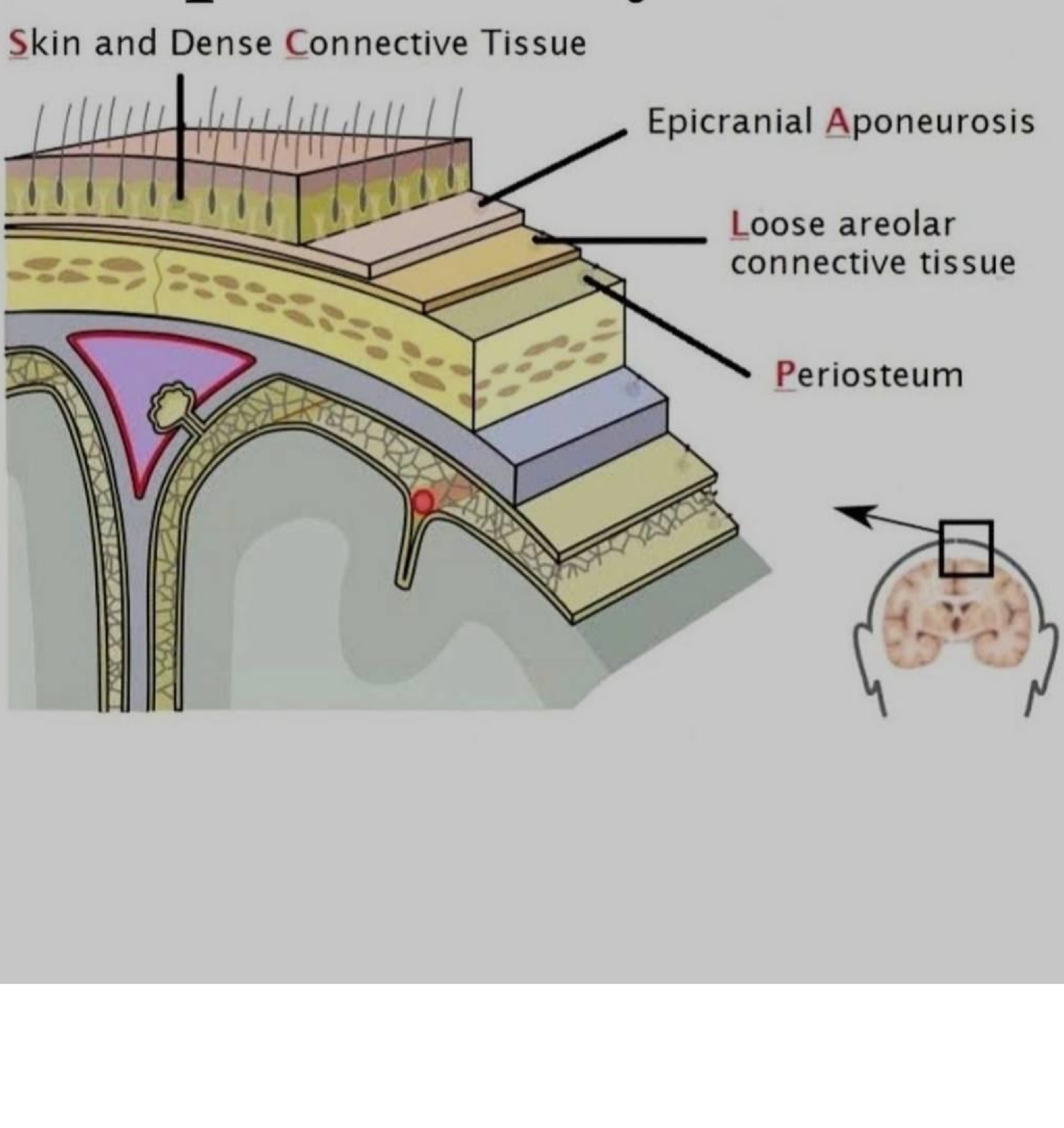
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WITH 3D ANATOMY MODEL AND HAND WRITTEN NOTES

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Scalp Anatomy Notes ->



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SCALP...

- Soft tissues covering - the cranial vault from scalp

Extent of scalp - Anteriorly - supra-orbital margin
- Posteriorly - external occipital ~~epo~~ protuberance
  - superior nuchal lines
- Each side - superior temporal lines

- Made up of five layer - skin
  - superficial fascia
  - Deep fascia or Galea Aponeurotica
  - DeLoose areolar tissue
  - Pericranium

- Skin - thick and hairy
  - It has sweet glands and sebaceous glands

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- Superficial fascia - more fibrous and dense in centre
  - contain - Blood vessel
  - It binds the skin → to Subjacent aponeurosis
  - and provide proper passage of vessel and nerve of skin

- Occipitofrontalis muscle :- has two bells → occipitalis → inserted → epicranial Aponeurosis
  → frontalis

  - Occipitalis - Small and separate
    - arise - lateral 2/3 of superior nuchal line
    supply by - Posterior auricular branch of facial nerve

  - frontalis - large and partly united
    - arise from - skin upper of eyelid and forehead
    - supply by - temporal branch of facial nerve

  - temparbparietalis - Present on lateral side
    - arise from - temporal fascia and fuse with epicranial aponeurosis.
    - supply by - temporal branch of facial nerve

- Epicranial aponeurosis :-
  - freely movable on pericranium
  - Attachment - Anteriorly → insertion - frontalis
    Posteriorly → external occipital protuberance
    - highest nuchal line b/w occipital bellies
    Side → superior temporal line
    expension - zygomatic arch

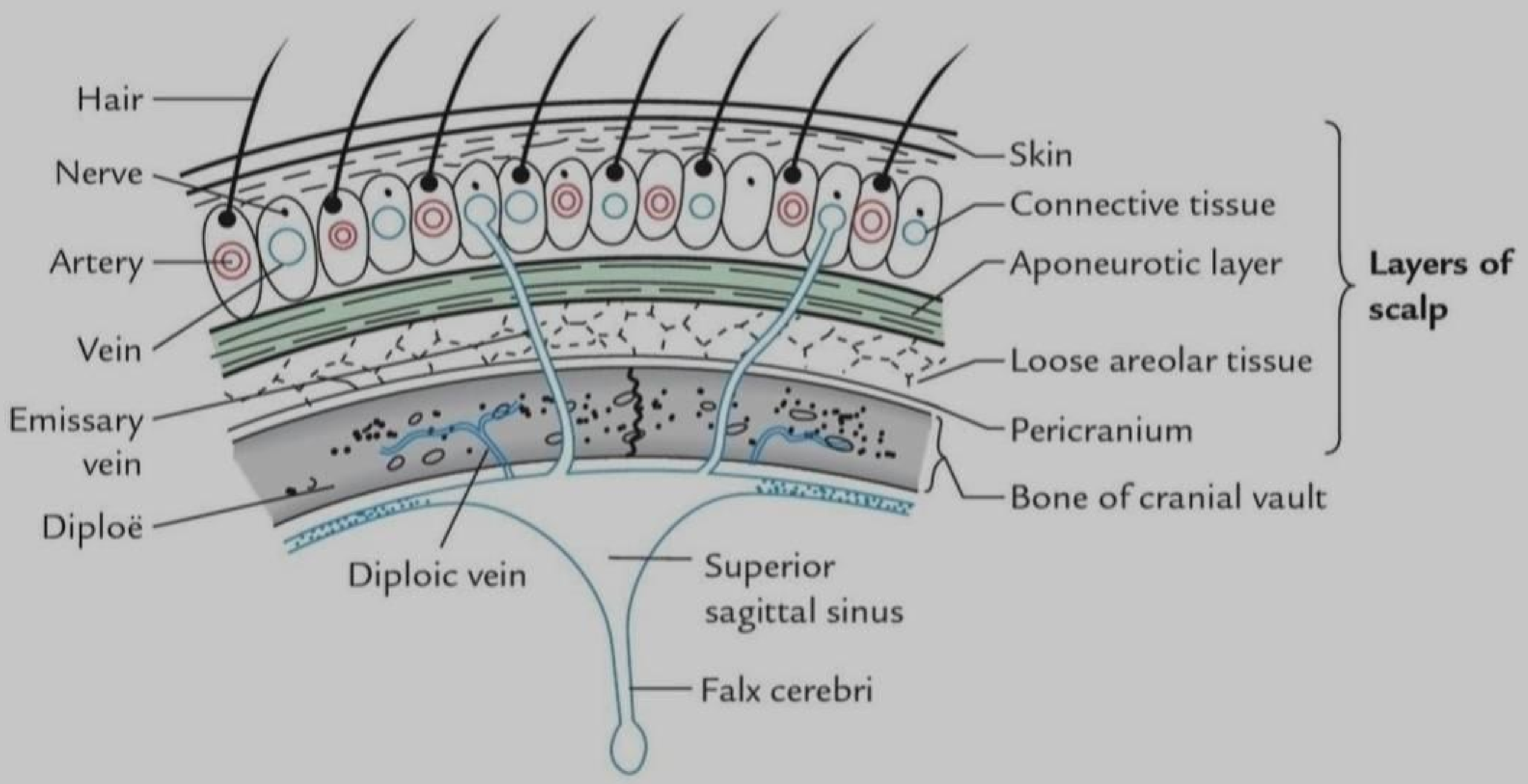
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loose areolar tissue :- Extend - Anteriorly - eyelids
Posteriorly - superior nuchal line
each side - superior temporal line

o- give passage to emissary vein which connect extracranial vein to intracranial venous sinuses

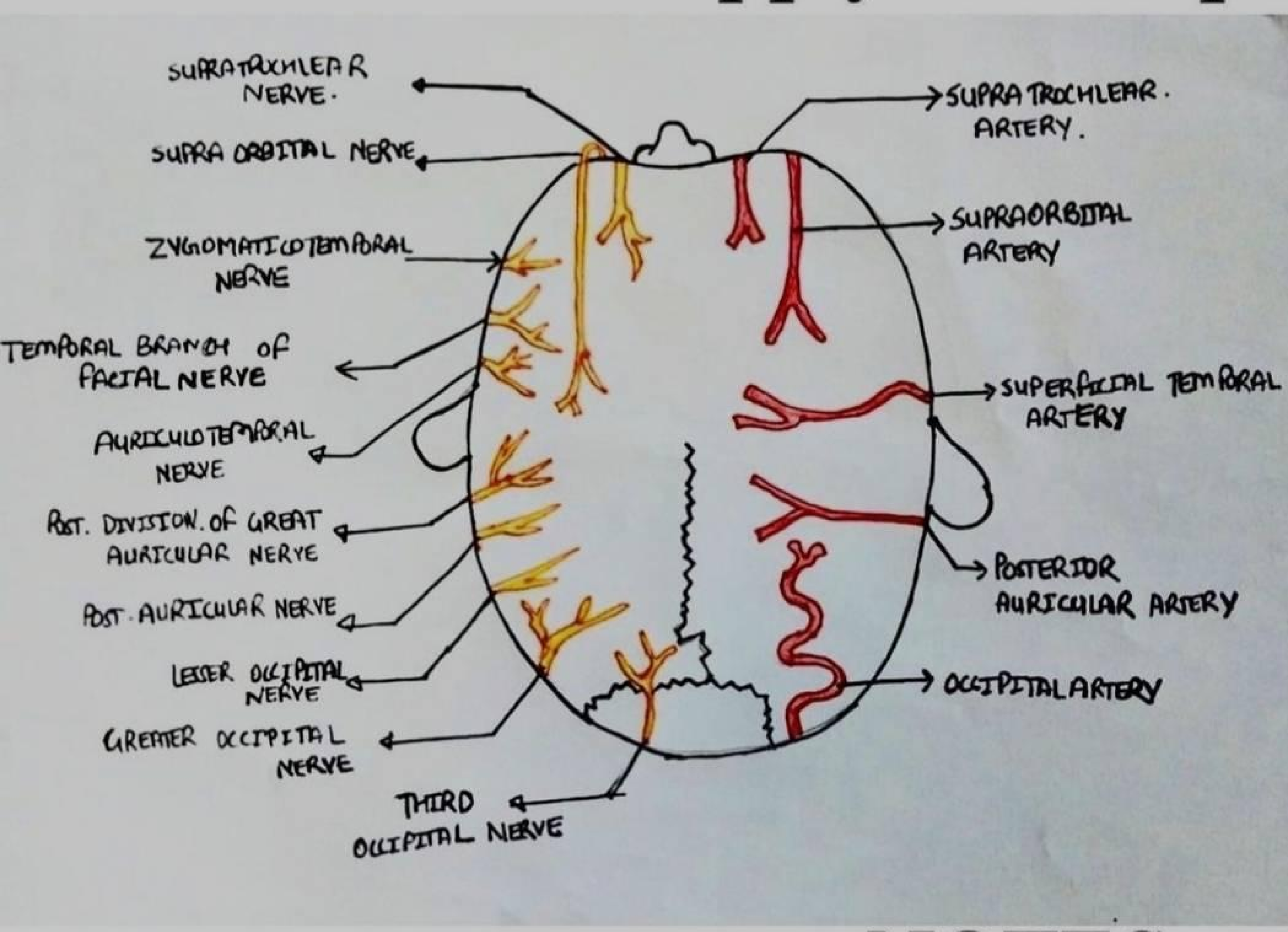
- Pericranium o- loosely attached to surface of bone
o- But is firmly adherent to suture.

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Blood & Nerve Supply Of Scalp



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NOTES->

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Arterial Supply :-

35 PO

- In front - Supratrochlear → branch → internal carotid artery
  - Supra-orbital → branch → internal carotid artery
  - Superficial temporal arteries → external carotid artery
- Behind - Posterior auricular → external carotid artery
  - Occipital → external carotid artery

Venous Drainage :-

- Supratrochlear and supraorbital vein - join each other to form angular vein → facial vein
- Superficial temporal vein - descend in front of tragus → to enter Parotid gland
  - join maxillary vein → form retromandibular vein
    - ↓ terminated
    - anterior & Posterior division
- Posterior auricular vein - It descend behind auricle and
  - unite the posterior division of retromandibular vein ↓ form external jugular vein
- Occipital vein - terminate - suboccipital venous plexus
- Vein of scalp communicate with intracranial dural venous sinuses through emissary vein
  - ↳ emissary vein - vein connecting vein outside the cranium with intracranial vein by passing through foramina in cranium
  - Parietal emissary vein → communicates → superior sagittal sinus
    - Passes → Parietal foramen
  - Mastoid emissary vein → -//- → sigmoid sinus
    - Passes → mastoid foramen

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Lymphatic Drainage :-

- Anterior part of scalp → drain → Preauricular or superficial parotid
- Posterior part of scalp → Posterior auricular or mastoid
  - occipital lymph node

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ive Supply :- supply by 10 nerve

- front of auricle - five nerve (4x sensory and 1x motor)
  - supraorbital
  - supratrochlear nerve
  - Zygomaticotemporal nerve

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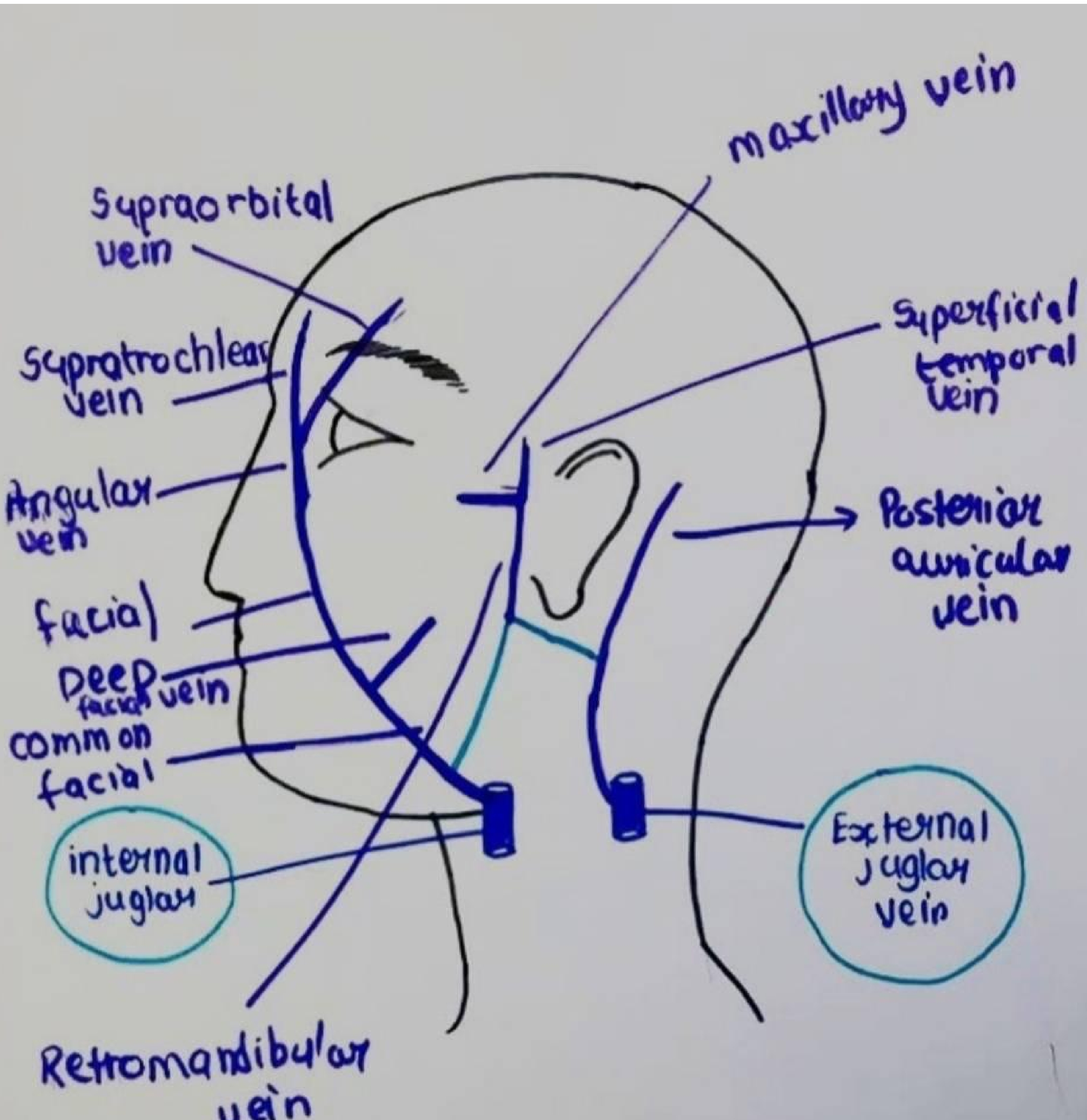
TATA

  - Temporal broth of facial nerve [motor]
  - Auriculotemporal nerve

- Behind auricle- five [again 4x sensory and 1 x motor]
  - Greater auricular nerve
  - Posterior auricular nerve (motor)
  - lesser occipital nerve
  - greater occipital nerve
  - Third occipital nerve

PT

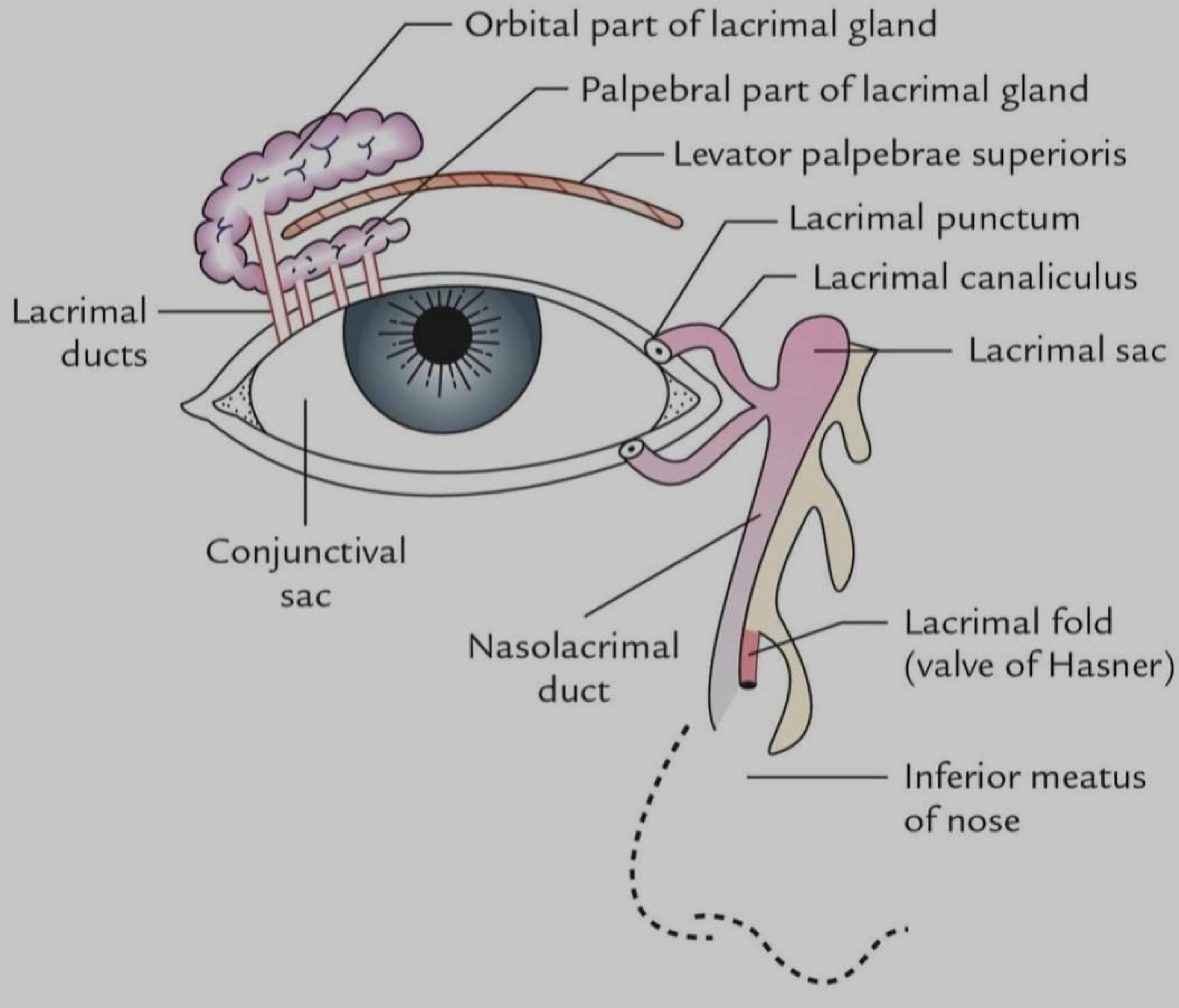
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Venous Drainage of the Head and Neck

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Lacrimal Apparatus Notes ->



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LACRIMAL APPARATUS... (Page 1)

Components :- Structure concerned with secretion and drainage of Lacrimeal or tear fluid - constitute the lacrimal apparatus

- made up - Lacrimal gland and its duct
  Conjunctival sac
  Lacrimal puncta and Lacrimal canaliculi
  Lacrimal sac
  Nasolacrimal duct

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- **Lacrimal gland:-** Serous gland
  - Situated - lacrimal fossa - anterolateral of roof of bony orbit
  - 'J' shaped gland - being indented by - tendon-of - Levator Palpebrae superior
    - ↳ orbital part - larger and deeper
    - ↳ Palpebral part - smaller and superficial
  - Dozen of ducts pierce - conjunctiva of upper lid and open into conjunctival sac
  - Artery supply → lacrimal branch of opthalmic artery
    nerve supply → lacrimal ~~gland~~ nerve
  - lacrimal fluid —secreted→ lacrimal gland —flow→ conjunctival sac
    ↓ Periodic blinking
    ↓ Spread fluid

- **Conjunctiva sac:-**
  - conjunctiva lining - deep surface - of eyelids —call→ Palpebral conjunctiva
    ↳ front of eyeball ⟶ bulbar conjunctiva
  - Potential space b/w palpebral and bulbar - called conjunctiva sac

- **Lacrimal Puncta & canaliculi:-**
  - Each lacrimal canaliculi - begins at Lacrimal punctum, 10mm long
  - Both canaliculi —open→ Lateral wall of Lacrimal sac
    —Behind→ medial palpebral ligament

- **Lacrimal sac :-** membranous sac - 12mm long and 5mm wide
  - situated in - Lacrimal groove
  - behind medial palpebral ligament
  - upper end - is blind
  - Lower end - continous with nasolacrimal duct

- **Nasolacrimal duct:-** membranous passage - 18mm long
  - begins → lower end of Lacrimal sac
  - run downward, backward - open into - inferior meatus of nose.

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Posterior Triangle of Neck Notes->

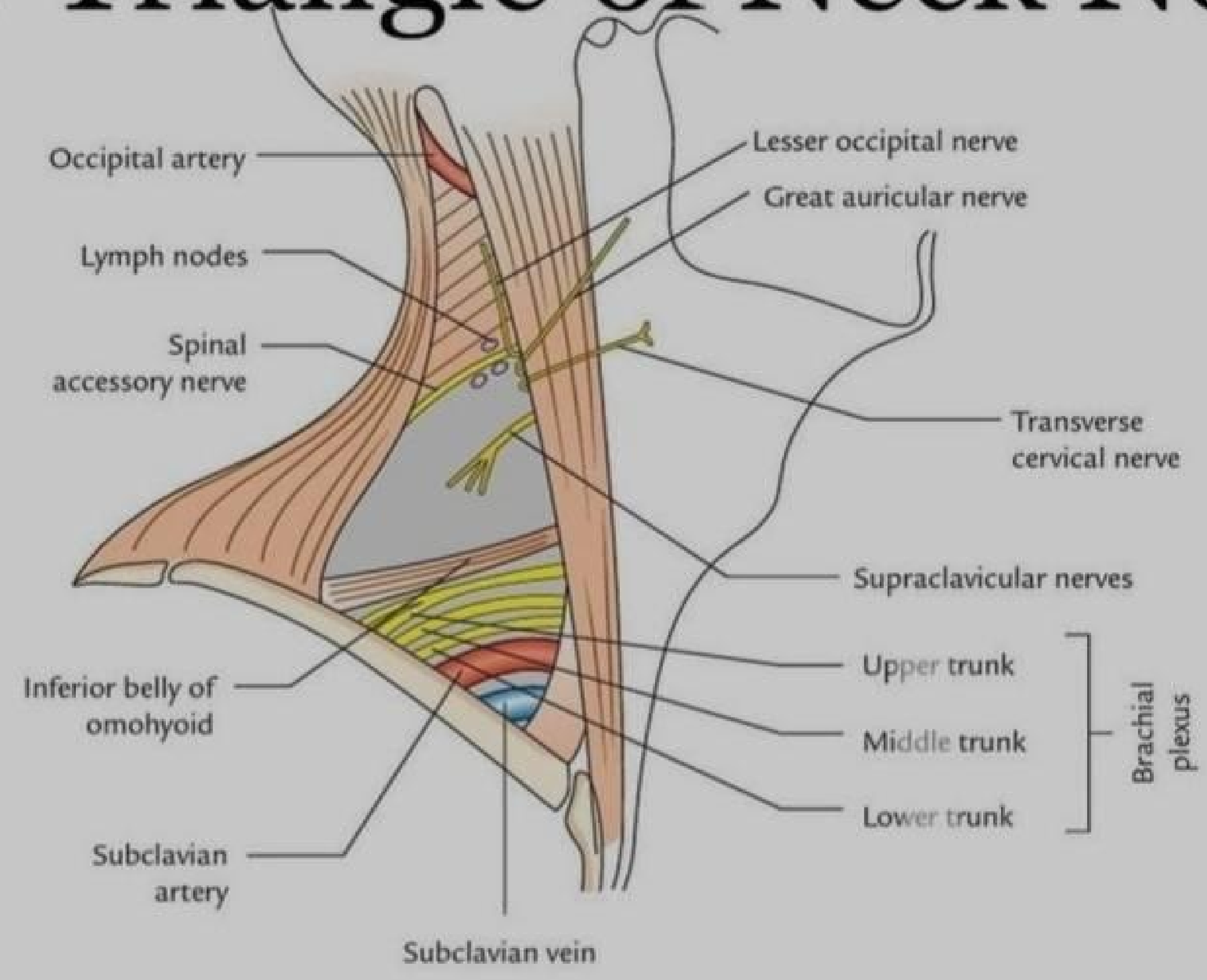
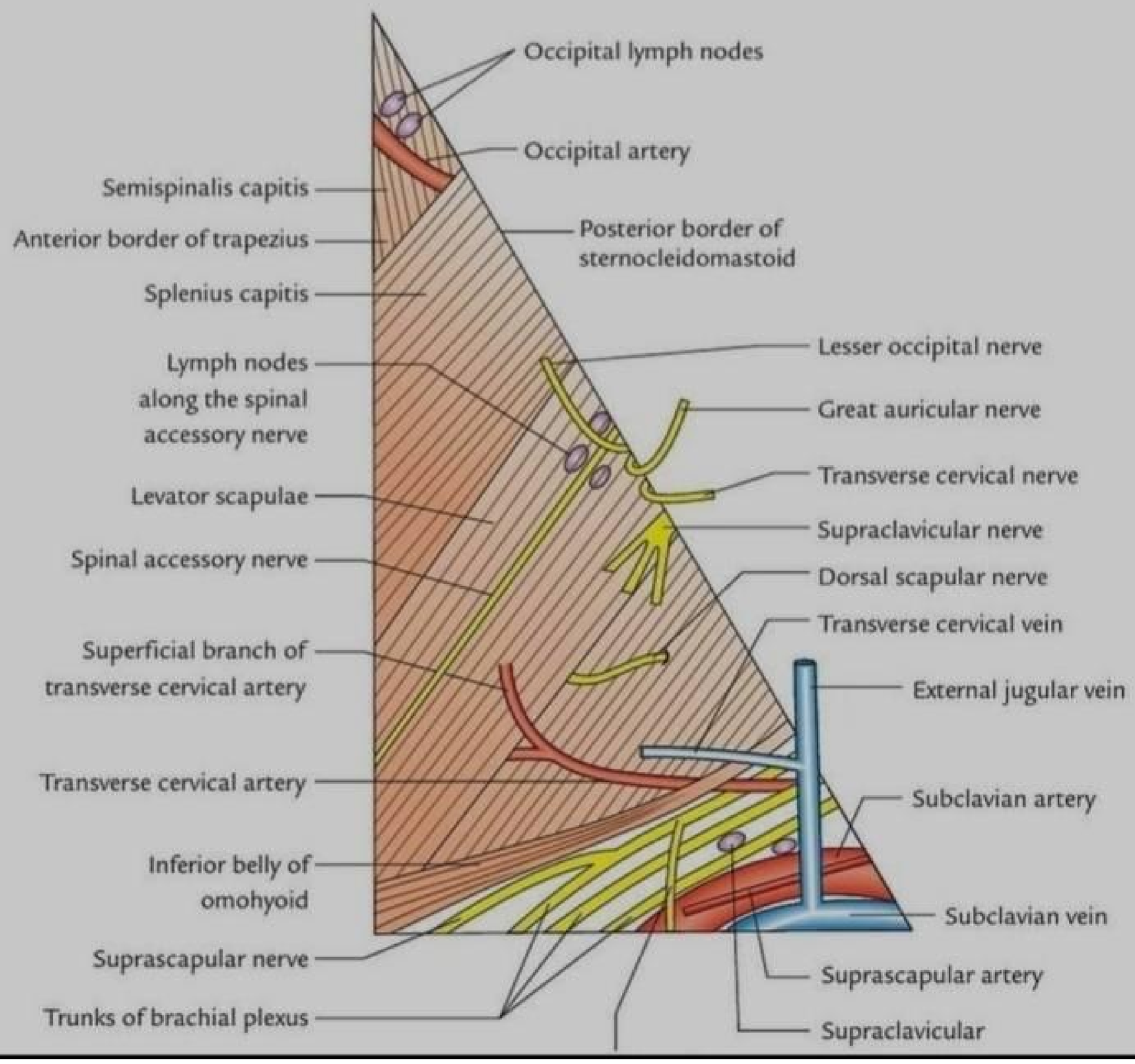


Fig. 5.7 Subdivisions and main contents of the posterior triangle.



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Posterior Triangle ...

eatures :- ◦- Space on the side of neck
• - Situated - behind - sternocleidomastoid muscle

Boundaries :-

* Anterior - Posterior border of sternocleidomastoid
* Posterior - Anterior border of Trapezius
* Base - middle 1/3 of clavicle
* Apex - Lies on superior nuchal line
◦- where trapezius and sternocleidomastoid meet

* floor - formed by - Prevertebral layer of deep cervical fascia
, covering following muscle ◦- Splenius capitis
◦- Levator scapulae
◦- Scalenus medius
◦- Semispinalis capitis

* Roof - Skin
Superficial fascia → contain → Platysma
investing layer of deep cervical fascia
◦- external jugular vein
◦- supraclavicular nerve
◦- greater auricular
◦- Transverse cutaneous
◦- lesser occipital nerve
◦- suprascapular artery
◦- occipital transverse cervical Artery
◦- Lymph vessel

Division of Posterior Δ :-
◦-sub-divided by - omohyoid muscle
◦- large upper part — Occipital Δ
◦- small lower part - subclavian Δ

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Content :-
Occipital Δ - nerve ◦- spinal accessary nerve
◦ 4x cutaneous branch → Transverse cutaneous nerve
◦- Supraclavicular nerve
◦- Great auricular nerve
◦- Lesser occipital nerve
◦ muscular branch → 2x levator scapulae
2x trapezius
Nerve to rhomboid

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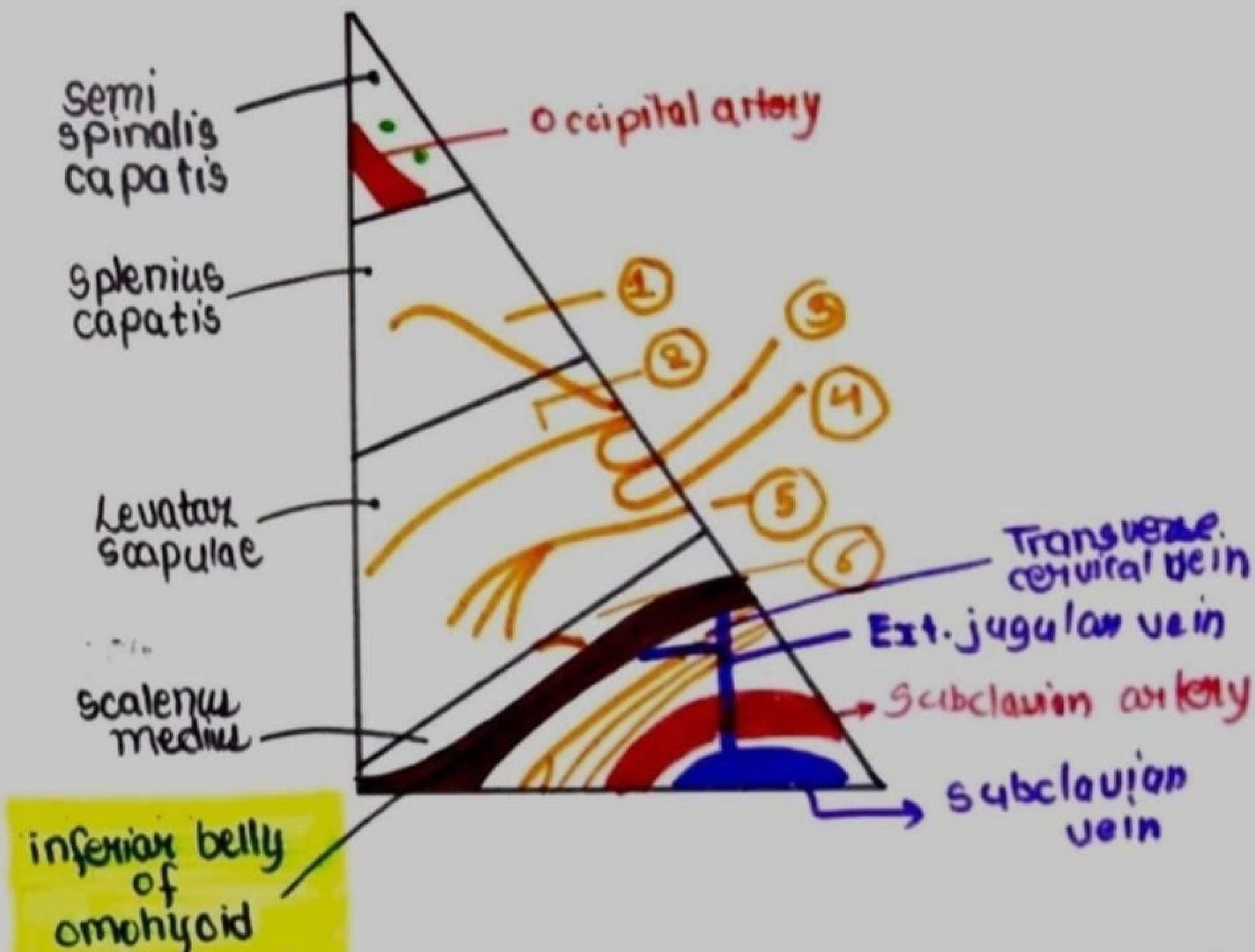
vessels - Transverse cervical artery and vein
- Occipital artery

Lymph node - supra-clavicular node
- occipital node

Subclavian Δ - nerve - Root and trunk brachial plexus
- nerve to subclavius
- nerve to serratus anterior
- suprascapular nerve

vessel - 3rd part of subclavian artery and vein
- Suprascapular artery
- Transverse cervical artery
- external jugular vein

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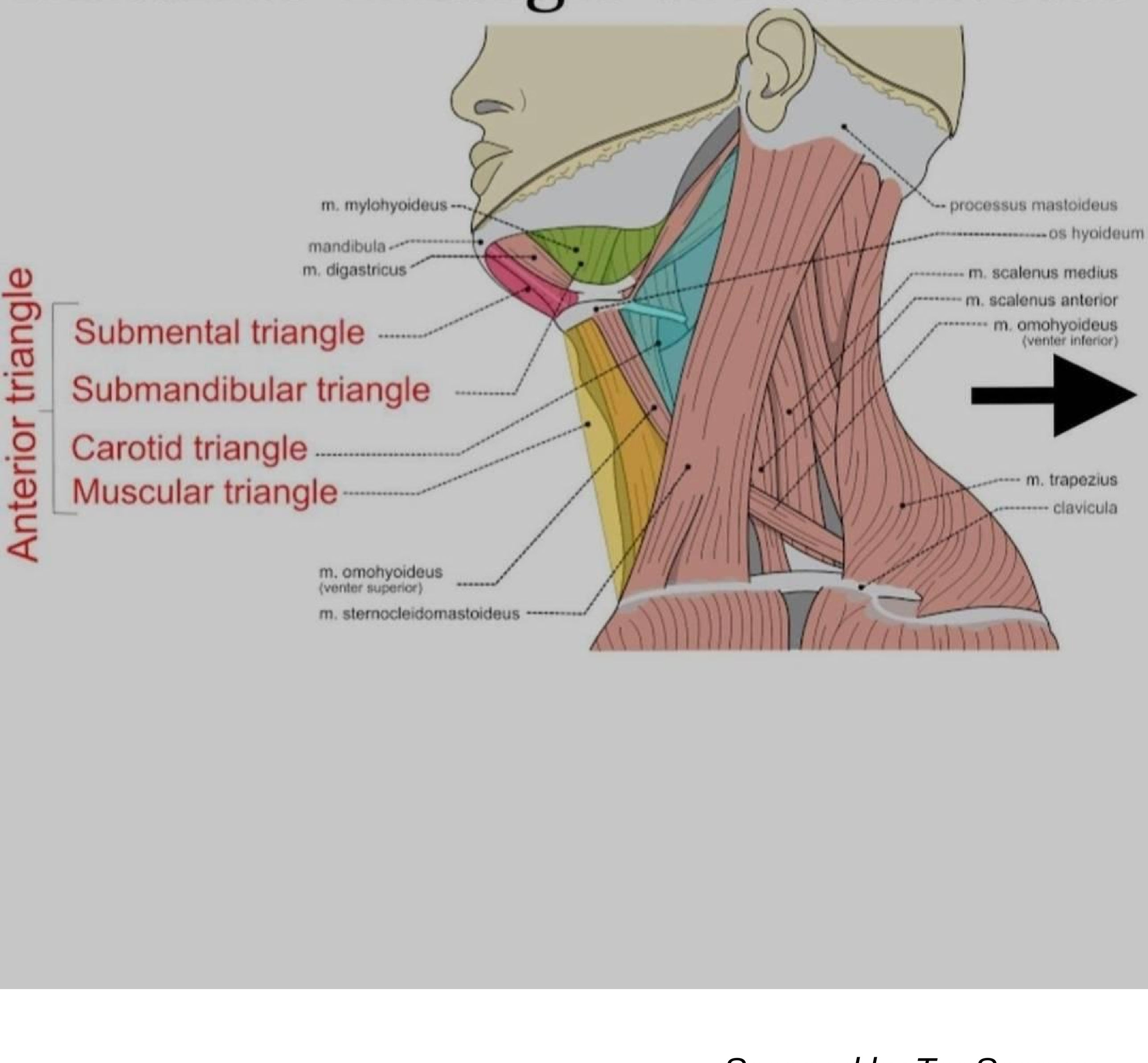


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Contents of Posterior Triangle of Neck

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Anterior Triangle of Neck Notes



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Anterior Δ ...

Boundaries :- medially - anterior median plane of neck
laterally - sternocleidomastoid
Base - A line joining the angle of mandible to mastoid process

Sub-division :- Sub-mental Δ
Digastric Δ
Carotid Δ
Muscular Δ

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Sub-mental Triangle ...

- median triangle

Boundaries - each side - Anterior belly of digastric muscle
Base - Hyoid bone
floor - Right and left myohyoid

Content - 4x submental lymph node
- Small sub-mental vein → join to form anterior jugular vein

Digastric Triangle ...

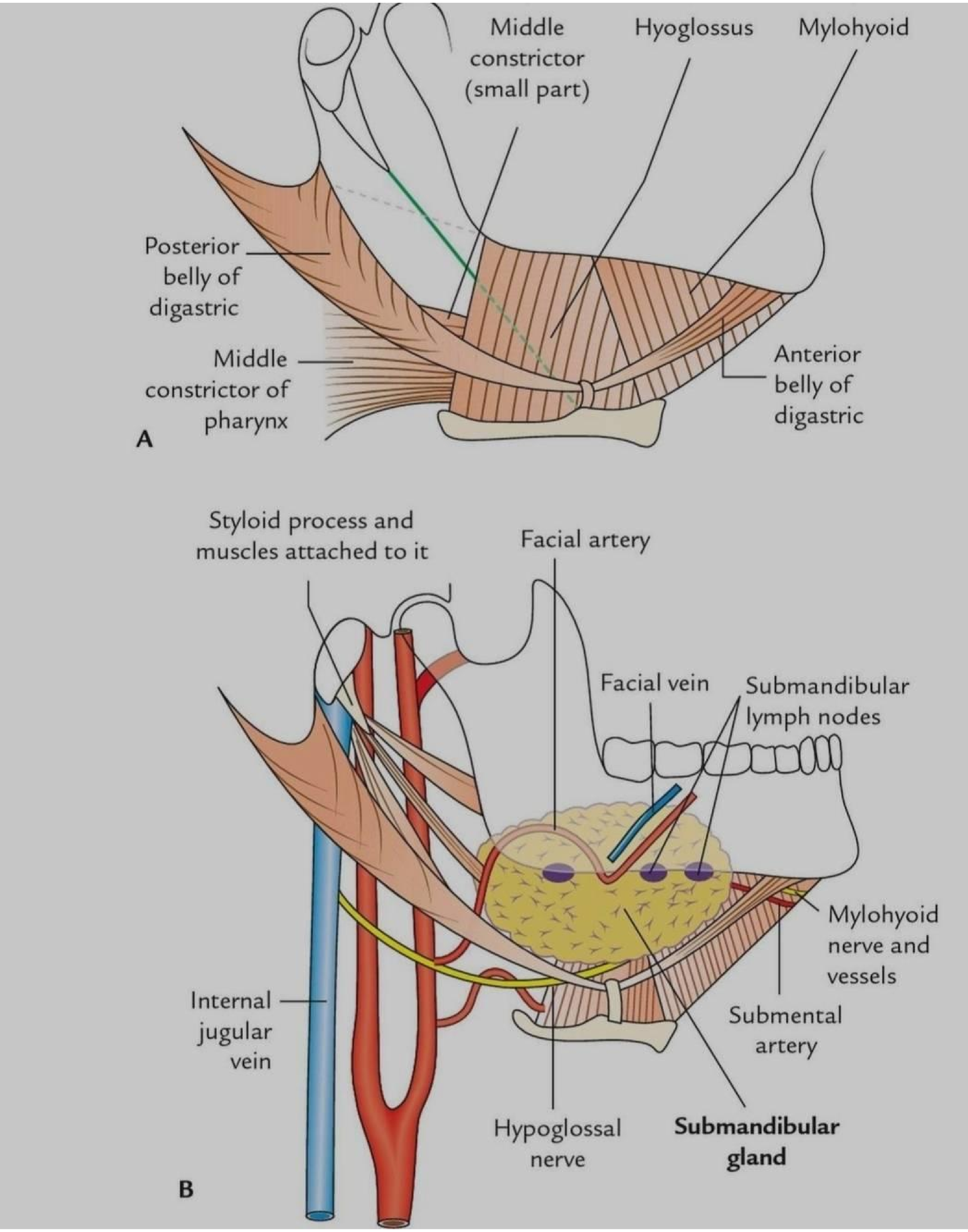
Boundaries - Antero-inferiorly - Anterior belly of digastric
Postero-inferiorly - Posterior belly of digastric stylohyoid
Base - Base of mandible

Roof - Skin
Superficial fascia → Platysma
Deep fascia
cervical branch of facial nerve
Anterior cutaneous nerve

floor - Anteriorly - mylohyoid muscle
Posteriorly - Hyoglossas muscle

Content → Anterior part of Δ
- Salivary gland
- submandibular lymph node
- facial artery and vein
- sub-mental ~~artet~~ artery
- Hypoglossal nerve
- Mylohyoid nerve

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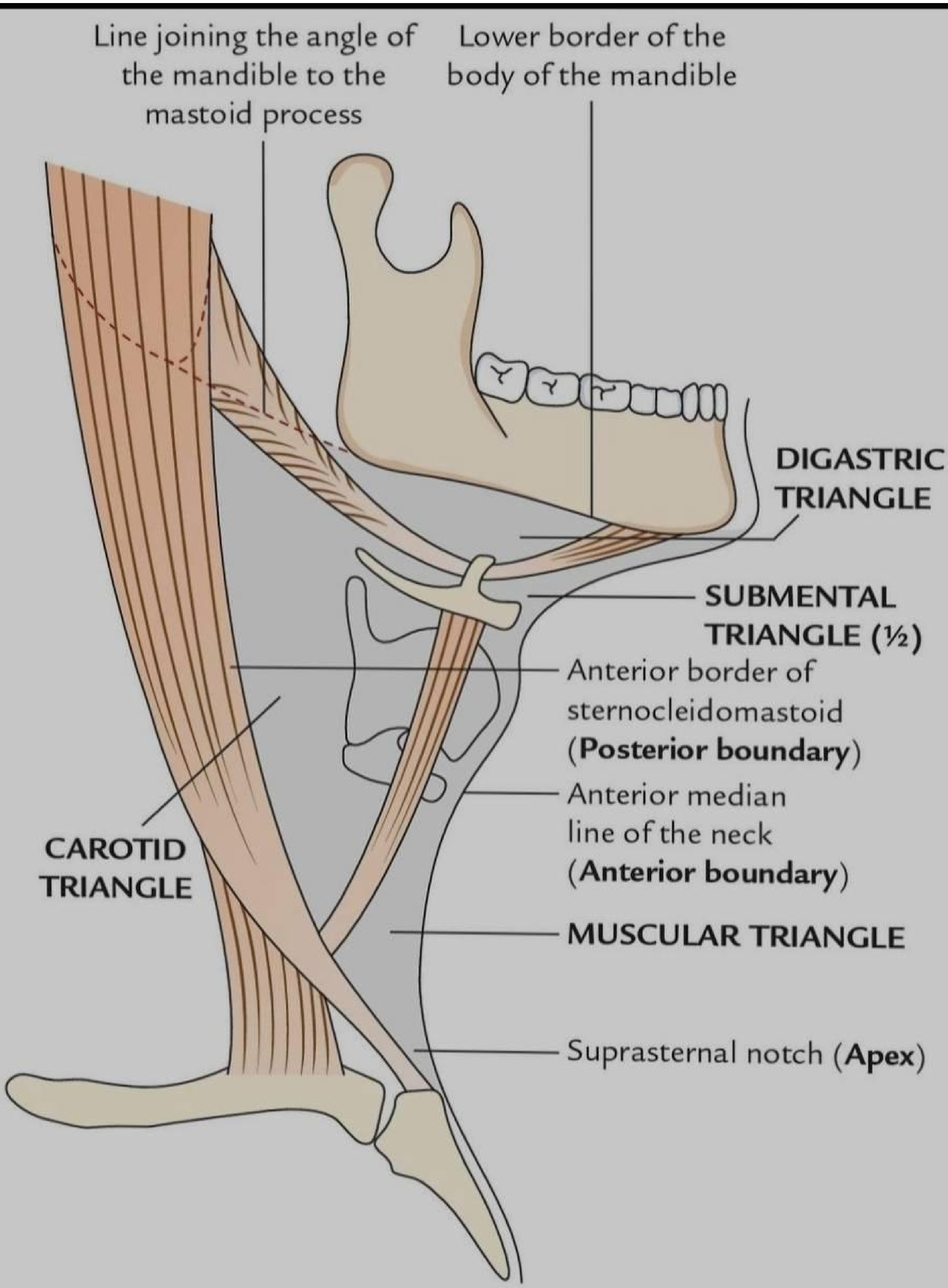
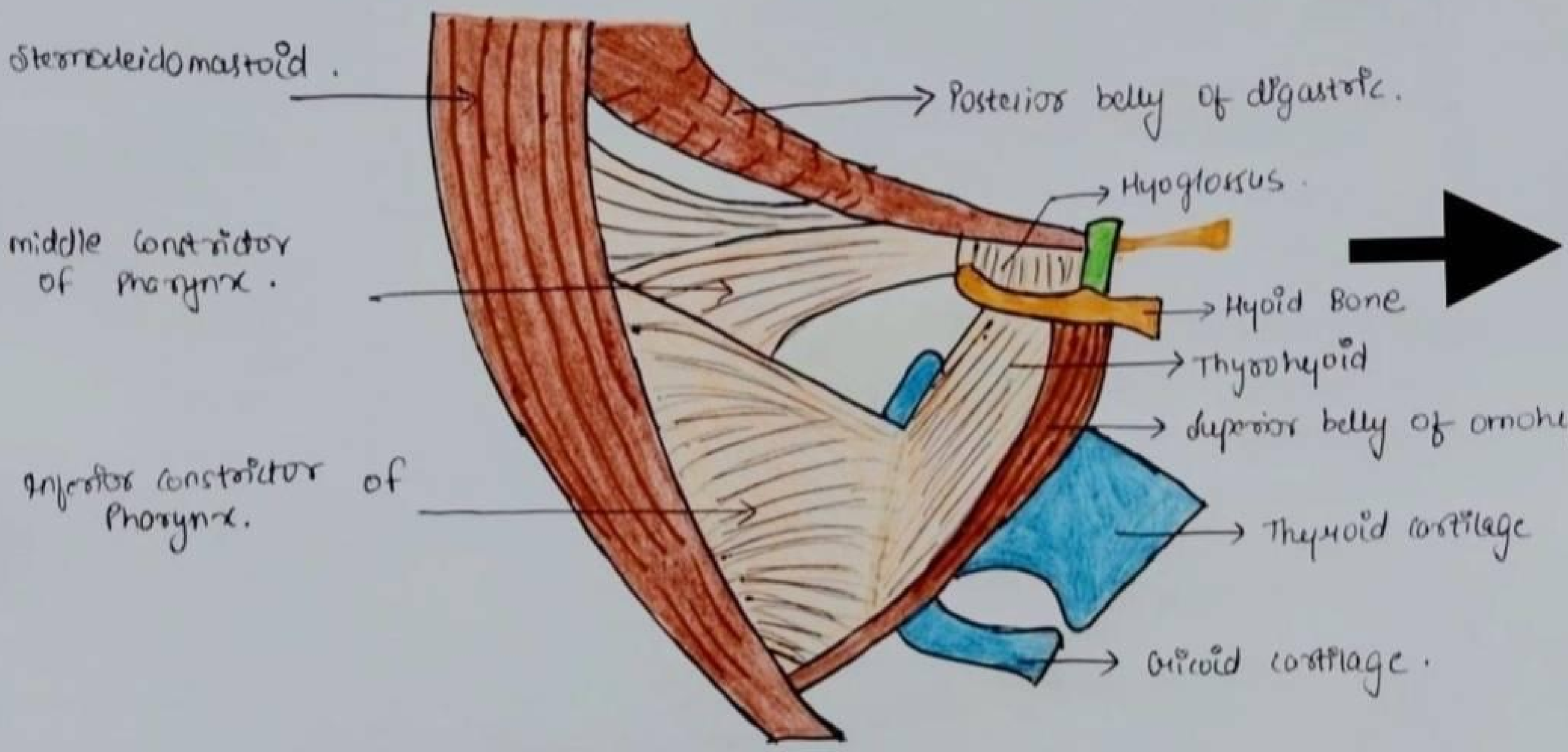


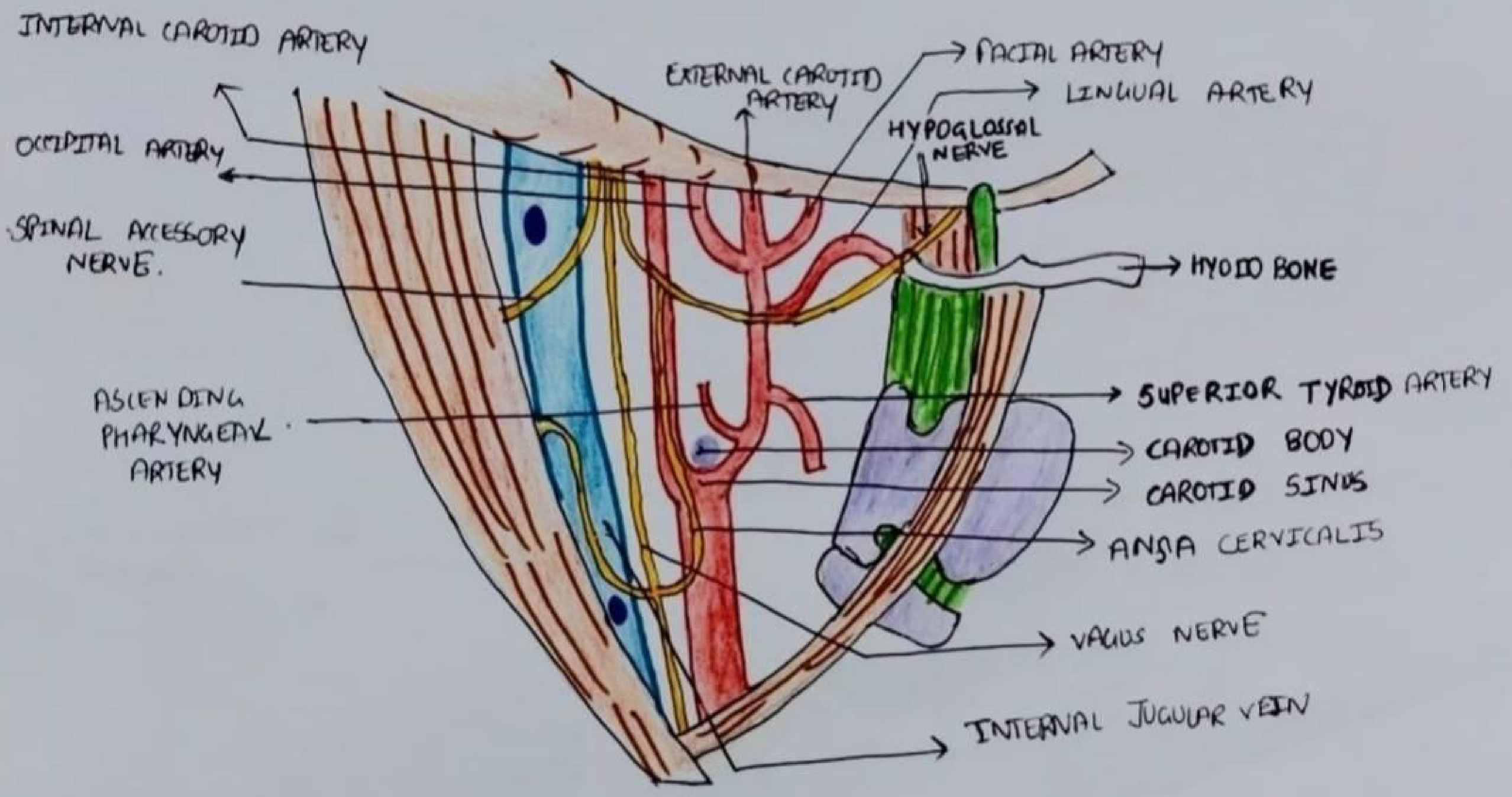
Fig. 6.3 Boundaries and subdivisions of the right anterior triangle.

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Carotid Triangle Notes



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CAROTID △ ...

Introduction :- Sub-division of - anterior triangle of neck

Boundaries :- Anterosuperiorly - Posterior belly of digastric muscle
- Stylohyoid

- Anteroinferiorly - Superior belly of omohyoid

- Roof - Skin
  - Superficial fascia → contain → Plastysma
  - Cervical branch of facial nerve
  - transverse cutaneous nerve
  - Deep cervical fascia (investing layer)

- floor - middle constrictor of Pharynx
  - inferior constrictor of Pharynx
  - Thyrohyoid membrane

- Posteriorly - Anterior border of sternocleidomastoid

Contents

Arteries - Common Carotid artery
- internal carotid artery
- external carotid artery - Superior thyroid
  - lingual
  - fascial
  - ascending pharyngeal
  - Occipital branches

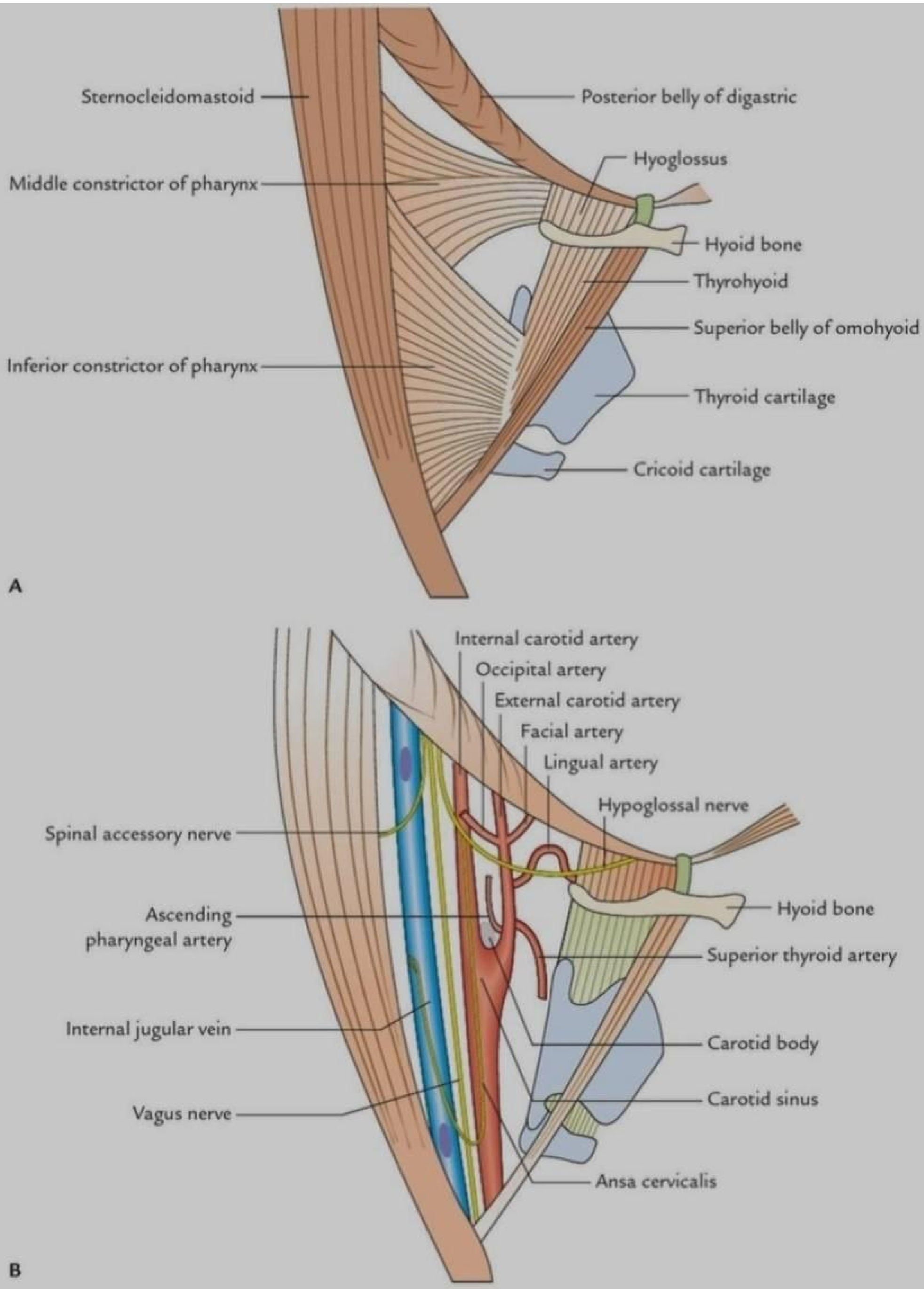
Vein - internal jugular vein
- common facial vein
- Pharyngeal vein
- lingual vein

Nerve - Vagus nerve
- Spinal accessory nerve
- Superior laryngeal branch → external laryngeal nerve
  - internal laryngeal nerve
- Hypoglossal nerve
- Sympathetic chain nerve

Lymph node - Deep cervical lymph node (internal jugular vein)
- jugulodigastric node
- jugulo-omohyoid node

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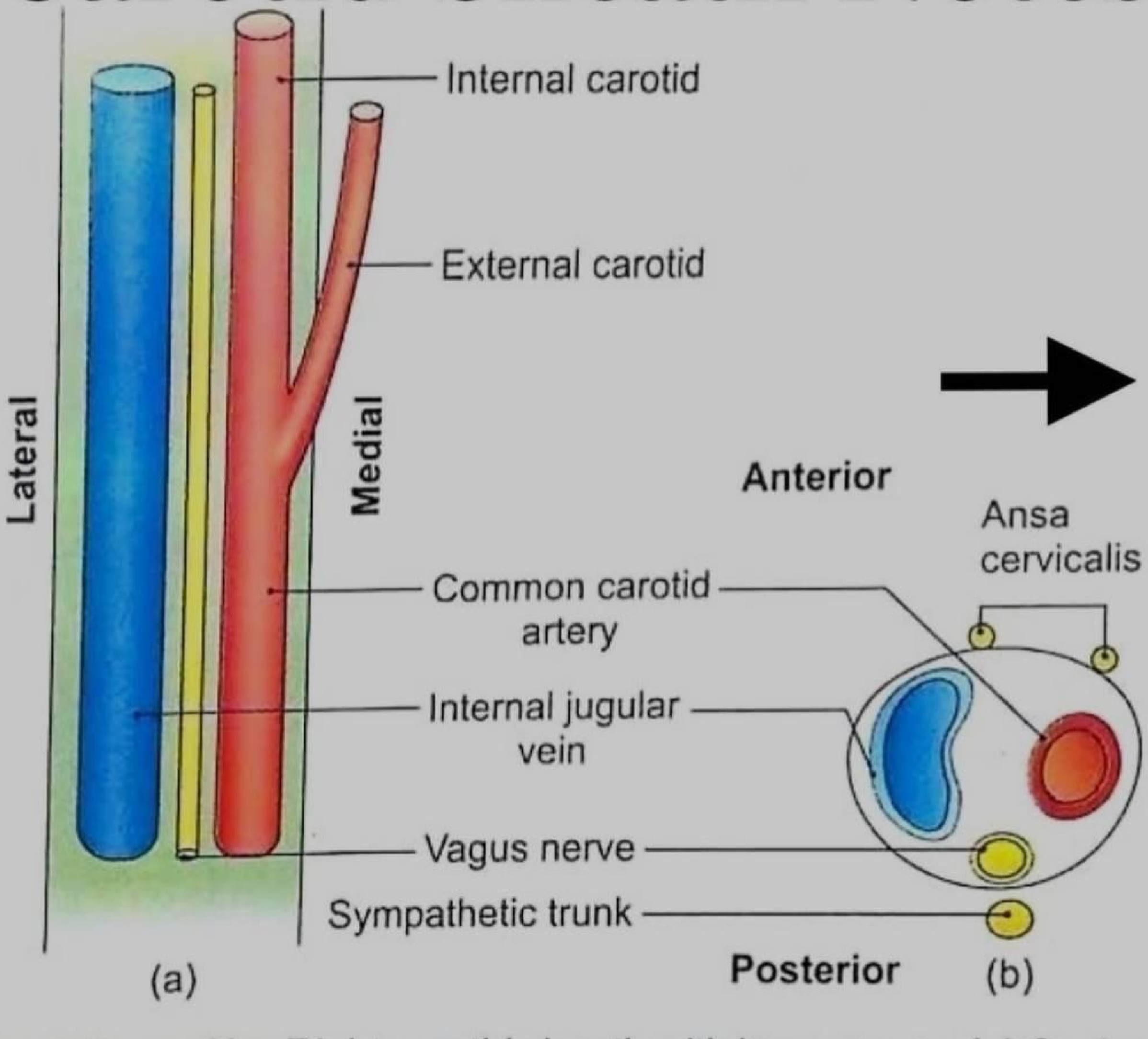
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Carotid triangle: A, boundaries and floor; B, boundaries and contents.

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Carotid Sheath Notes



Figs 3.8a and b: Right carotid sheath with its contents: (a) Surface view, and (b) sectional view

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CAROTID Sheath...

- Condensation of fibroareolar tissue - around main vessel of neck

formation :- Anterior - Pretracheal fascia
Posterior - Prevertebral fascia

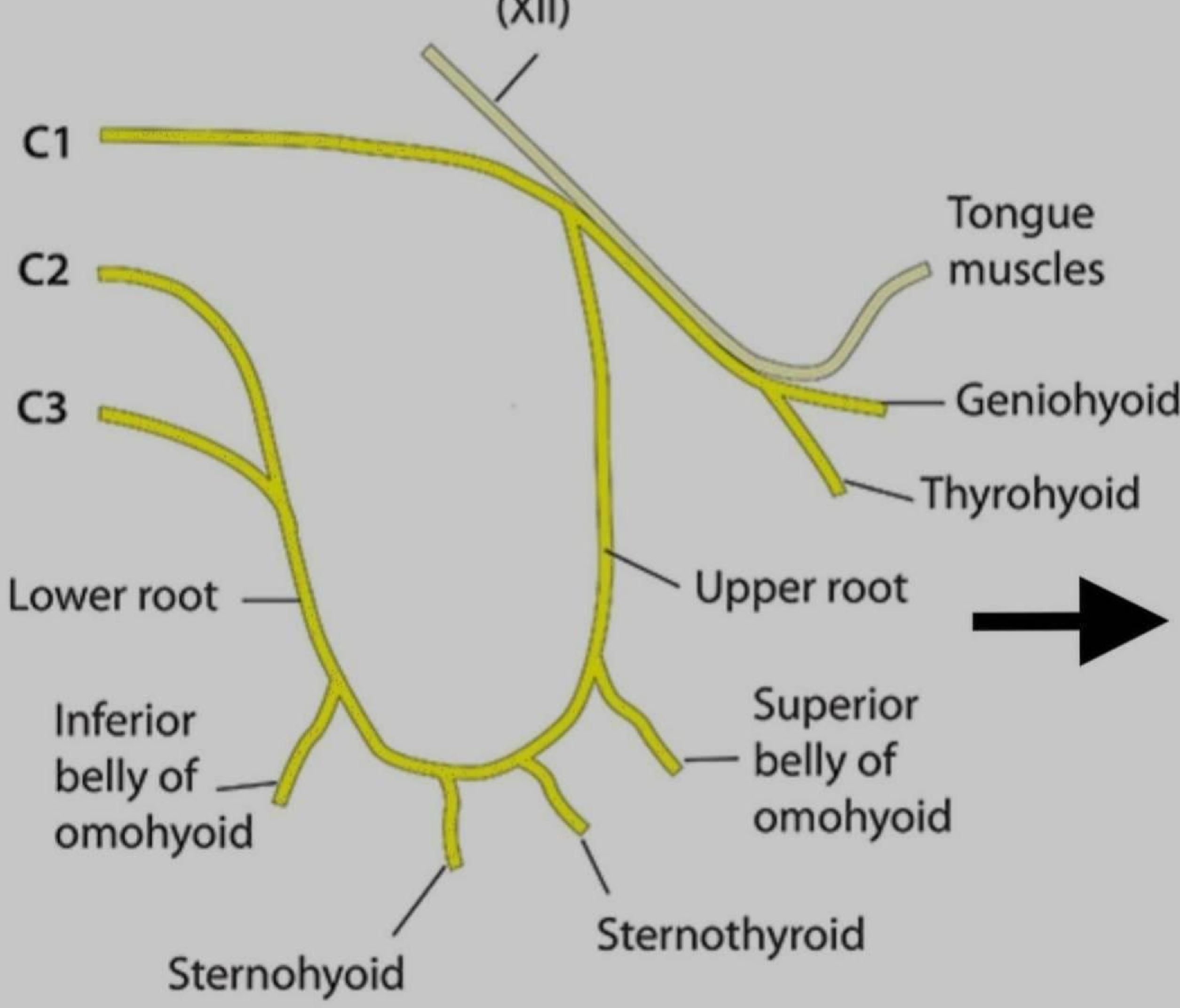
Content :- ∘ - Internal carotid arteries
∘ - Internal jugular vein
∘ - Vagus nerve
∘ - In upper part of sheath - IX, XI, XII nerve

Relation → Anterior wall - Ansa cervicalis. lie embedded
∘ - Posterior wall - sympathetic chain lie behind
∘ - overlapped by - sternocleidomastoid muscle
∘ - fused to - Deep cervical fascia

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ANSA CERVICALIS Notes



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Ansa Cervicalis or Ansa Hypoglossi ...

- Thin nerve loop (C_1, C_2 and C_3)
- lie embedded - in anterior wall of carotid sheath
- Supplies - infrahyoid muscles → Thyrohyoid
  → omohyoid
  → Sternothyroid and Sternohyoid

formation - formed by - superior and inferior root

- Superior root - fibres are derived - from first cervical nerve
  - continuation of descending branch of hypoglossal nerve
  - Descend over - internal carotid artery
    - common carotid artery

- Inferior root - Derived from 2nd and 3rd cervical spinal nerve
  - Descends → it wind round - internal jugular vein
  - Continues - anteroinferiorly - to join superior root
    ↓ in front
    Common Carotid artery

Distribution -

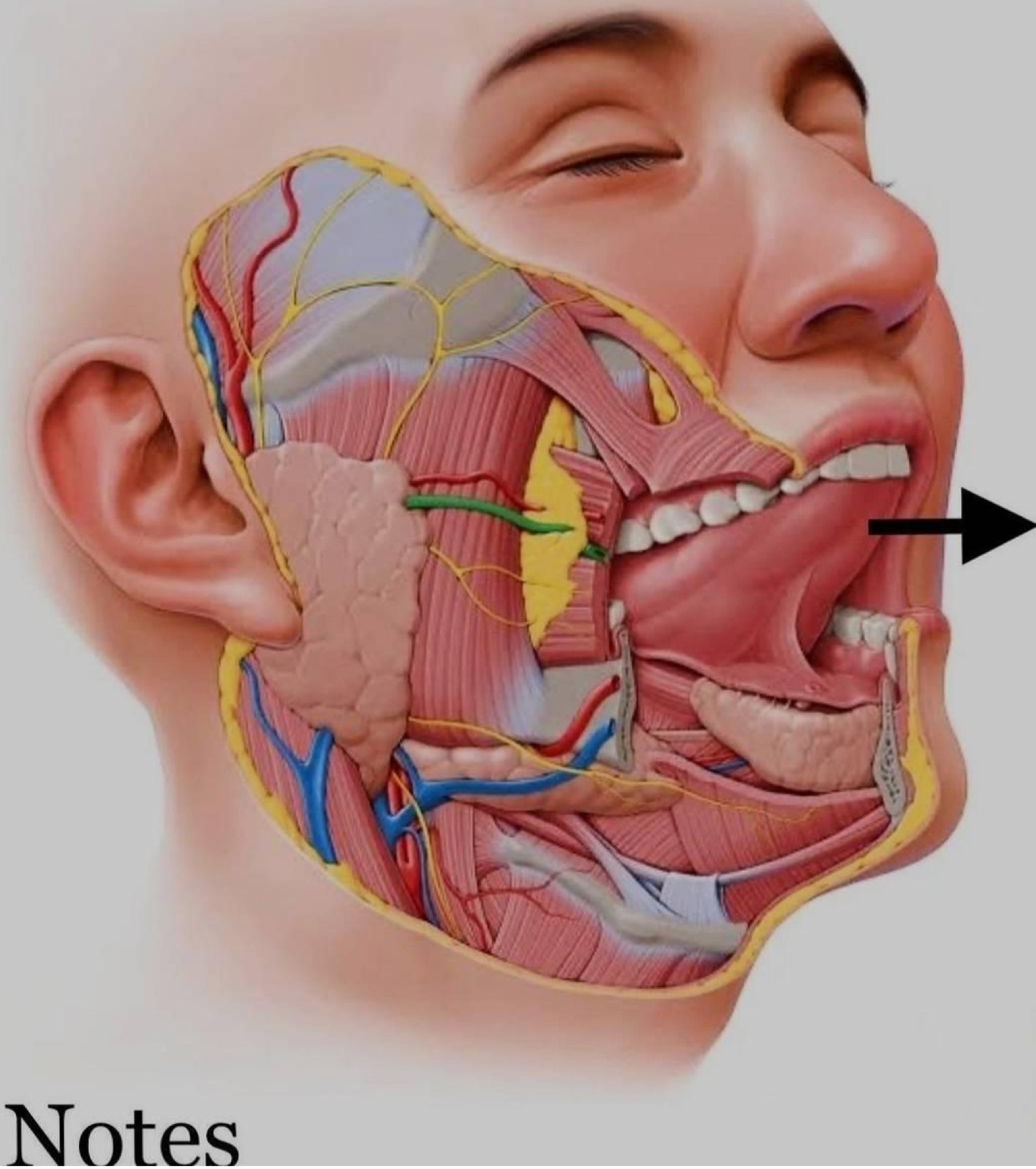
- Superior root - superior belly of omohyoid
- Inferior root - inferior belly of omohyoid
- Ansa cervicalis - Sternohyoid
  - Sternothyroid

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- Thirohyoid and geniohyoid supplied by - separate branches from 1st cervical nerve

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Parotid Gland Anatomy PART-1



Notes

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Parotid Gland...

Introduction ∘- largest salivary gland - 25g
∘- Situated- below external accustic meatus.
B/w ramus of mandible and sternocleidomastoid
- Anteriorly - overlap - masseter muscle

Capsule ∘- investing layer of deep cervical fascia - form capsule for gland
∘- Supplied - by auricular nerve
∘- fascia splits - enclose gland
∘- superior lamina - thick & adherent to gland
- Attached above - zygomatic arch

∘- Deep lamina - thin
- Attached - ~~styloid process~~ Tympani plate
- Portion of D.L extend b/w styloid process and mandible - is thickened ~~form~~ → "Stylomandibular ligament"

External feature ∘- 3x side pyramid
∘- Apex - downward
∘- 4x surface - superior
∘- Superficial
∘- Anteromedial
∘- Posteromedial

∘- 3x border ∘- Anterior
∘- Posterior
∘- Medial

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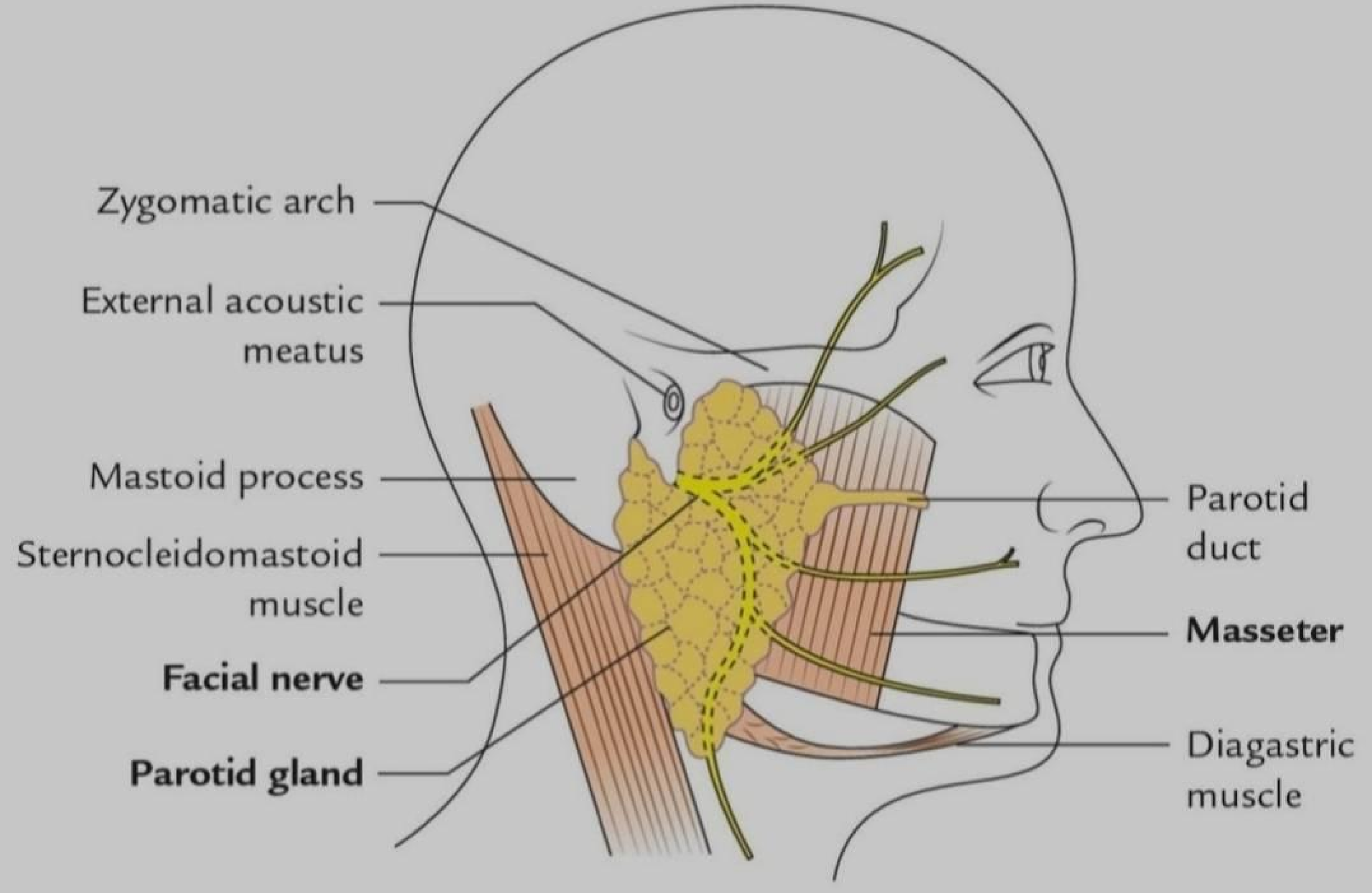


Fig. 8.1 Main features of the parotid region.

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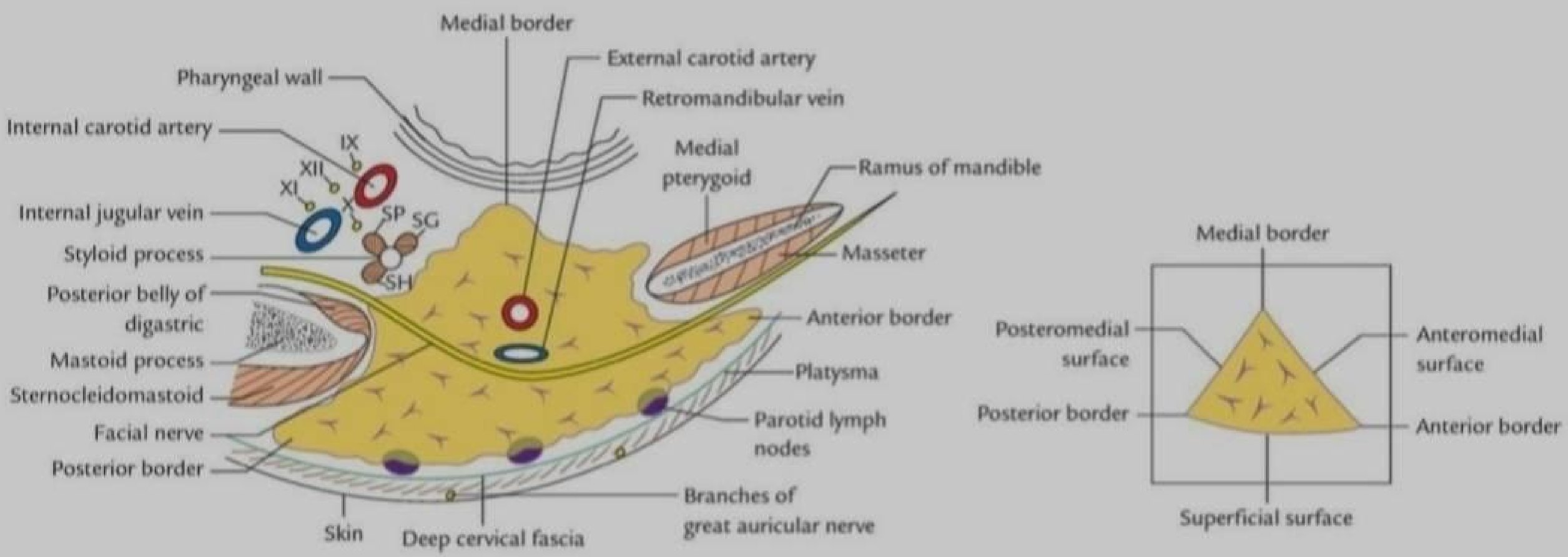


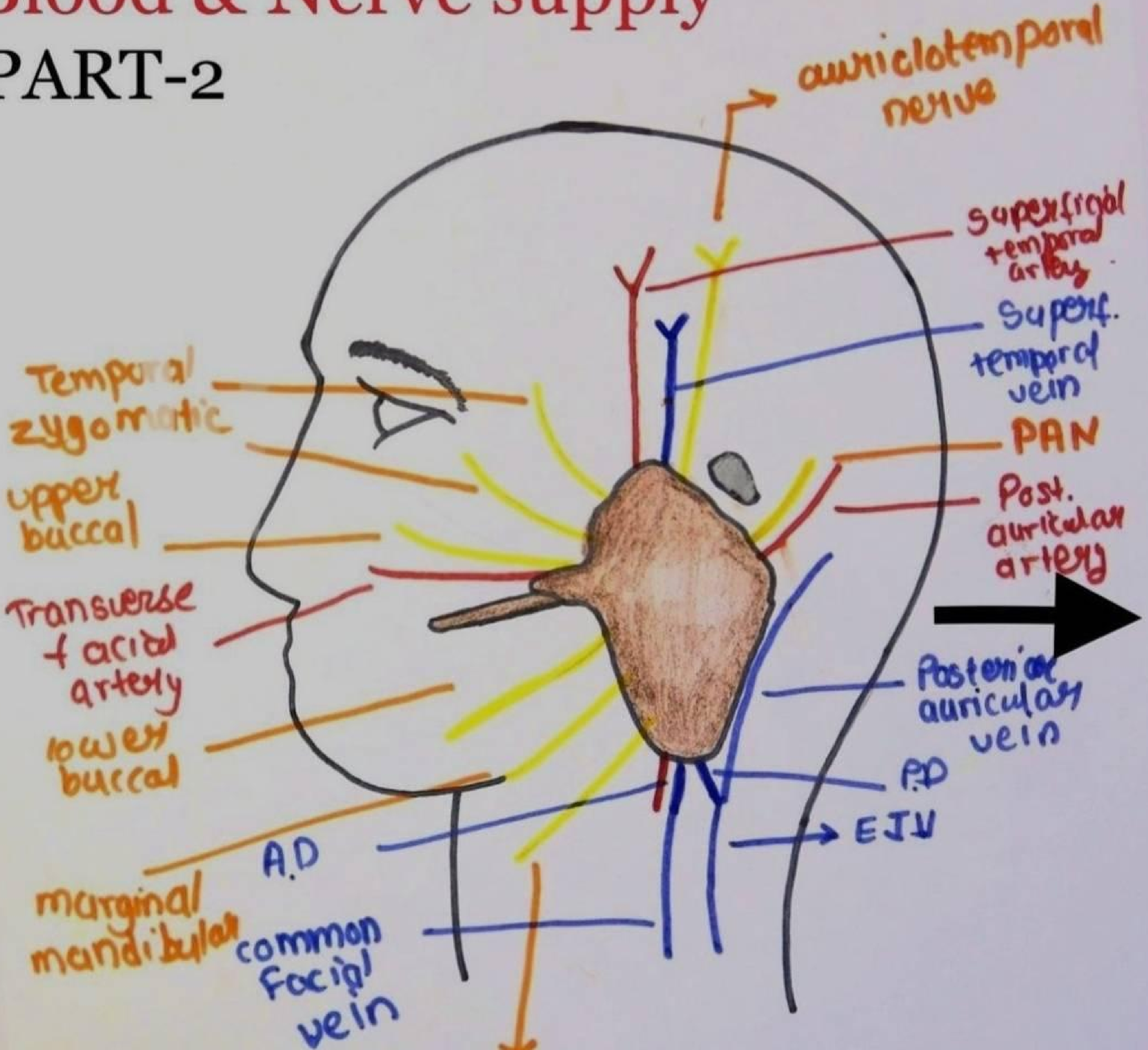
Fig. 8.4 Horizontal section through parotid gland showing its relations and the structures passing through it. The inset figure shows borders and surfaces of the parotid gland (SG = styloglossus muscle, SH = stylohyoid muscle, SP = stylopharyngeus muscle).

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Relations of Parotid Gland

Blood & Nerve supply

PART-2



NOTES

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ıation :-

- Apex
  - Posterior belly digastric
  - cervical branch of facial nerve
  - 2x division of retromandibular vein

- Surface :-
  - Superior Surface - superficial temporal vessel
    - auriculotemporal nerve
    - external acoustic meatus
    - Posterior surface of TMJ
  - Superficial surface → Skin
    - Superficial fascia
    - Parotid fascia
    - Deep parotid lymph node
  - Anteromedial surface - ramus of mandible
    - masseter
    - medial pterygoid
    - lateral surface of TMJ
  - Posteromedial surface - mastoid process
    - Styloid process

- Border :-
  - Anterior border - Terminal branch of facial nerve
    - Parotid duct
    - Transverse facial vessel
  - Posterior border separate superficial surface to Posteromedial surface
  - Medal border separate Anteromedial to Posteromedial surface

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Structure within the Parotid gland :-

- Arteries :- "ECA" enter → Posteromedial surface
  - ↳ maxillary A. leave → anteromedial surface
  - ↳ TFA → anterior part → superior surface

- Vein :- Retromandibular vein → formed in gland with by → STV + MV
  - → lower part → Anterior, Posterior division emerge → Apex

- Nerve :- facial nerve
  - enter → Posteromedial surface
  - leave → anteromedial surface
  - branch → Temporal branch
    - zygomatic branch
    - Buccal branch
    - marginal mandibular branch
    - Cervical branch

- Parotid Duct emerge → middle of anterior border
  - run → forward downward on masseter
  - Pierces in - buccinator muscle
  - finally duct open → vestibule of mouth

Blood → Artery - External carotid artery
- Vein - EJV and IJV

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Nerve - sensory nerve - auriculotemporal nerve
- Sympathetic nerve - vasomotor - Plexus around - middle meningeal artery
- Parasympathic nerve - secretomotor

Inferior salivatory nucleus
↓
glossopharyngeal nerve
↓
Tympanic branch
↓
Tympanic nerve
↓
lesser petrosal nerve
↓
otic ganglion
↓
Post-ganglionic fibre through auriculotemporal nerve

IG TT LAG AT

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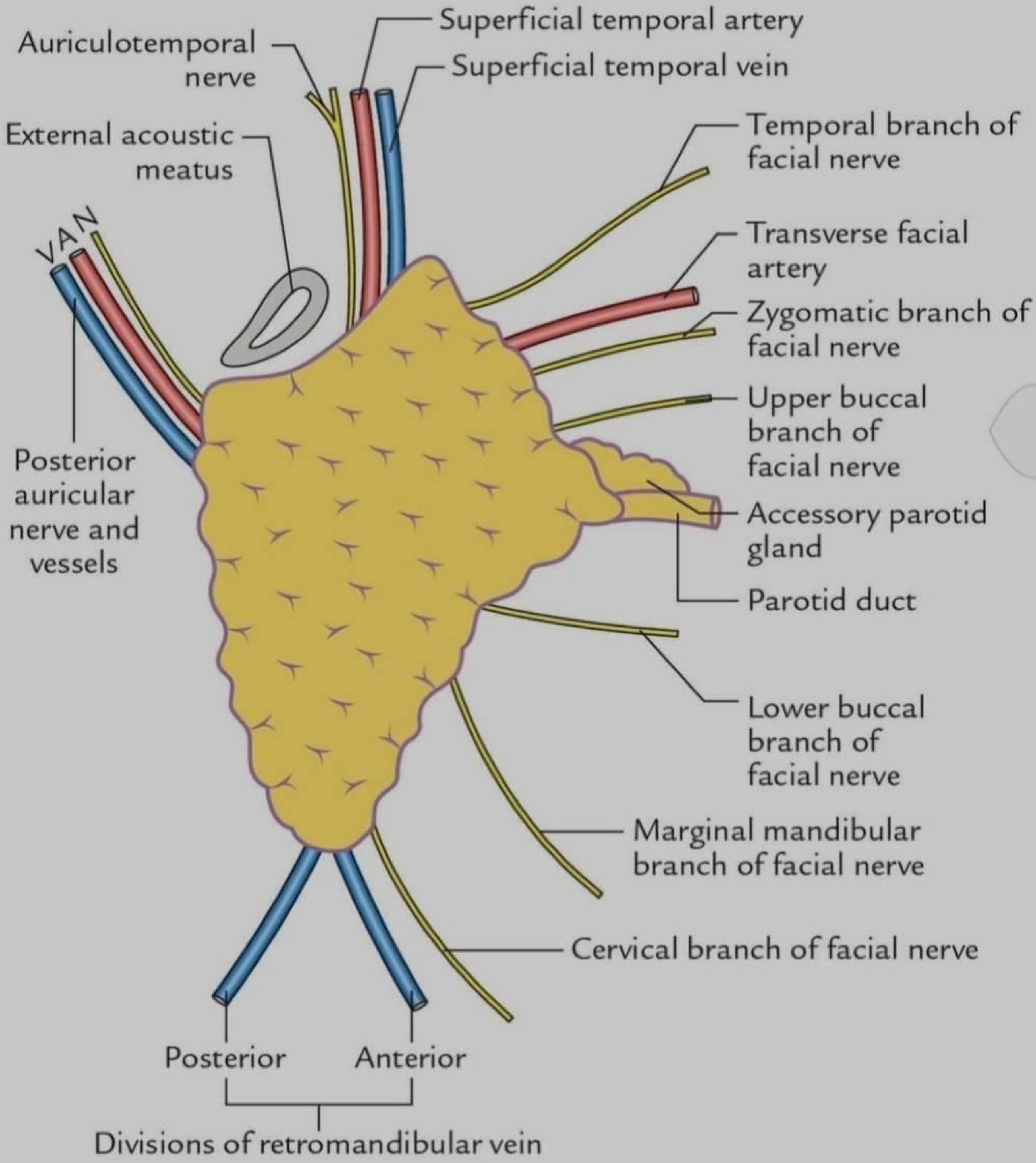
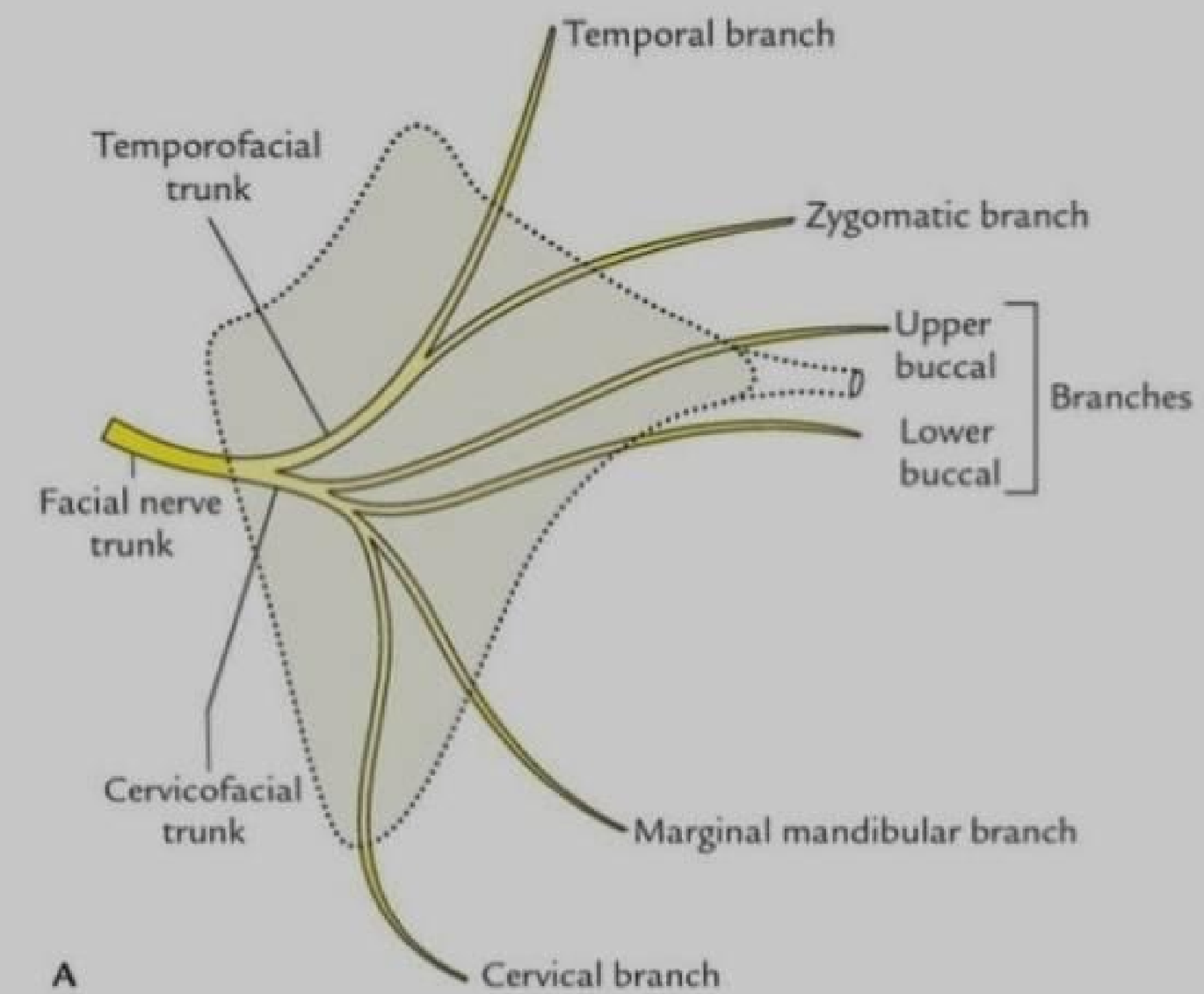
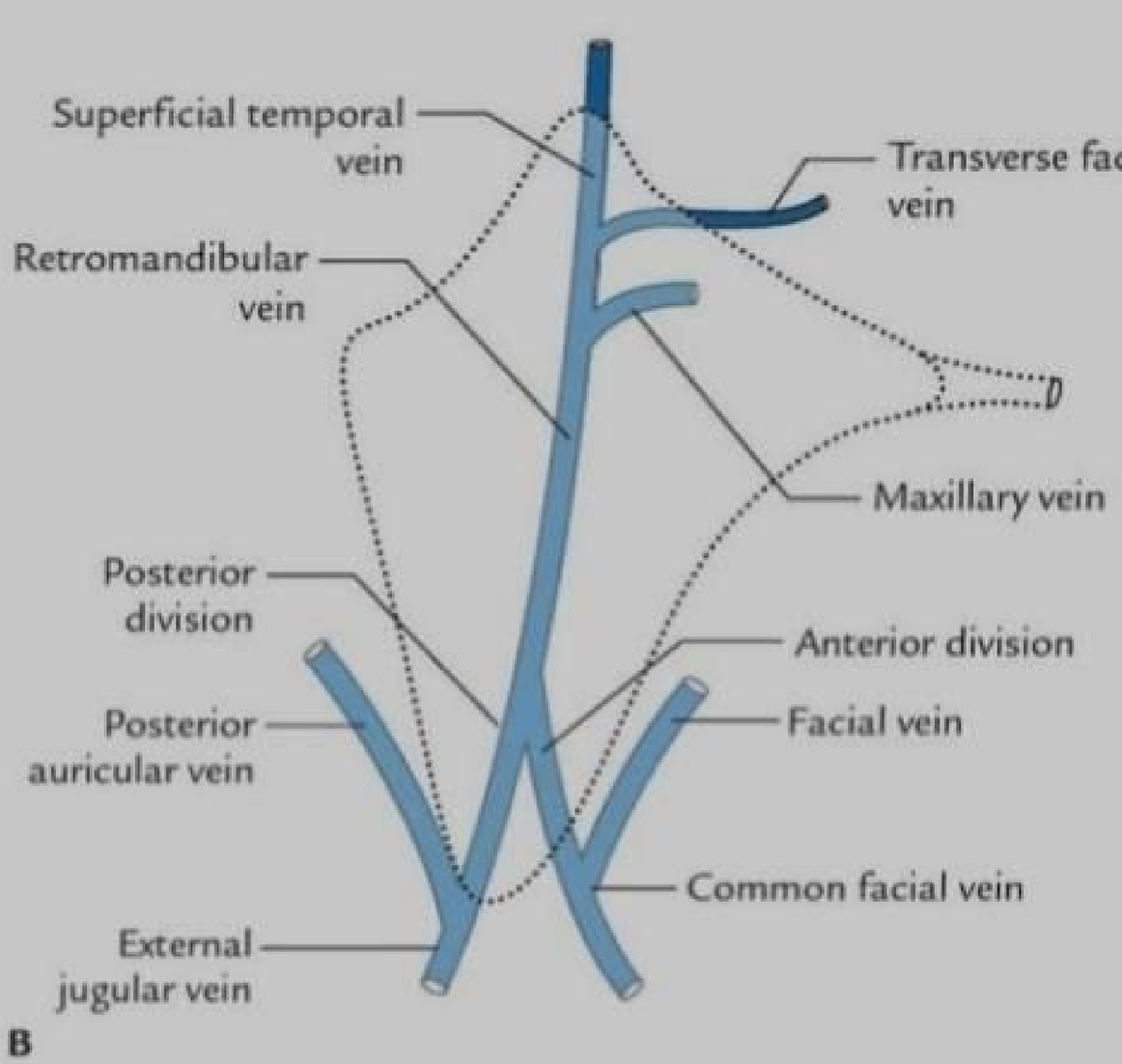
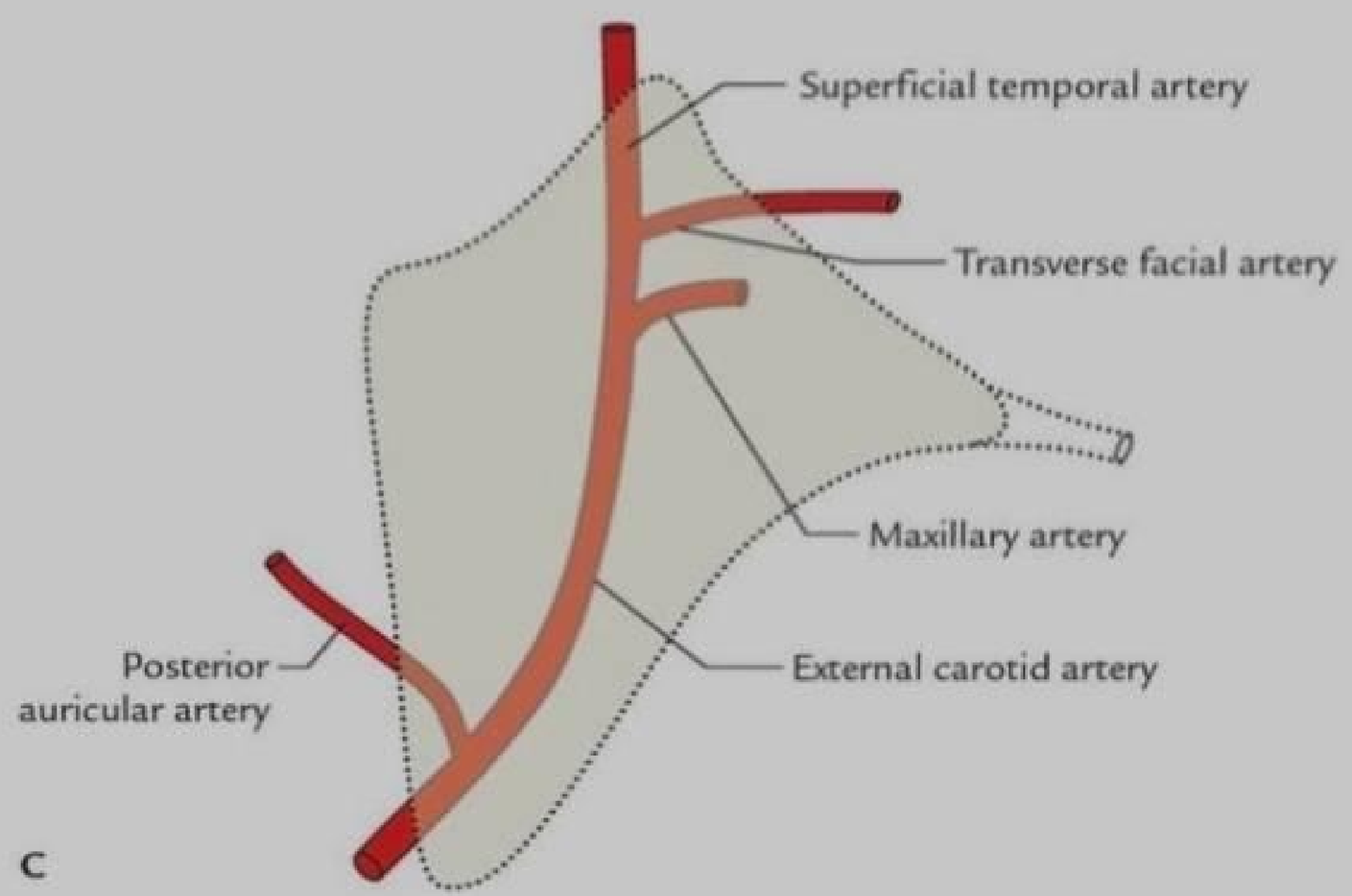


Fig. 8.5 The structures emerging at the periphery of the parotid gland.

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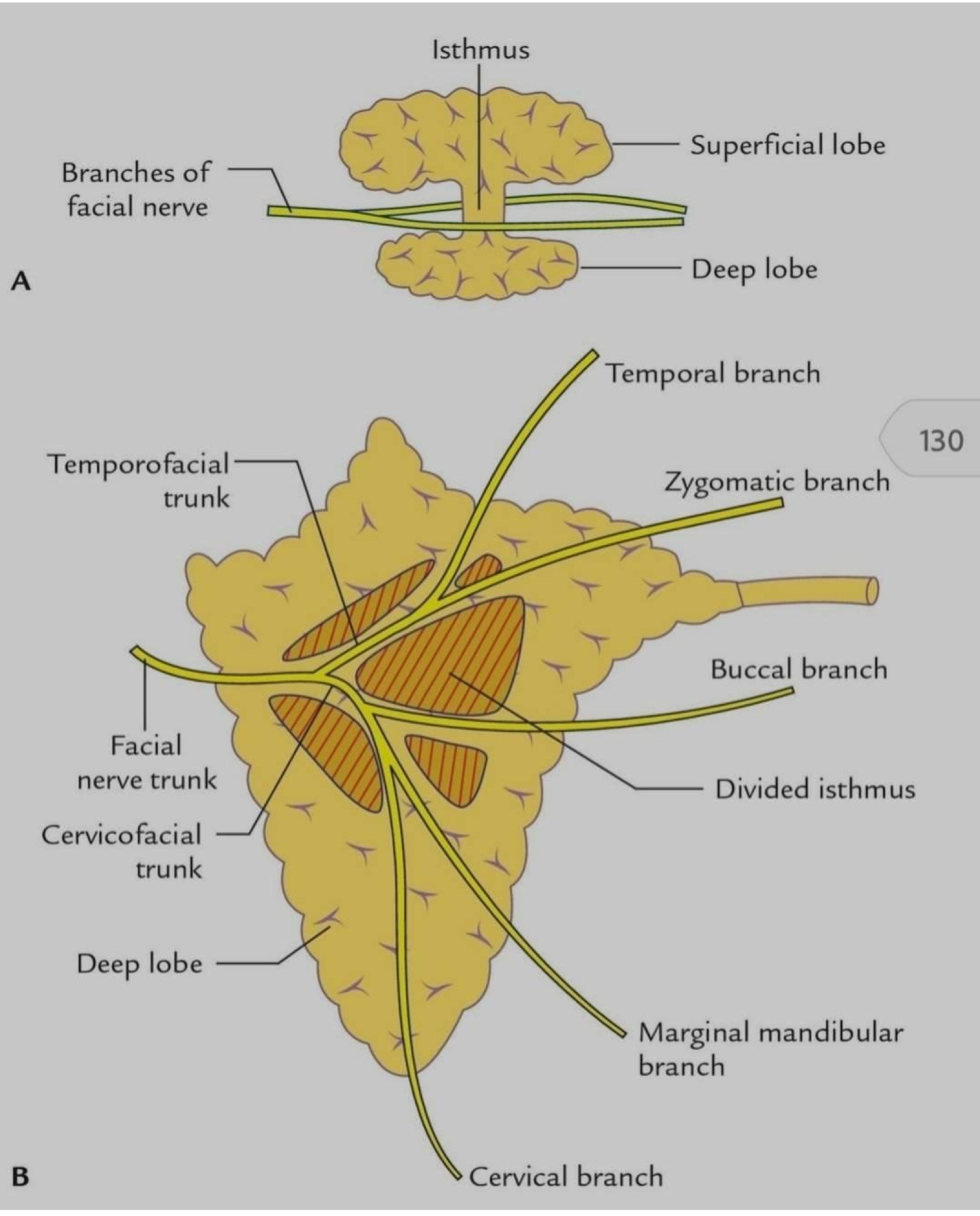






ig. 8.6 The structures traversing the parotid gland: A, facial nerve; B, retromandibular vein; C, external carotid artery.

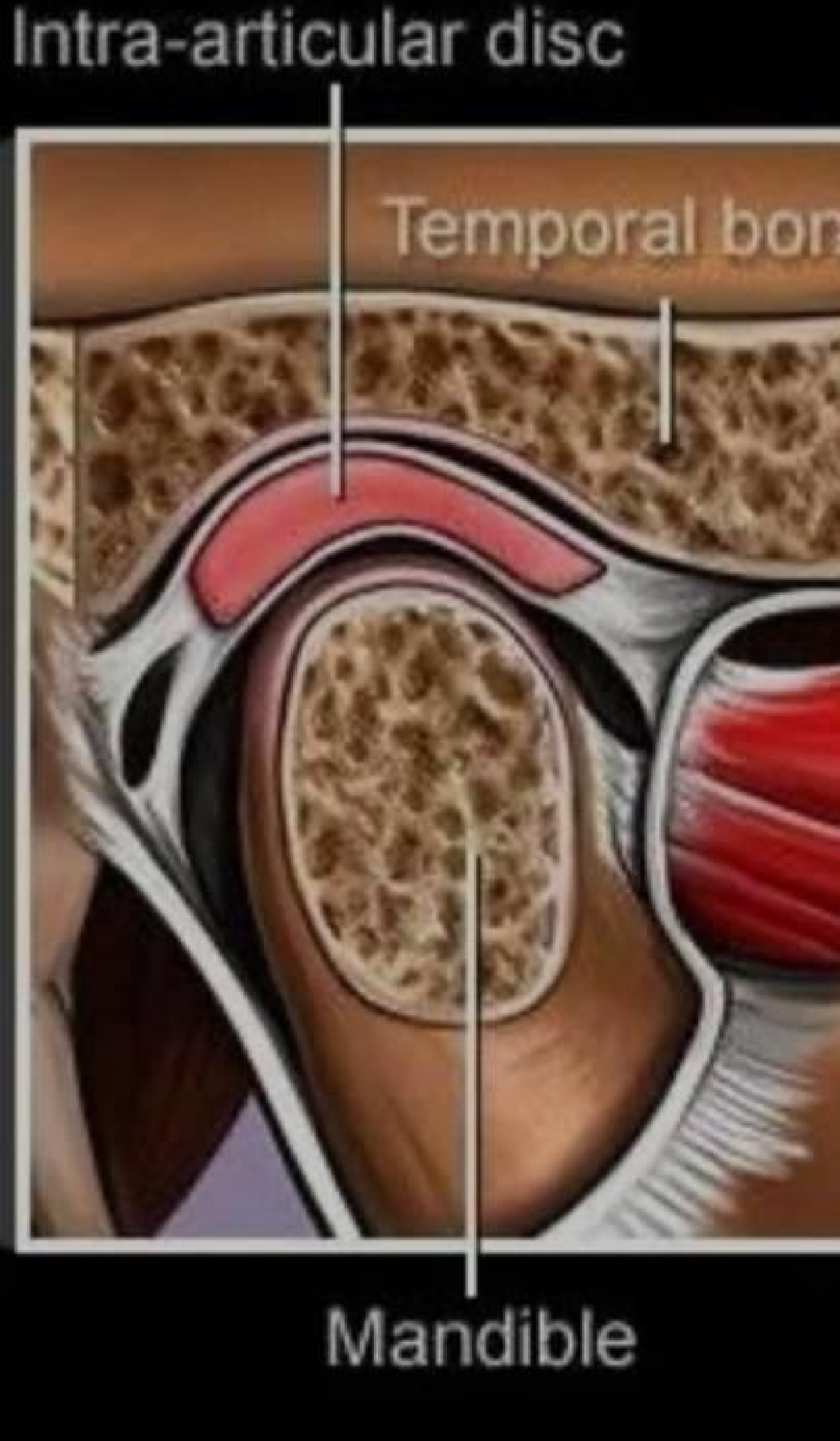
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Temporomandibular Joint Anatomy





Notes ->

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Temporomandibular Joint ...

Type of joint :- Synovial joint of condylar variety

articular surface - upper articular surface is formed by - temporal bone
- Articular tubercle
- mandibular fossa
- tympanic plate

- inferior articular surface → head of mandible

Ligament :-

- fibrous capsule - attach - above - articular tubercle
  circumference of mandibular fossa
  - Behind - squamotympanic fissure
  - Below - neck of mandible
- synovial membrane line - fibrous capsule

- lateral temporomandibular ligament - attach - above - articular tubercle
  Below - Posterolateral aspect of neck of mandible
- Strengthens → lateral part - capsule

- Spheno-mandibular ligament - attach → superiorly - spine of sphenoid
  inferiorly - lingula of mandibular foramen
  - accessory ligament
  - Remnant → Dorsal Part of "Meckel" Cartilage

- Stylomandibular ligament - attach → above - styloid process
  Below - Angle and Posterior border of ramus of mandible
  - accessory ligament
  - thickend - Deep cervical fascia

Articular Disc :- oval predominantly fibrous plate - divide joint → upper / lower compartment
- composed of - Anterior extension
  - Anterior band
    intermediate zone
  - Posterior band
    Bilaminar region
- Disc represents - degenerated primitive insertion of lateral Pterygoid
- Prevents friction between the articulating surfaces
- shock absorption
- "Proprioceptive fibre" - help → regulate movements of joint

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Temporomandibular Joint ... 2

Relations :-

- **Lateral**
  - skin
  - superficial fascia
  - Parotid gland
  - Temporal branch of facial nerve
- **medial**
  - tympanic plate
  - spine of sphenoid } bone
  - Auriculotemporal and chorda-tympani nerve
  - middle meningeal artery

AC MM

- **Anterior**
  - lateral pterygoid
  - masseteric nerve and artery
- **Posterior**
  - Parotid gland
  - superficial temporal vessel
  - Auriculo-temporal nerve
- **superior**
  - middle cranial fossa
- **inferior**
  - maxillary artery

Blood supply :- superficial temporal artery and vein
Maxillary arteries and vein

Nerve supply :- Auriculo-temporal nerve
masseteric nerve

movements :-

- Depression (open mouth) → lateral Pterygoid
  Digastric & Geniohyoid, Mylohy
- Elevation (close mouth) → Masseter and Medial Pterygoid
- Protrusion → medial & lateral Pterygoid
- Retrusion → Temporalis & masseter
- Lateral or side to side movement during chewing

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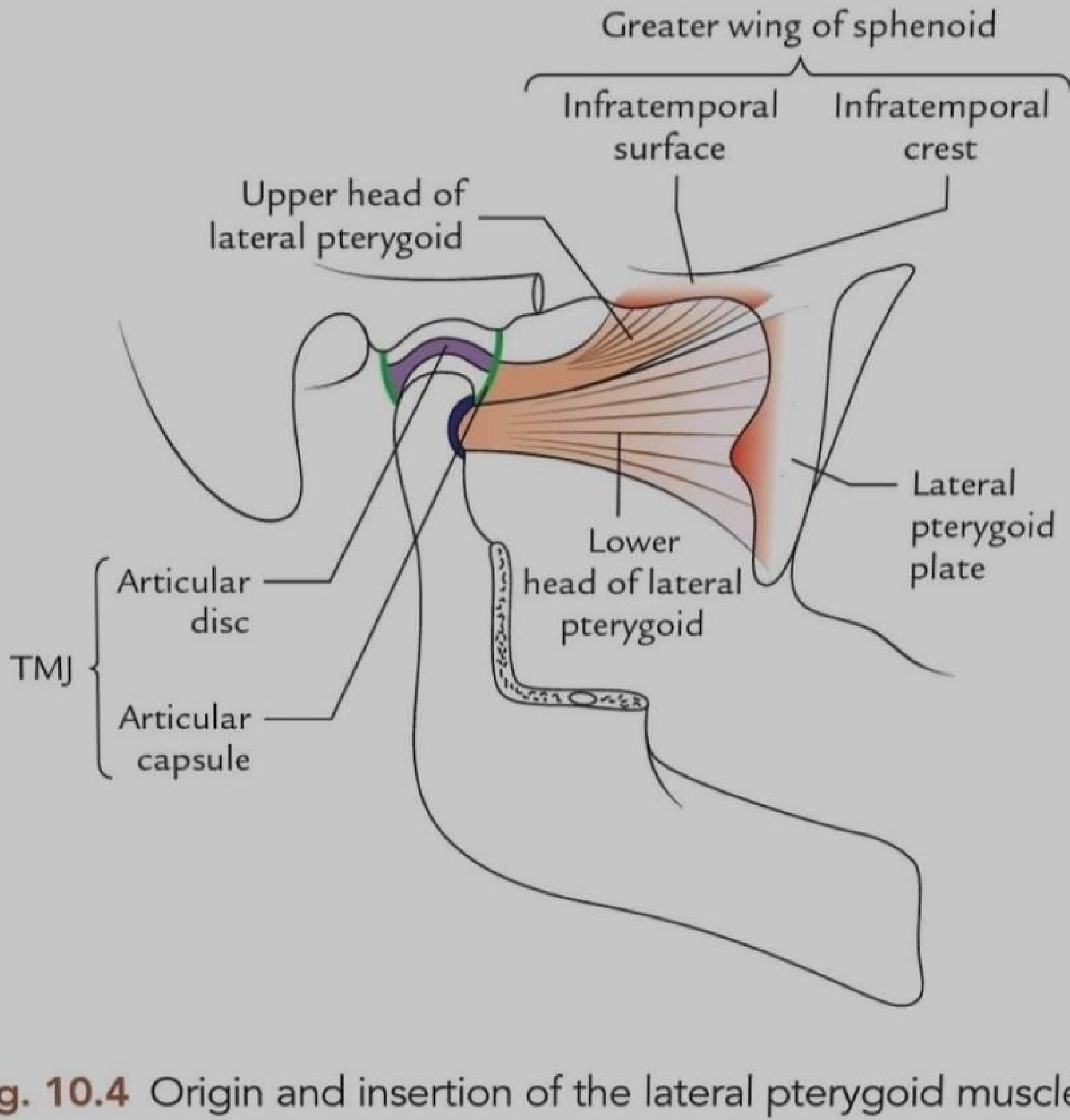


Fig. 10.4 Origin and insertion of the lateral pterygoid muscle.

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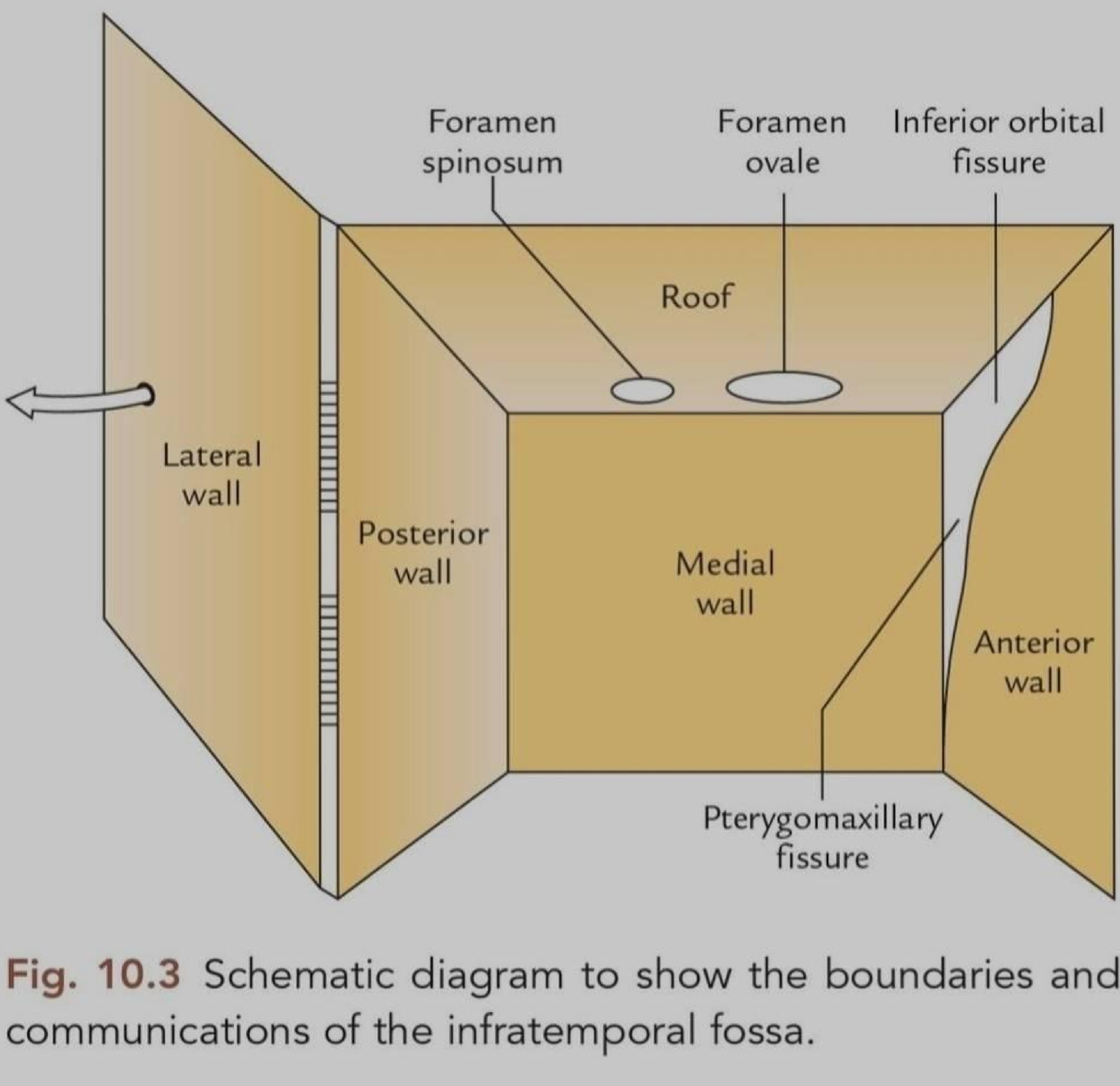
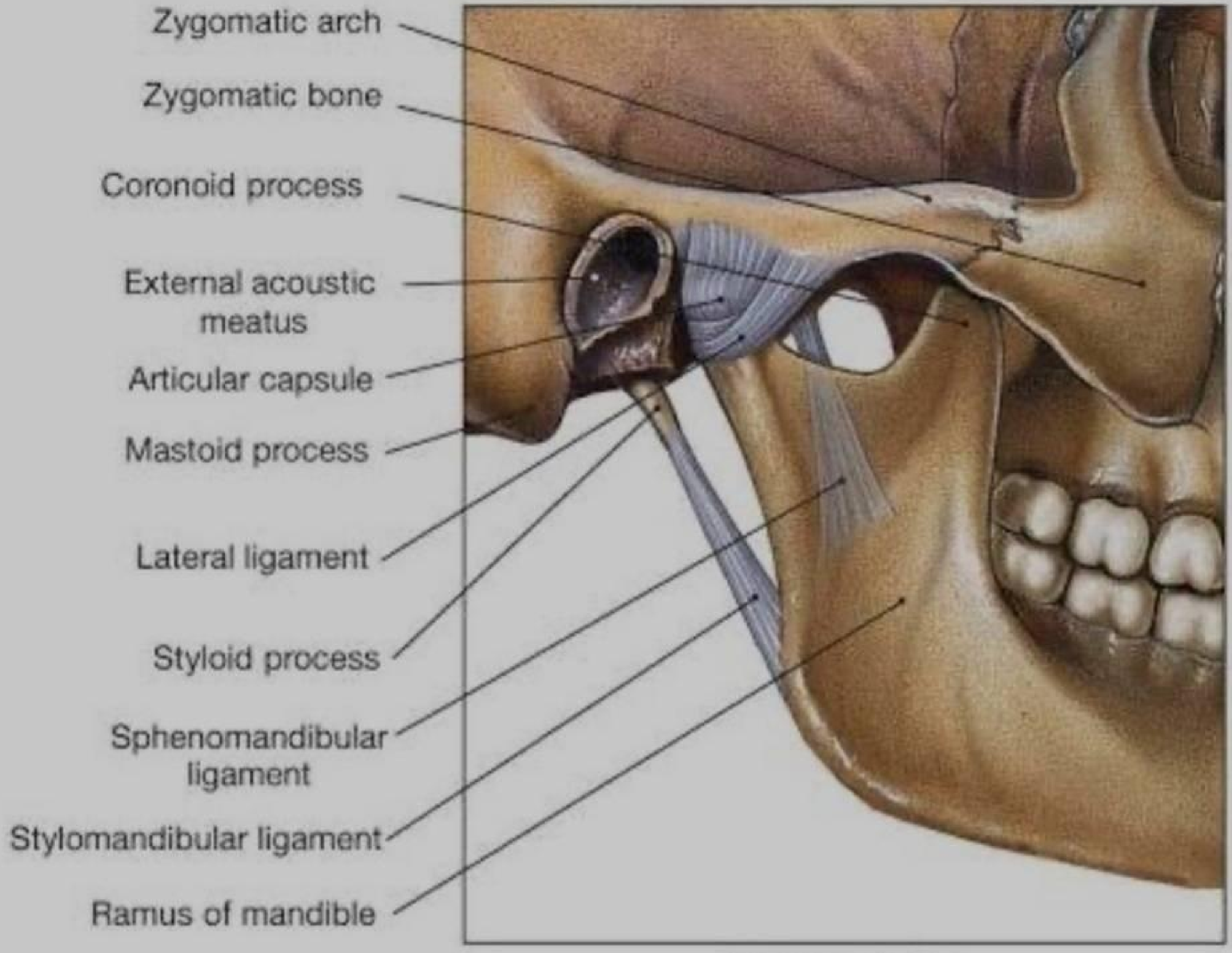


Fig. 10.3 Schematic diagram to show the boundaries and communications of the infratemporal fossa.

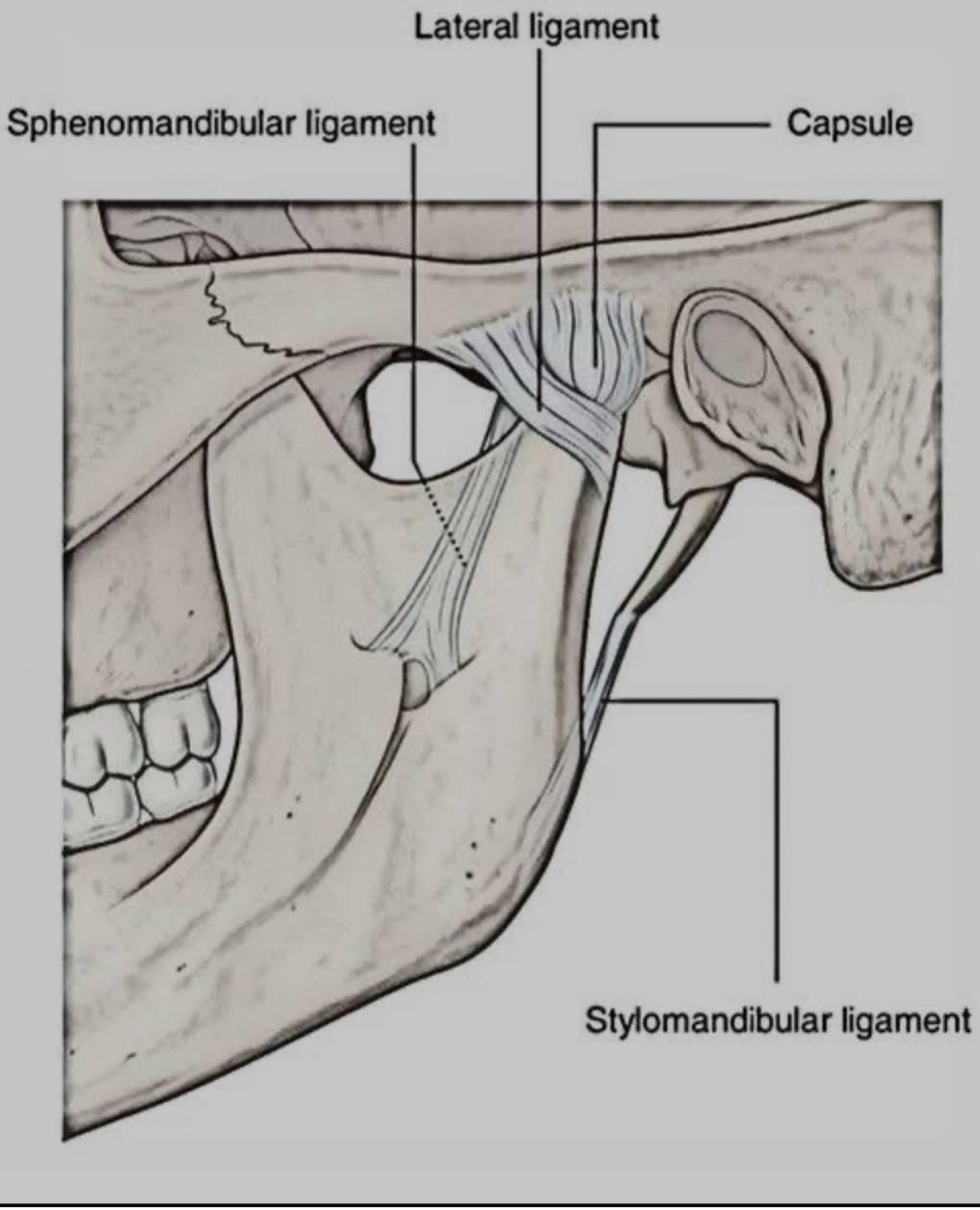
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(a) Lateral view

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Thyroid Gland Anatomy Notes->

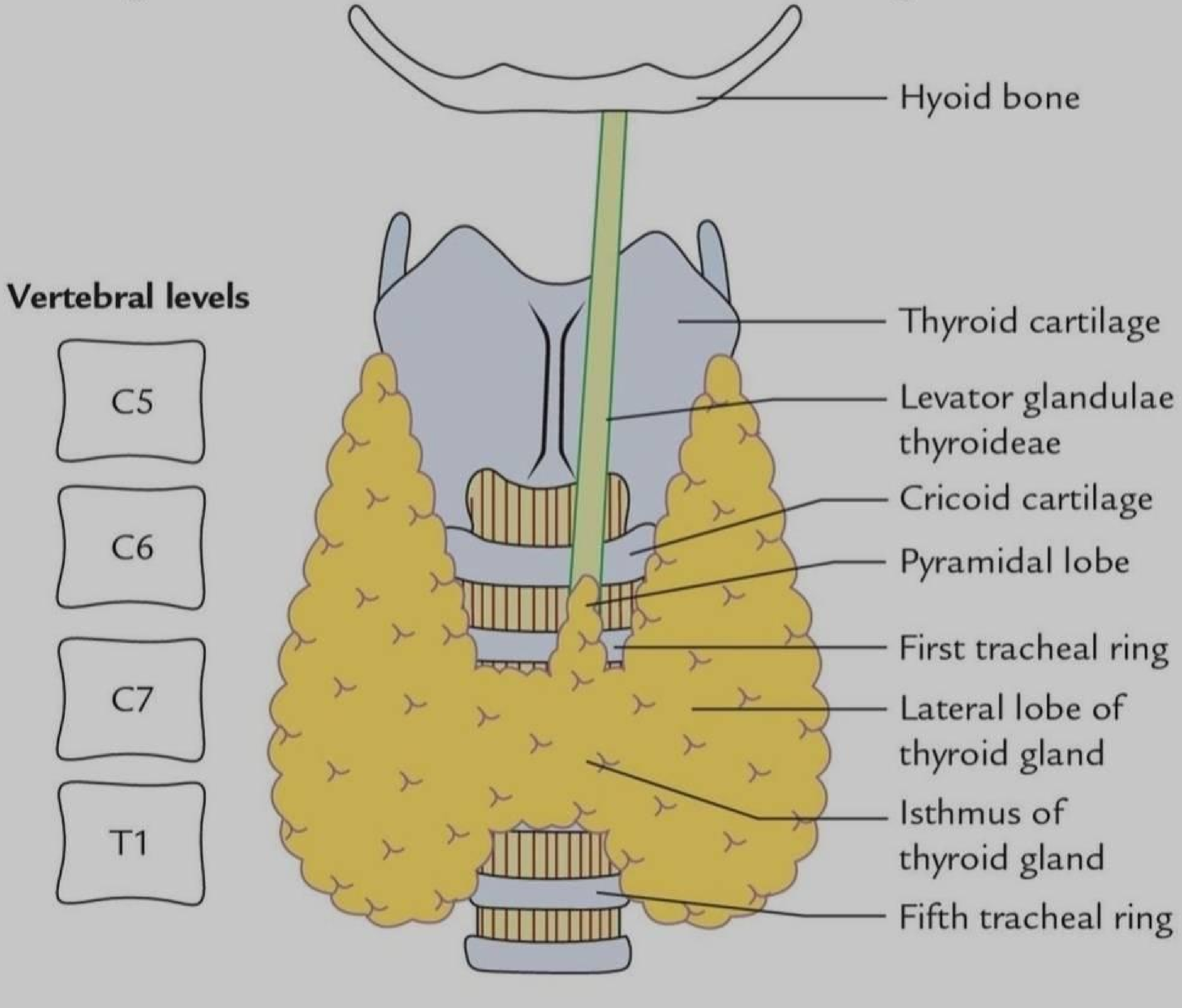


Fig. 11.2 Parts and extent of the thyroid gland.

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Thyroid Gland...

- largest endocrine gland
- lower part of front and side of neck
- function- Basal metabolic rate
  Calcium metabolism
  natural iodine synthesis

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- consist of - 2x lobes → right lobe, left lobe → joined to each other by - isthmus
- Situation - gland lies against - C_5-C_7, T_1 vertebrae
  - each lobe extend → middle of thyroid cartilage ↓ to fifth tracheal ring
  - isthmus extend → second to fourth tracheal ring

Capsule of Thyroid :-

- True capsule → Peripheral condensation of connective tissue of gland
  - dense capillary plexus → is present deep to true capsule
- false capsule → Derived from - Pretracheal layer of deep cervical fascia
  - thin in Posterior border of lobe
  - thick inner surface of gland ↓ to form Suspensory ligament (of Berry)

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Thyroid gland - 2 ...

Parts and Relations :-

- lobes :- Apex
  - Base
  - 3 x surfaces - Lateral
    medial
    Posterolateral
  - 2 x borders - Anterior
    Posterior
- isthmus :-
  - 2 x surface - Anterior
    Posterior
  - 2 x border - Superior
    inferior

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- Thyroid gland Relation :-
  - lobes :- Apex - Sternothyroid muscle
    - superior thyroid artery
    - external laryngeal nerve

    Base - - inferior thyroid artery
    - recurrent laryngeal nerve

    surface :- lateral - sternothyroid
    sternohyoid
    superior belly of omohyoid
    sternocleidomastoid

    Medial - 2 x tube - trachea
    oesophagus

    2 x muscle - inferior constrictor
    cricothyroid

    2 x nerve - external
    recurrent
    } laryngeal nerve

    Posterolateral - Carotid sheath

Border :- Anterior border - thin
- Superior thyroid artery

Posterior border - thick
- inferior thyroid artery
- Parathyroid gland
- thoracic duct

Isthmus :- Anterior surface - sternothyroid
sternohyoid

Posterior surface - 2nd to 4th tracheal rings

Upper surface - superior thyroid arteries

lower surface - inferior thyroid vein

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Arterial supply :-

- Superiar thyroid artery
  - first branch of ECA
  - Downward, forwards run
  - relation - External laryngeal nerve
  - Apex of lobe
- inferiar thyroid artery
  - branch of - thyrocervical trunck
    ↓
    subclavian artery
  - upward, medially run
  - relation - recurrent laryngeal nerve
  - Base of gland
- Accessory thyroid artery - arise from tracheal and oesophageal artery

Venous drainage :-

- Superiar thyroid vein → drain → internal jugular vein
- middle thyroid vein → internal jugular vein
- inferiar thyroid vein → left brachiocephalic vein

Lymphatic :- Deep cervical lymph nodes

Nerve :- derived from - meddle cervical ganglion
Partly from - Superiar and inferiar cervical ganglia

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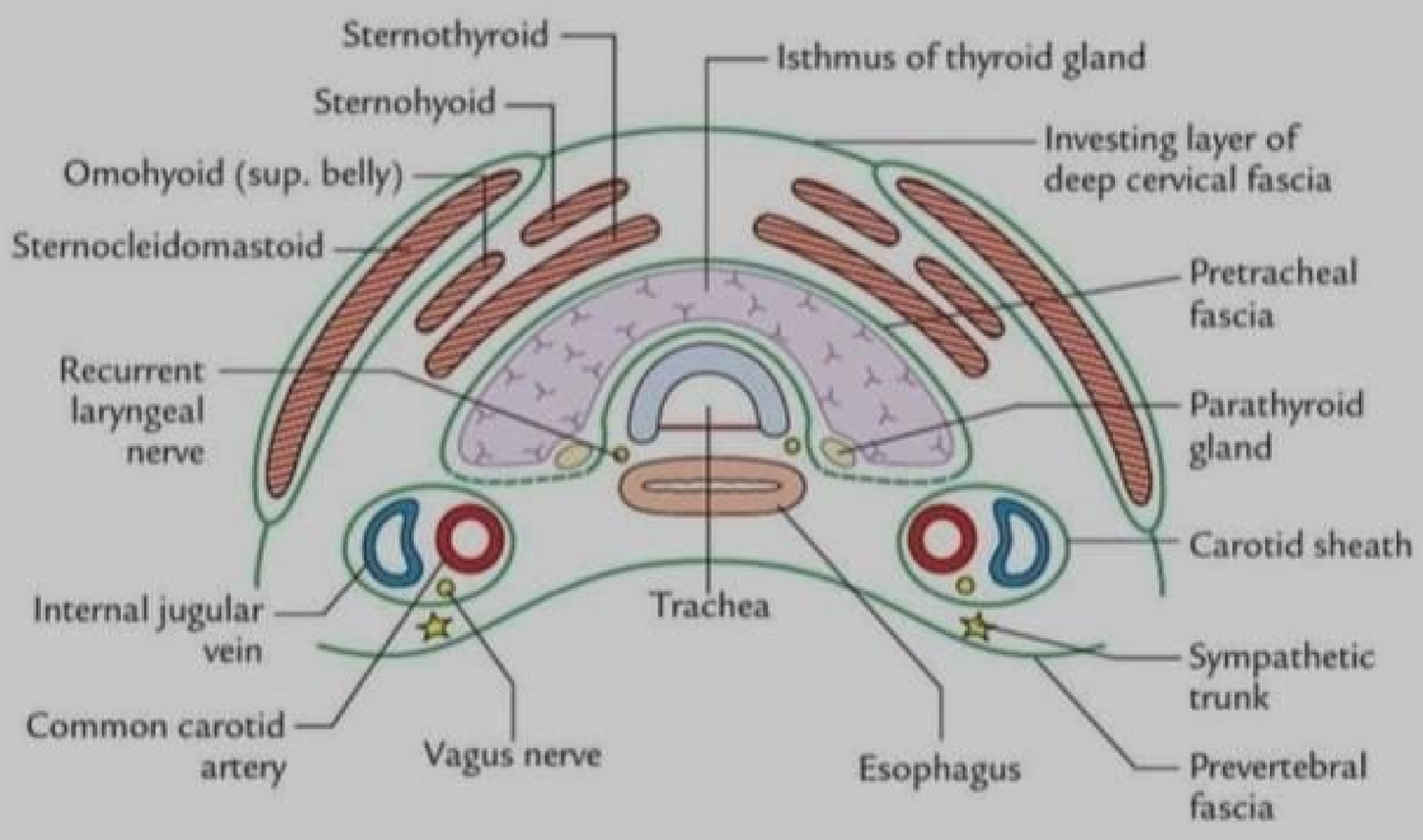
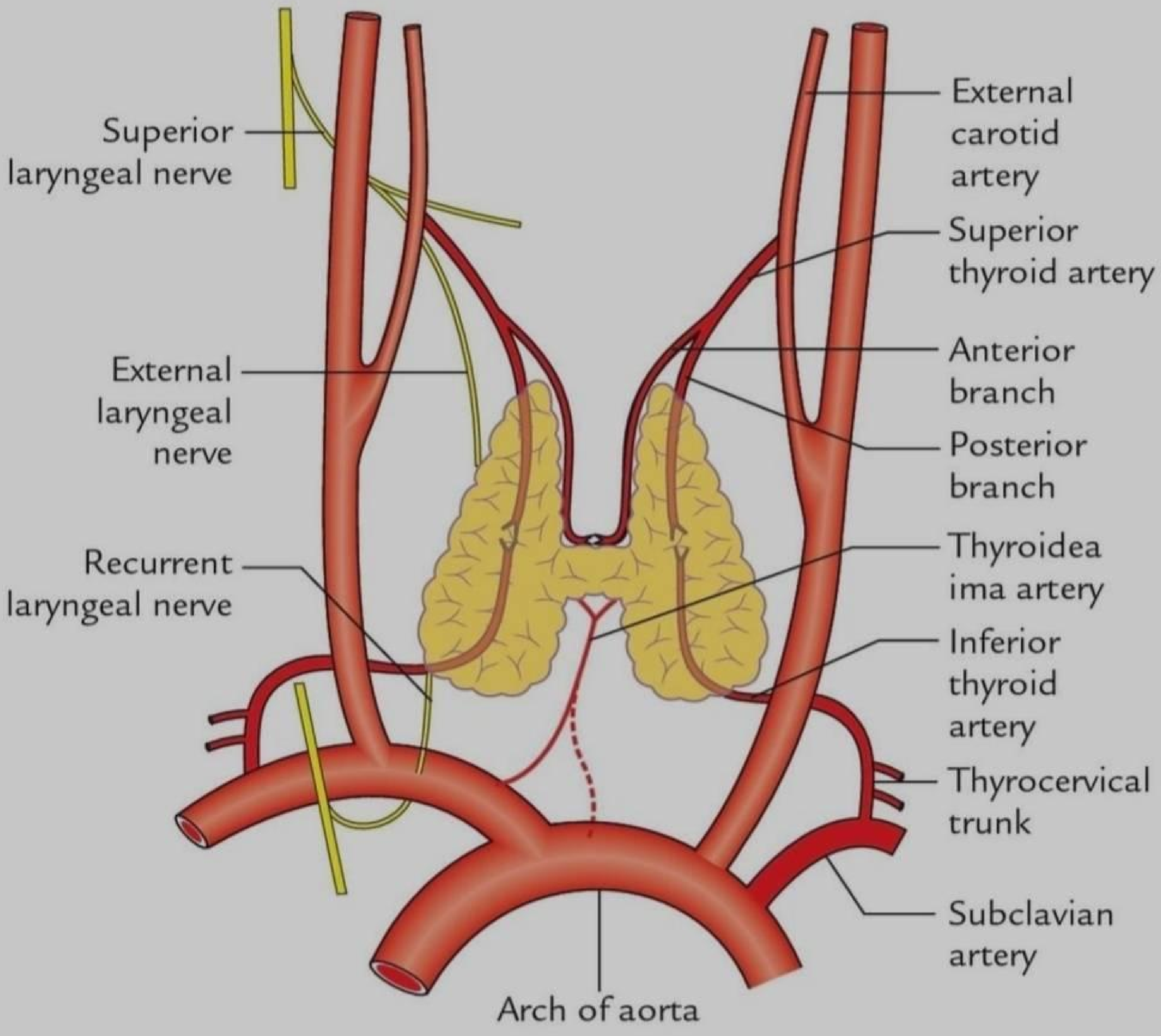
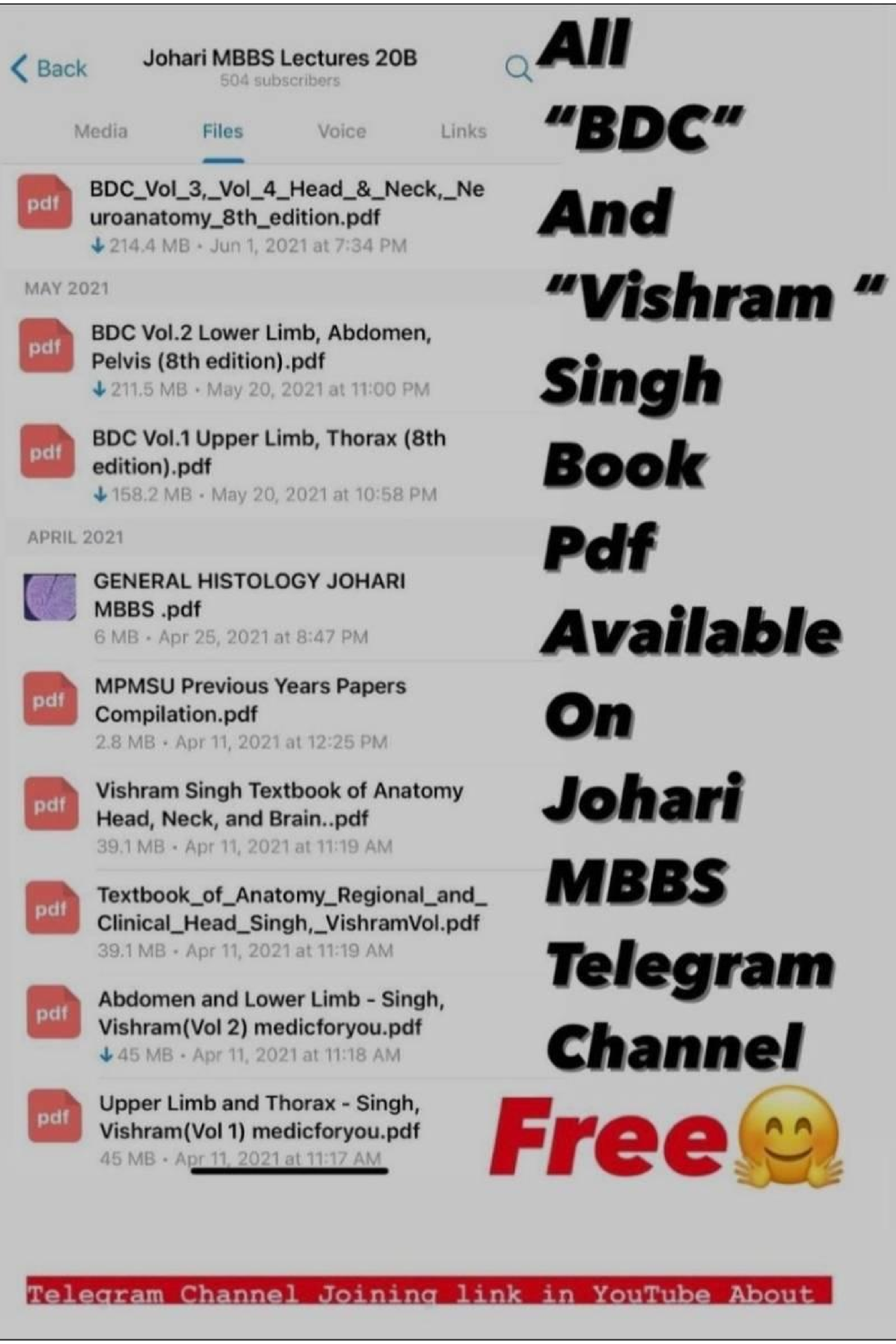


Fig. 11.5 Transverse section of the anterior part of the neck at the level of thyroid isthmus, showing relations of the thyroid gland.

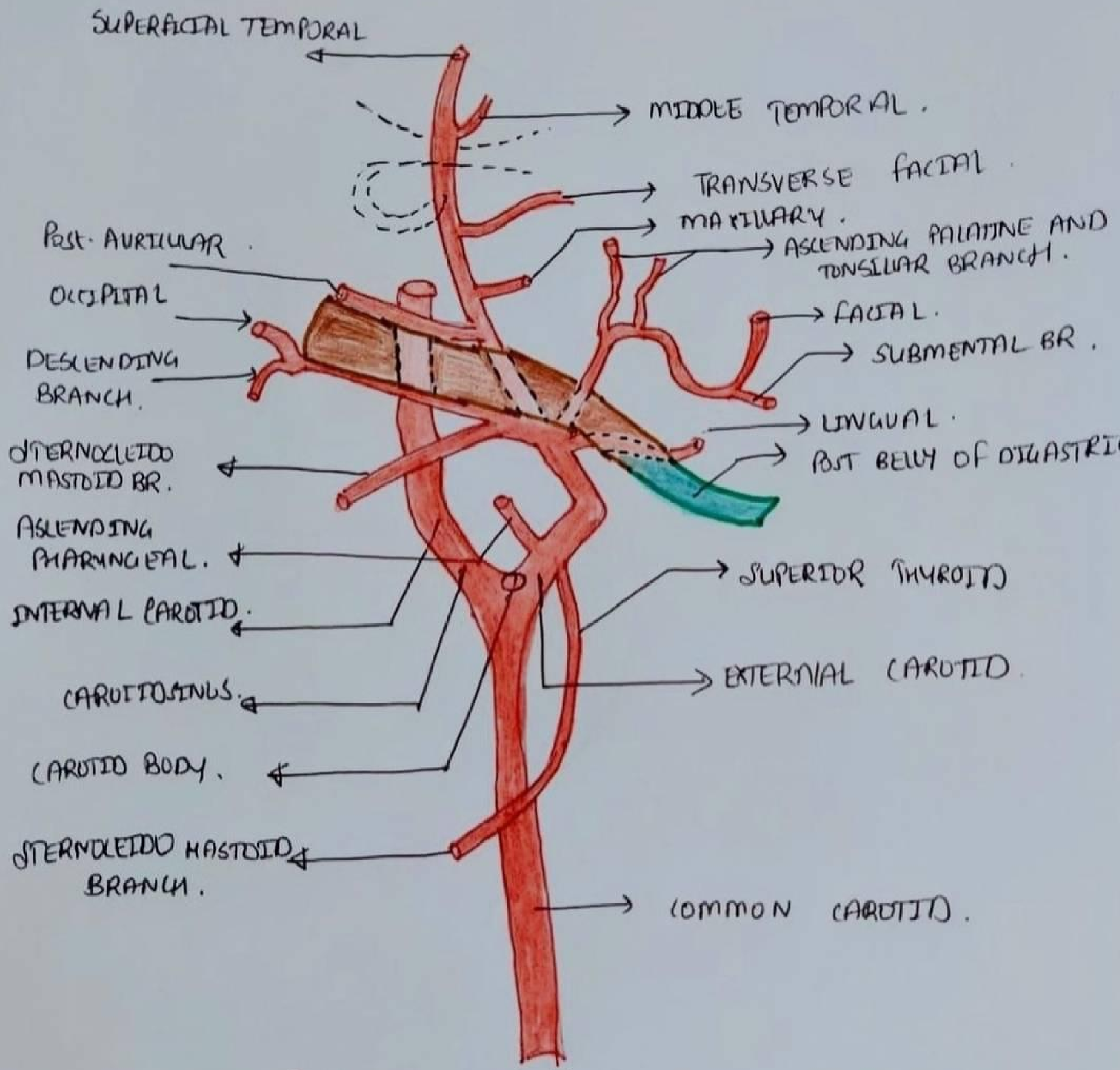
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External Carotid Artery Notes->

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External Carotid Artery ...

ECA is terminal branch of common carotid artery
. chief artery of supply → front of neck and face

Course :-

- Begin in → carotid triangle at level of upper border of thyroid cartilage
  - opposite to C_3 and C_4
  - run upward, backward and laterally

- terminate → behind neck of mandible
  - dividing → maxillary and superficial temporal arteries.
- slightly curved course → lower anteromedial → to ICA
  upper anterolateral
- In carotid triangle → is superficial
  lie → upper cover of anterior border of sternocleidomastoid muscle
- Above carotid → lie deep in parotid gland
  related → retro-mandibular vein and facial nerve

Branches :- Anterior — Superior thyroid artery
Lingual
facial

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Posterior — occipital
Posterior auricular artery

medial — Ascending pharyngeal artery

Terminal → maxillary artery
superficial temporal artery

- **Superior Thyroid artery :-**
  - Branch of ECA → just below greater cornua of hyoid bone
  - run - downward, forward parallel to lateral lobe of thyroid
  - supply - Thyroid gland terminal branch → Superior laryngeal artery

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- Lingual artery : arise → opposite the tip of greater cornua
  - branches — suprahyoid
    Dorsal lingual
    sub-lingual

- facial artery :- SAME as video

- Occipital artery :
  - opposite origin of facial artery
  - crossed at it's origin hypoglossal nerve

- Posterior auricular artery :
  - above posterior belly of digastric
  - upward backward → deep → Parotid gland and Styloid Process
  - supply → skin → over mastoid process and back of scalp

- Ascending pharyngeal artery :
  - smallest branch
  - arises → medial side of ECA
  - vertically upward → b/w side wall of pharynx tonsil and medial wall of middle ear

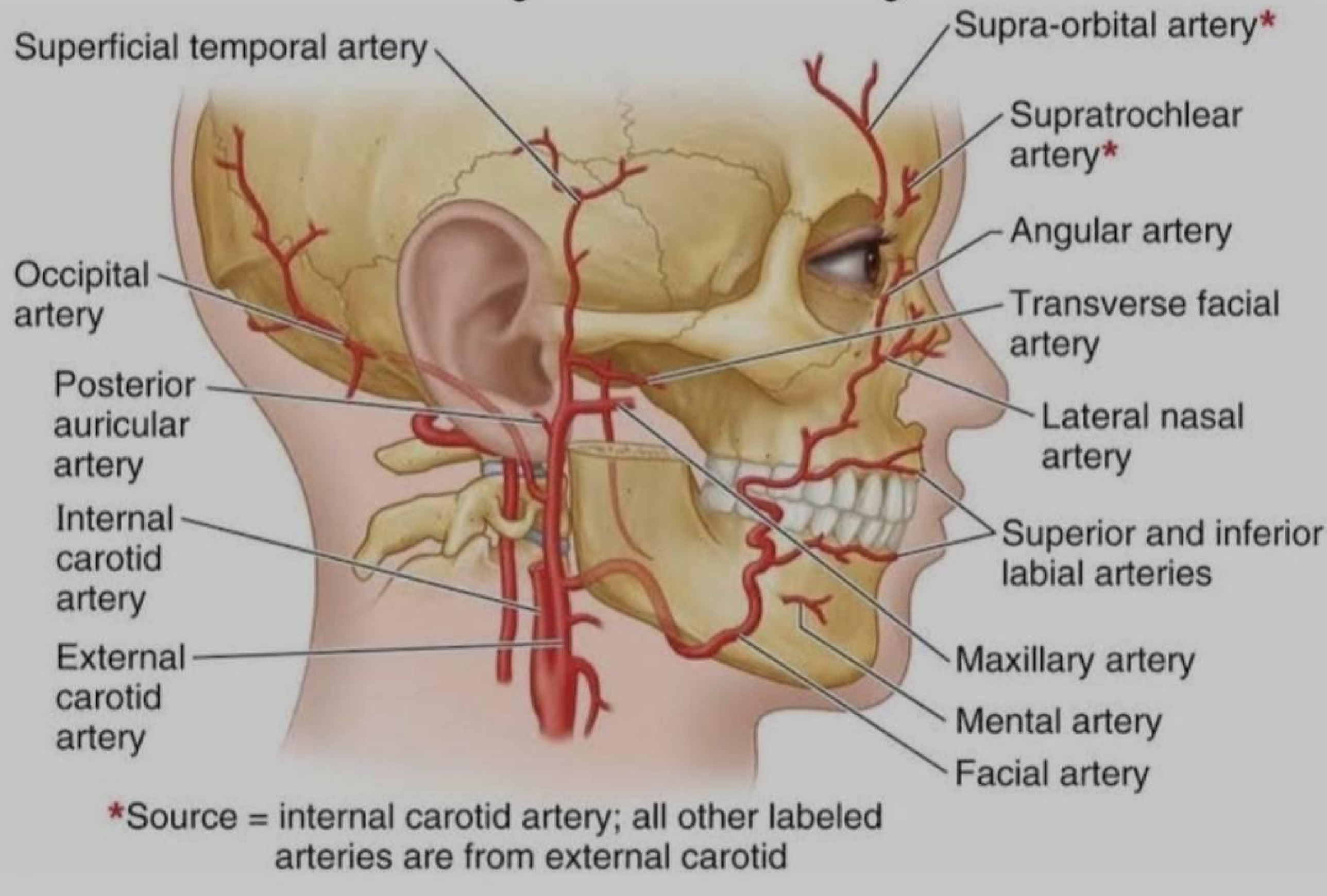
- maxillary artery :
  - largest branch
  - behind neck of mandible under parotid gland
  - Below auriculo-temporal nerve

- superficial temporal artery :
  - smaller terminal branch
  - behind neck of mandible
  - divided → anterior and posterior branch
    ↓ supply
    temple and scalp

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Facial Artery Anatomy Notes->



*Source = internal carotid artery; all other labeled arteries are from external carotid

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FACIAL Artery...

Head (2)

- Chief artery of face
- Branch → of external carotid artery $\xrightarrow{\text{give off in}}$ Carotid triangle $\xrightarrow{\text{just above}}$ greater cornua of hyoid bone

Course :-

- Enter face → winding around mandible $\xrightarrow{\text{and}}$ Piercing → deep cervical fascia
- first run upward forward → 1.25 cm lateral to angle of mouth
- ascends by side of nose $\xrightarrow{\text{up to}}$ angle of eye
- terminate by supplying lacrimal sac $\xrightarrow{\text{by}}$ anastomosing with dorsal nasal branch
- lie between superficial and deep muscle of ~~fascia~~ face

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Branches :-

Anterior branch :- inferior labial
superior labial
lateral ~~labiat~~ nasal

Posterior branch :- small and unnamed

Anastomoses :-

large anterior $\xrightarrow{\text{anastomoses}}$ similar branch of opposite side
smaller posterior → transverse facial
infra-orbital arteries

terminal branch of facial artery → Ophthalamic artery.

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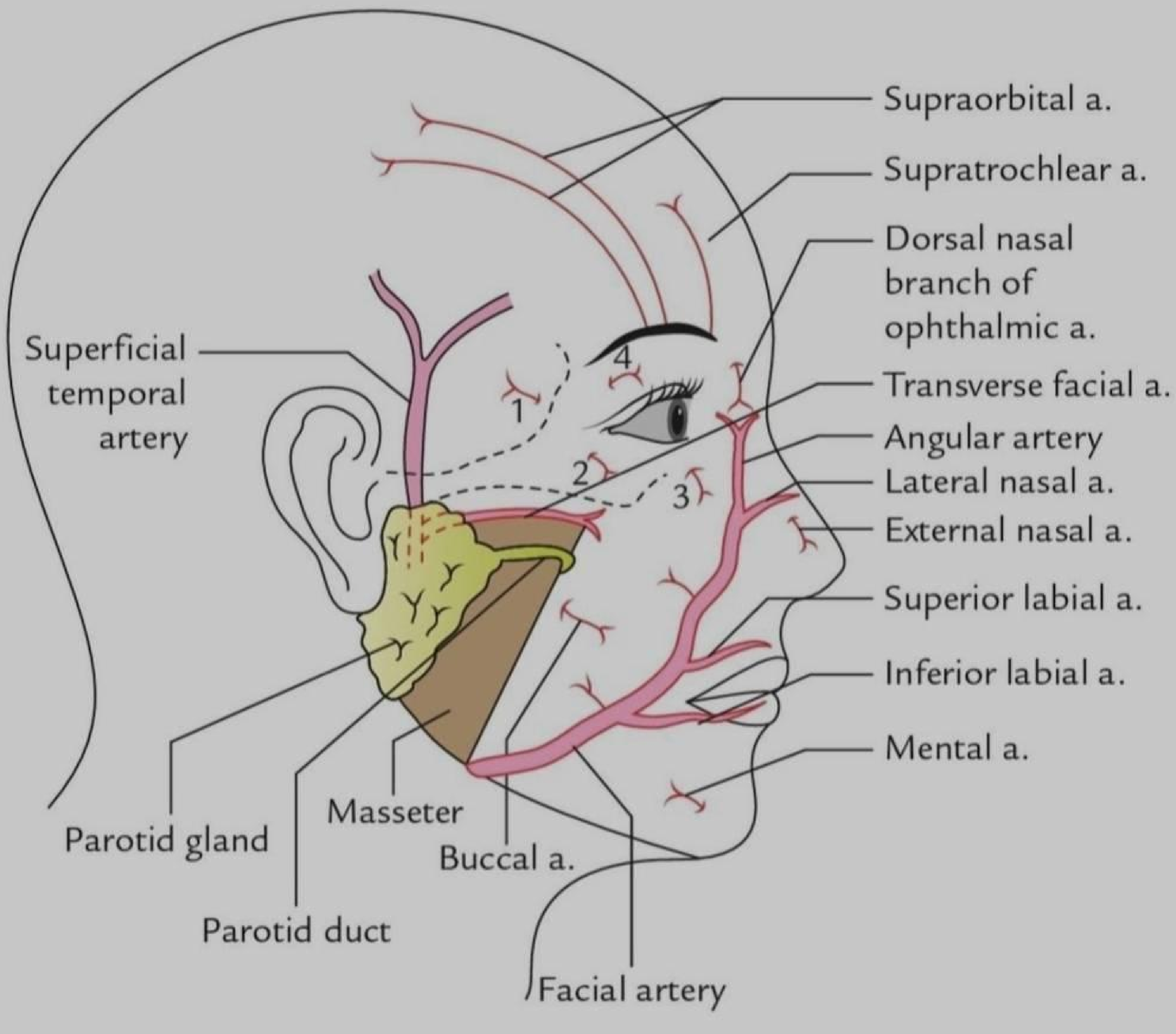
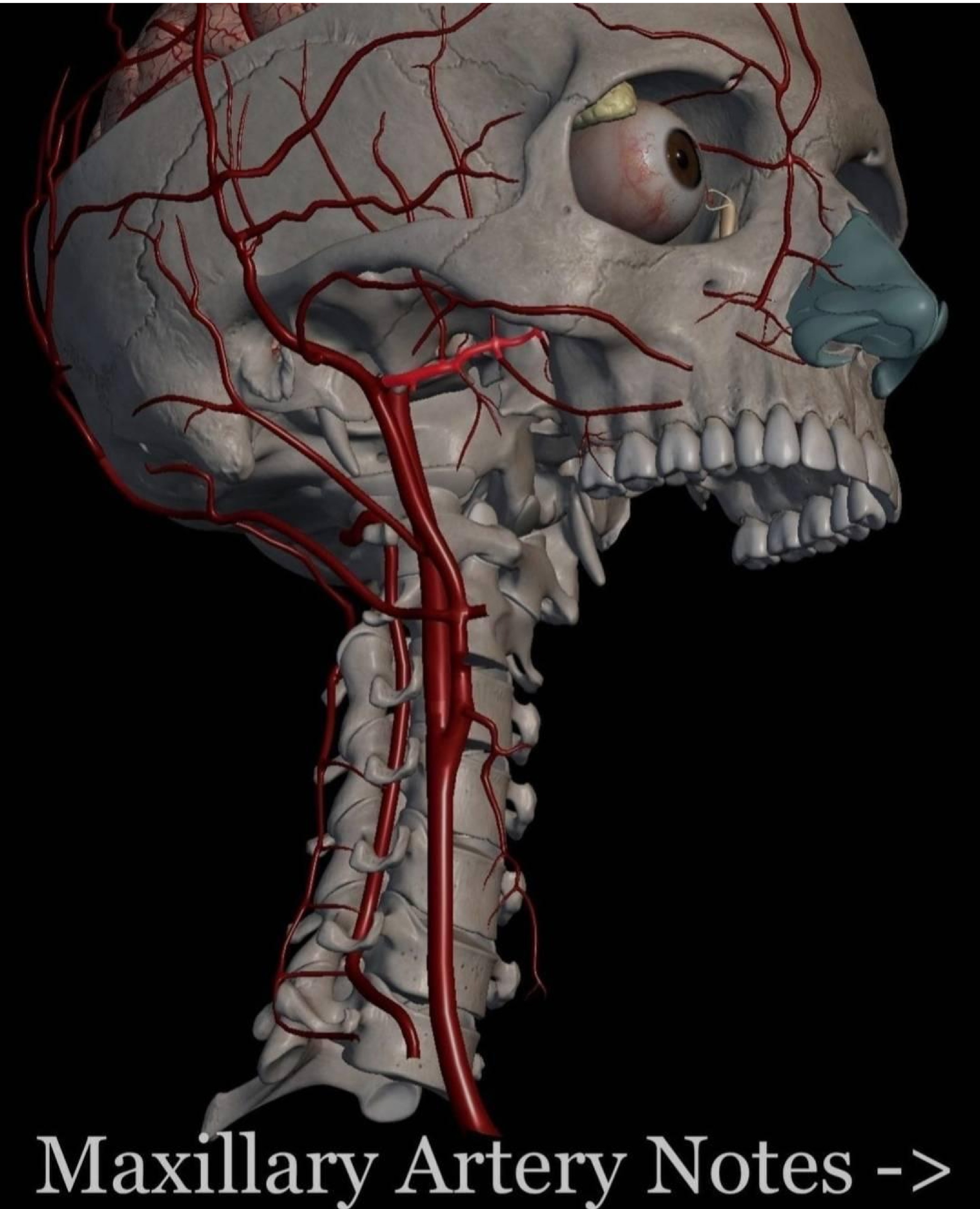


Fig. 3.17 Arterial supply of the face (1 = zygomaticotemporal, 2 = zygomaticofacial, 3 = infraorbital, 4 = palpebral branch of lacrimal artery).

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MAXILLARY Artery ...

- largest terminal branch of ECA $\xrightarrow[\text{off}]{\text{given}}$ behind the neck of mandible
- supplies :- external and middle ear
  dura mater
  muscle of temporal and infra-temporal region
  nose and para-nasal air sinuses
  Palate
  roof of pharynx

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Course and Relation - Divided into 3x parts :-

- **first (mandibular) part** :- - run horizontally forwards
  - B/w neck of mandible and spheno-mandibular ligament
- **second (pterygoid) part** :- - run upward and forwards
  - superficial to lower head of lateral pterygoid
- **Third (Pterygopalatine) part** :- - passes b/w two head of lateral pterygoid
  - through pterygomaxillary fissure $\xrightarrow{\text{to}}$ enter ↓ Pterygo palatine fossa

Branches :-

Part-1 (5)

- Deep auricular artery $\xrightarrow{\text{supply}}$ external acoustic meatus, tympanic membrane
- Anterior tympanic $\xrightarrow{\text{supply}}$ middle ear, medial surface of tympanic membrane
- middle meningeal artery $\xrightarrow{\text{enter}}$ foramen spinosum ↓ reach middle cranial fossa
  - $\xrightarrow{\text{supply}}$ large frontal branch & small parietal
- Accessary meningeal artery $\xrightarrow{\text{enter}}$ foramen ovale ↓ reach middle cranial fossa
  - $\xrightarrow{\text{supply}}$ infra-temporal fossa
- Inferior alveolar artery $\xrightarrow{\text{enter}}$ mandibular foramen
  - $\xrightarrow{\text{supply}}$ mandible and root of tooth

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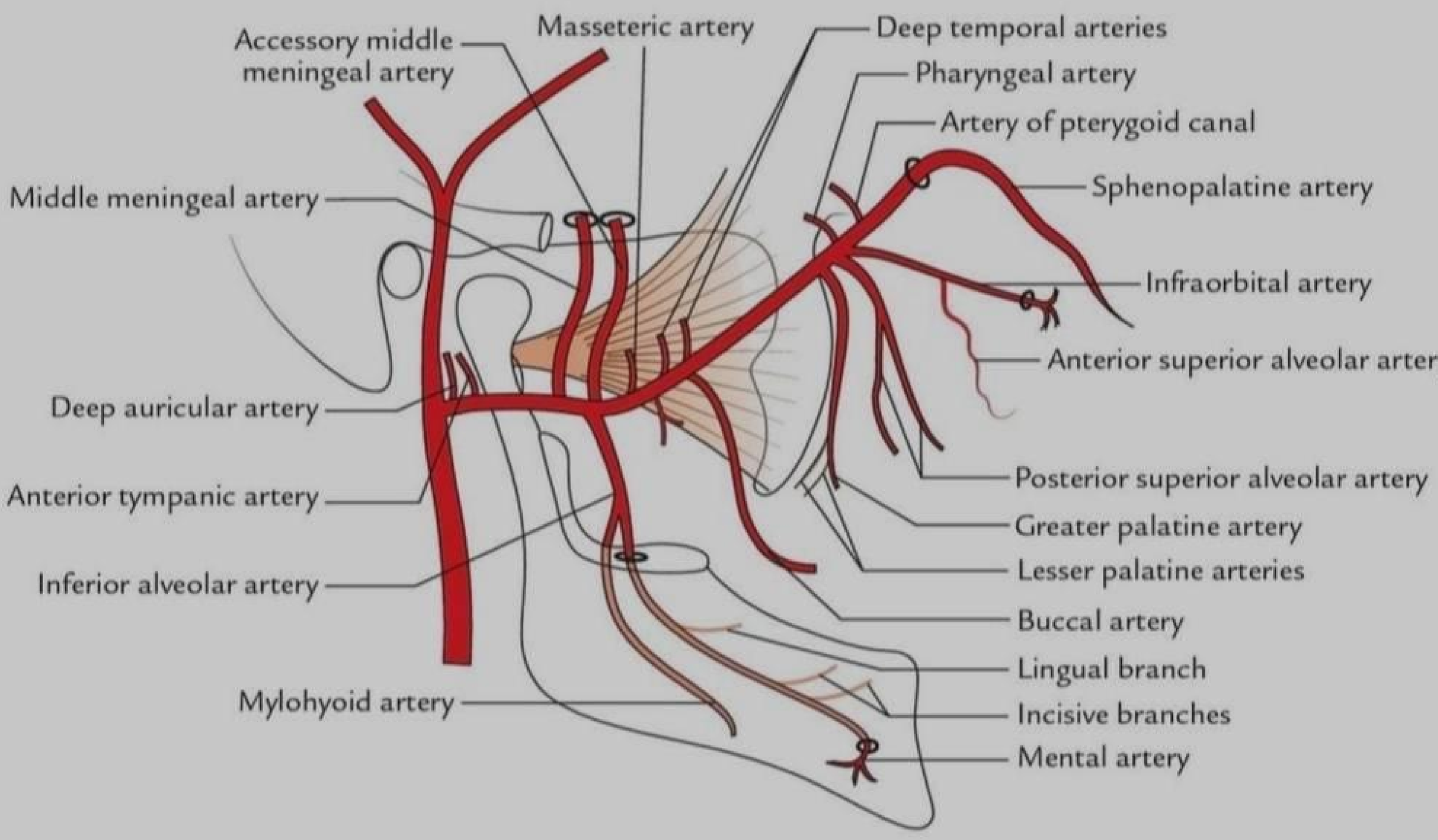
PART-2 (4):
- masseteric, Deep temporal] enter → lateral → Deep to temporalis muscle
- To pterygoid muscle, Buccal branch] supply → skin over cheek

PART-3 (5):
- Posterior superior alveolar artery → enter → Pterygomaxillary fissure
  - supply → molar and premolar teeth, maxillary air sinus
- Infra-orbital artery → enter → Pterygomaxillary fissure → inferior-orbital fissure → enter orbit
  - supply → orbital branches, middle superior alveolar branch, anterior superior alveolar branch

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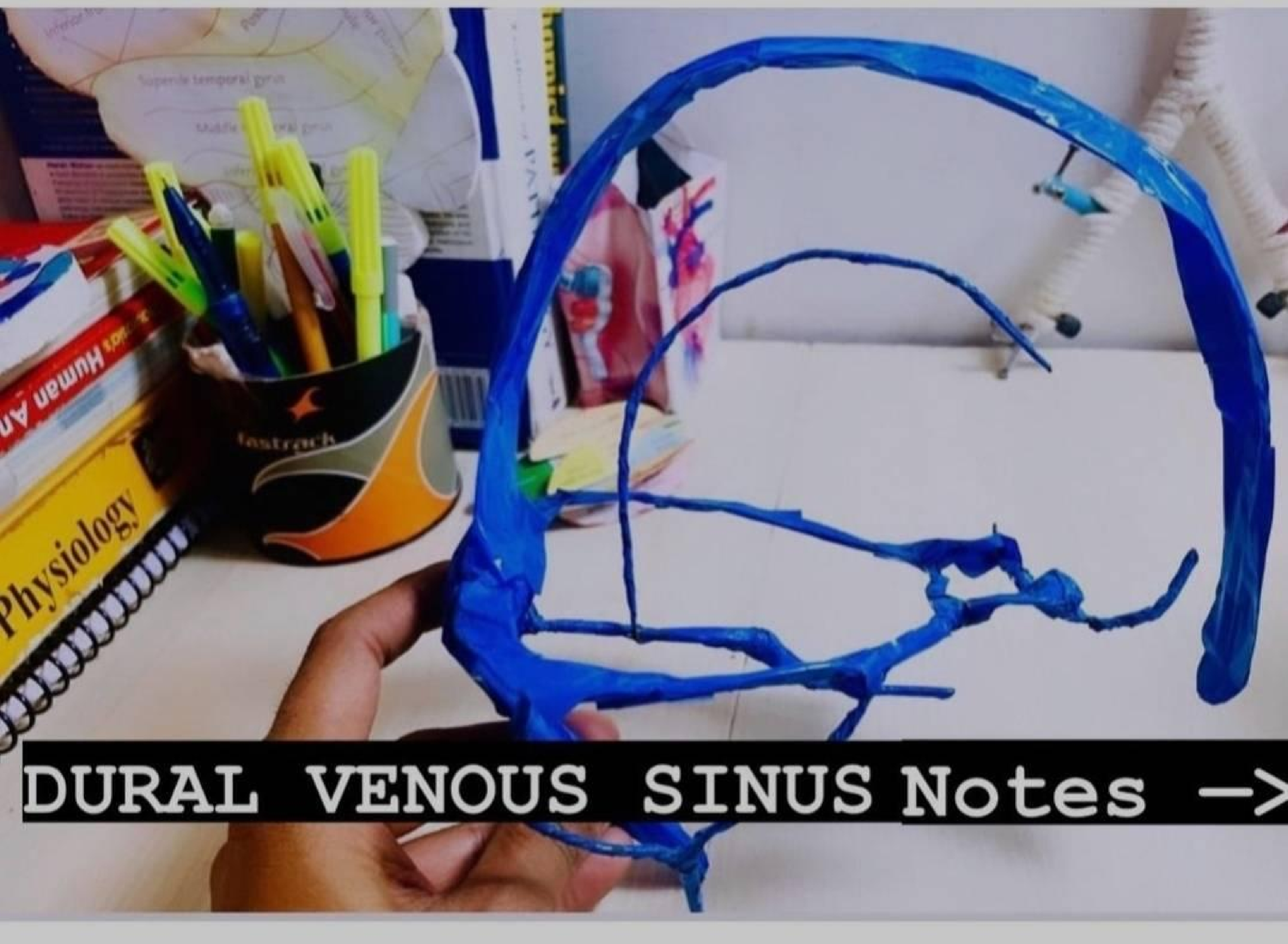
- Greater Palatine artery → enter → greater palatine foramen → greater palatine canal
  - supply → terminal branch enter nasal cavity
    - Palate gums
- Pharyngeal branch → enter → pharyngeal canal
  - supply → nasopharynx, auditory tube
- Anterior artery of Pterygoid canal → Pterygoid canal
  - supply → Pharynx, tympani cavity
- Sphenopalatine artery → Sphenopalatine foramen
  - Posteriolateral, Posteriomedial] nasal branch

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Branches of the maxillary artery.

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Venous Sinuses of Dura Mater ...

- Venous space in → wall which are formed by - dura mater
  - ↳ no muscle in their wall
  - They have no valve
- Venous sinuses receive → venous blood from → Brain meninges Bone of skull
- Cranial venous sinuses communicated → with outside skull through → emissary veins
- communications help to keep the pressure of blood constant
- 23 venous sinuses → 8 paired
  - ↳ 7 unpaired

Paired venous Sinuses :-

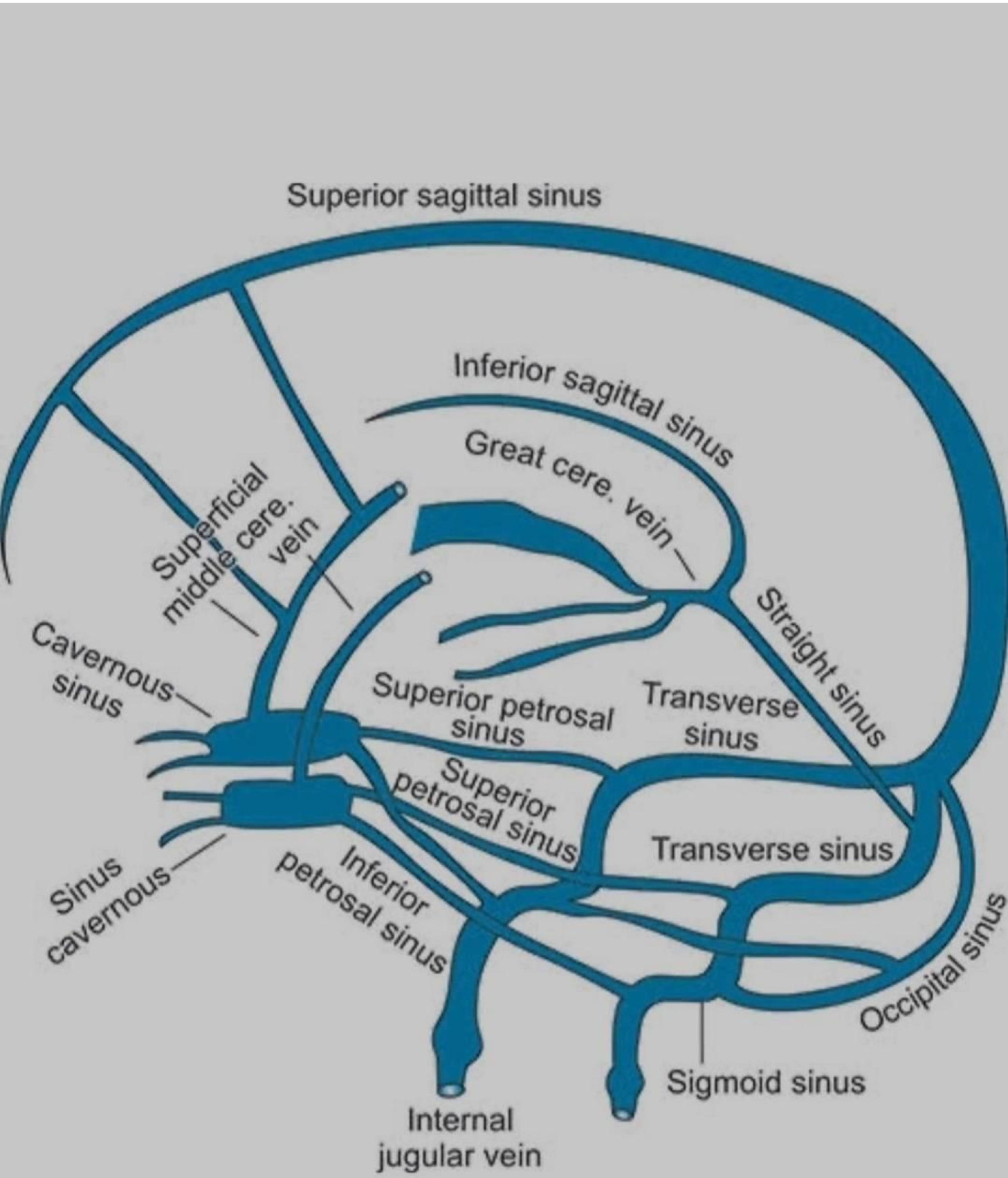
- Cavernous sinus
- Superior petrosal sinus
- inferior petrosal sinus
- Transverse sinus
- Sigmoid sinus
- Spheno-parietal sinus
- Petro-squamous sinus
- middle meningeal sinus

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unpaired venous sinuses :-

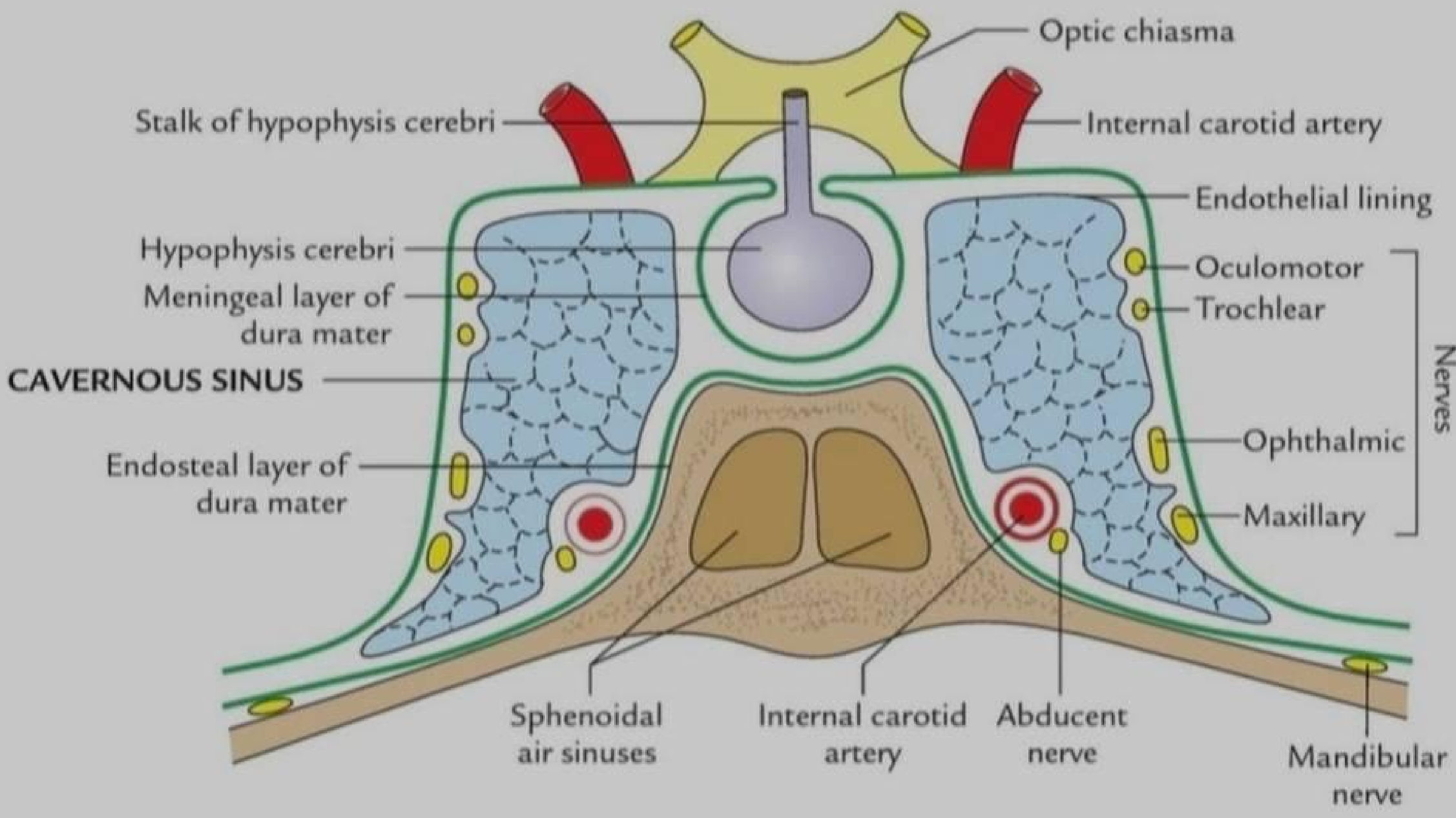
- Superior sagitch sinus
- inferior sagital sinus
- Straight sinus
- Occipital sinus
- Anterior intercavernous sinus
- Posterior intercavernous sinus
- Basilar Plexus of vein

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Cavernous sinus Notes->



onal section through the middle cranial fossa showing relations and contents of cavernous sinuse

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CAVERNOUS Sinus ...

- large venous space - situated in middle cranial fossa either side body of sphenoid bone.
  - floor / Medial wall → endosteal dura matter
  - Roof / lateral wall → meningeal dura matter
  - extend - Anteriorly - superior orbital fissure
    Posteriorly - Apex of Petrous temporal bone

Relations:-

Superiorly - Optic tract
optic chiasma
~~opt~~ olfactory tract
internal carotid artery

inferiorly - foramen lacerum
greater wing of sphenoid bone

medially - Hypophysis cerebri
sphenoidal air sinus

Laterally - Temporal lobe

Anteriorly - Superior orbital fissure

Posteriorly - Apex of Petrous temporal

Structure within lateral Wall :-

- Oculomotor nerve
- Trochlear nerve
- Ophthalmic nerve
- Maxillary nerve
- Trigeminal ganglion

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Structure passing through medial aspect of sinus :-

- internal carotid artery
- Abducent nerve

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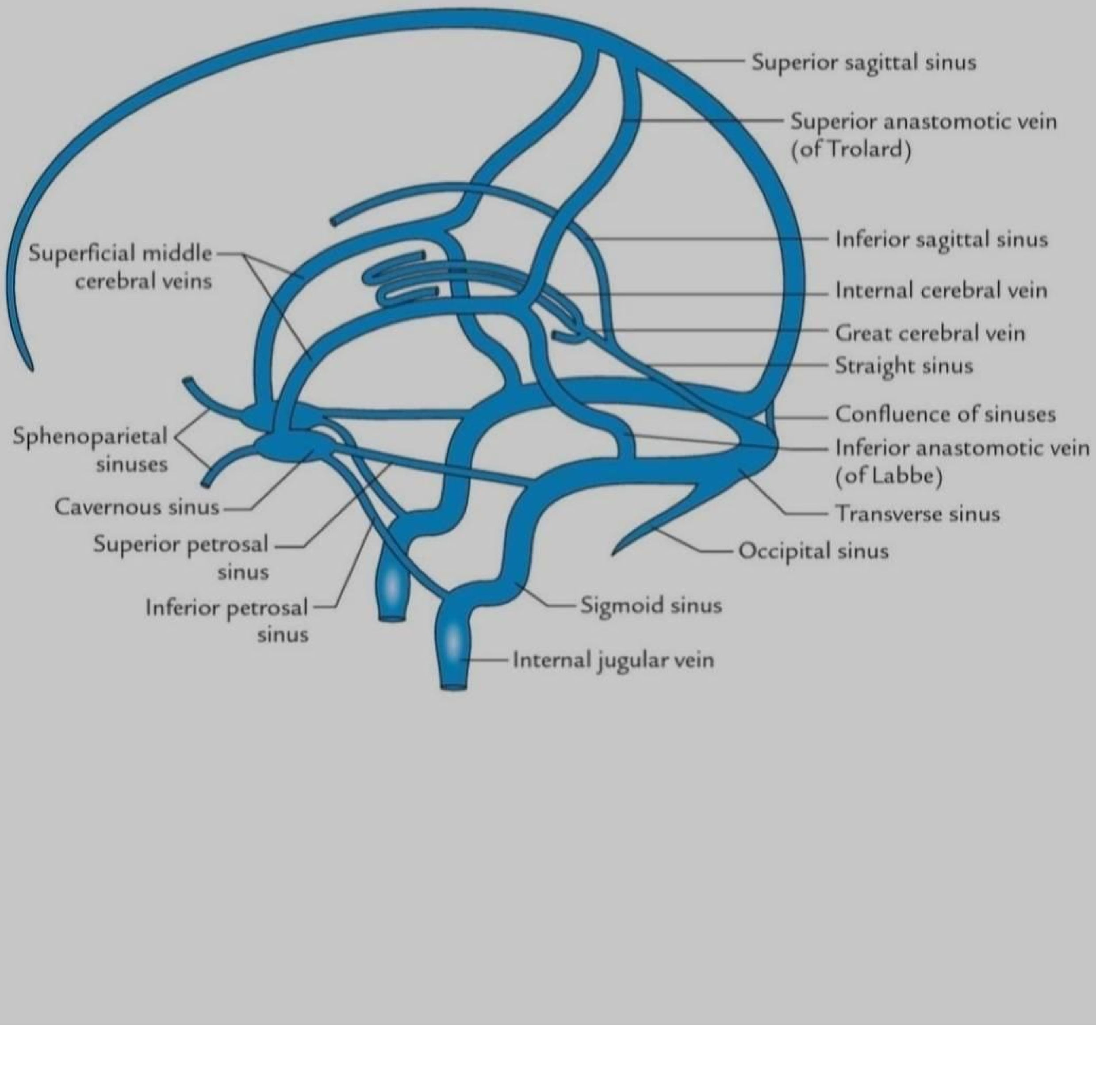
ributories or incoming channel :-

- from Orbit
  - Superior ophthalmic vein
  - inferior ophthalmic vein
  - central vein of retina
- from brain
  - Superficial middle cerebral vein
  - inferior cerebral vein
- from meninges
  - Spheno-parietal sinus
  - middle meningeal vein

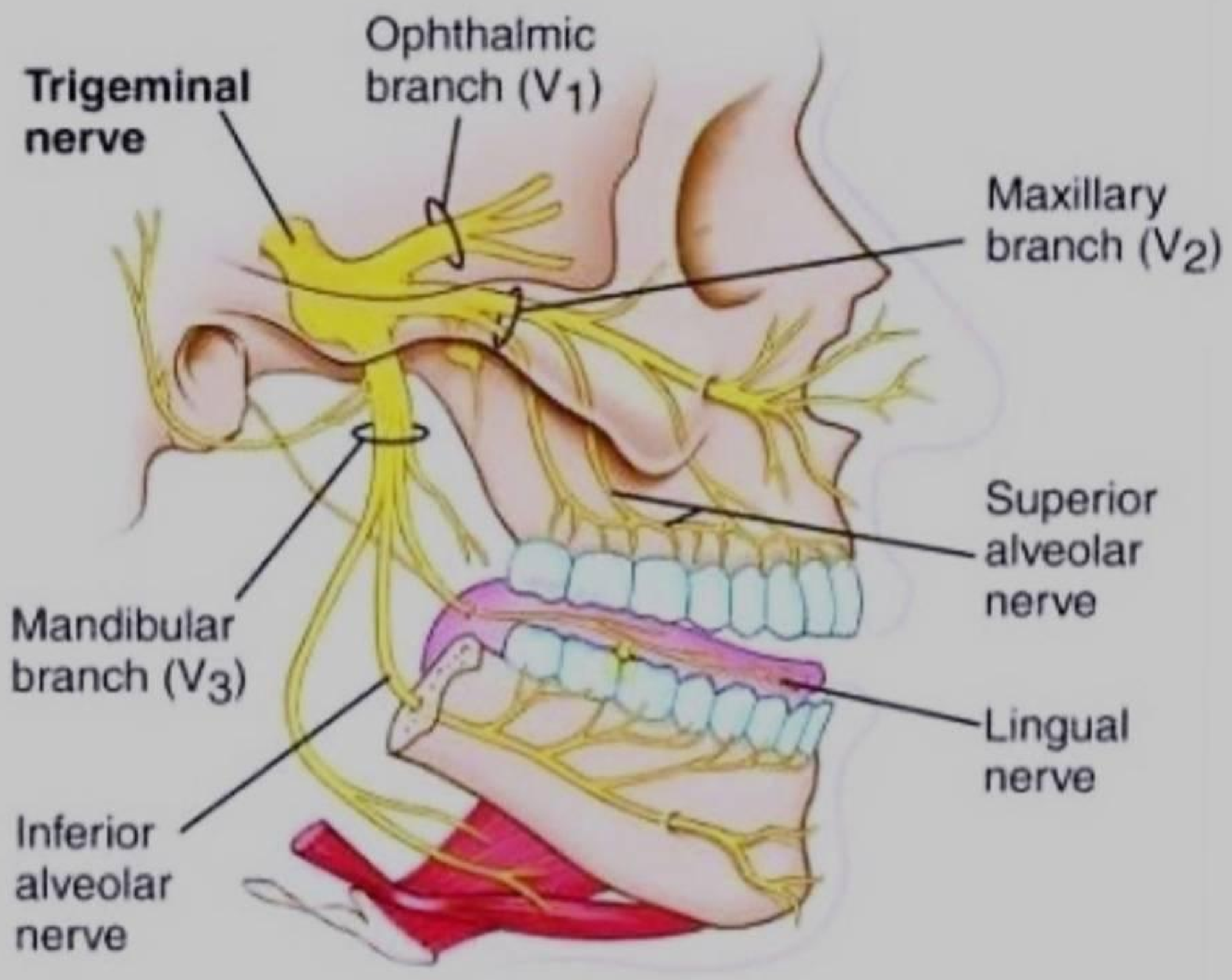
Draining channel :-

- Transverse sinus —through→ Superior petrosal sinus
  - internal jugular vein → internal carotid artery
  - Pterygoid plexus → emissary vein
  - facial vein → Superior ophthalmic vein
  - Right and cavernous sinuses → Anterior, Posterior → intercavernous sinuses

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Mandibular Nerve Notes ->

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MANDIBULAR NERVE ...

- largest mixed branch of trigeminal nerve
- Otic and sub-mandibular ganglia → associated with this nerve

Course :-

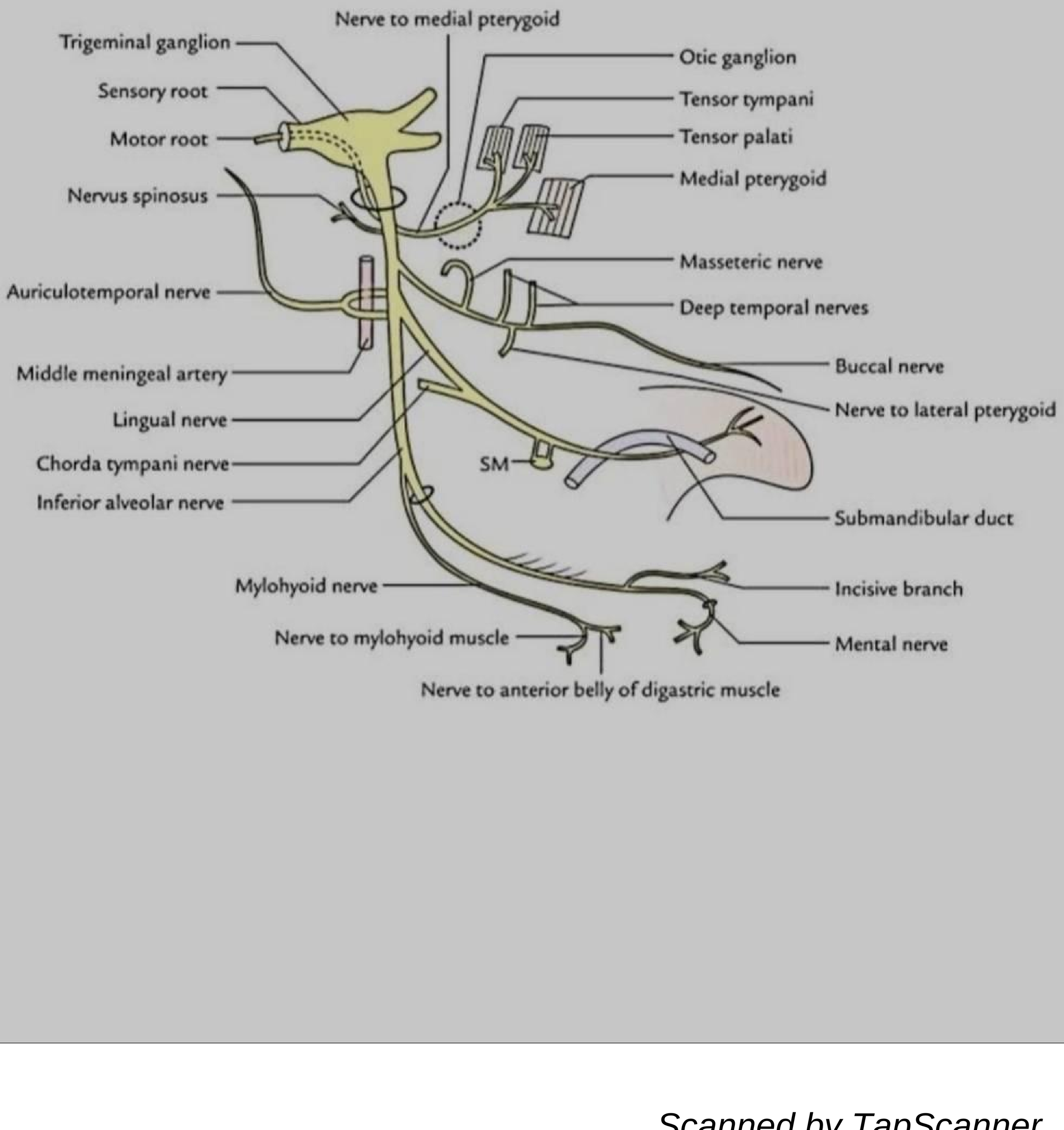
- Begin in middle cranial fossa —through→ large sensory root
  small motor root
- large sensory root —arise from→ lateral part of Trigeminal ganglion
  ↓ leave cranial cavity
  foramen ovale
- small motor root —lie→ deep to Trigeminal ganglion
  ↓ leave cranial cavity
  foramen ovale
- "Both root join → below foramen —form→ main trunk"
- Main trunk in infratemporal fossa —deep to→ lateral Pterygoid
- main trunk —divided→ small → Anterior trunk
  large → Posterior trunk

Branches :-

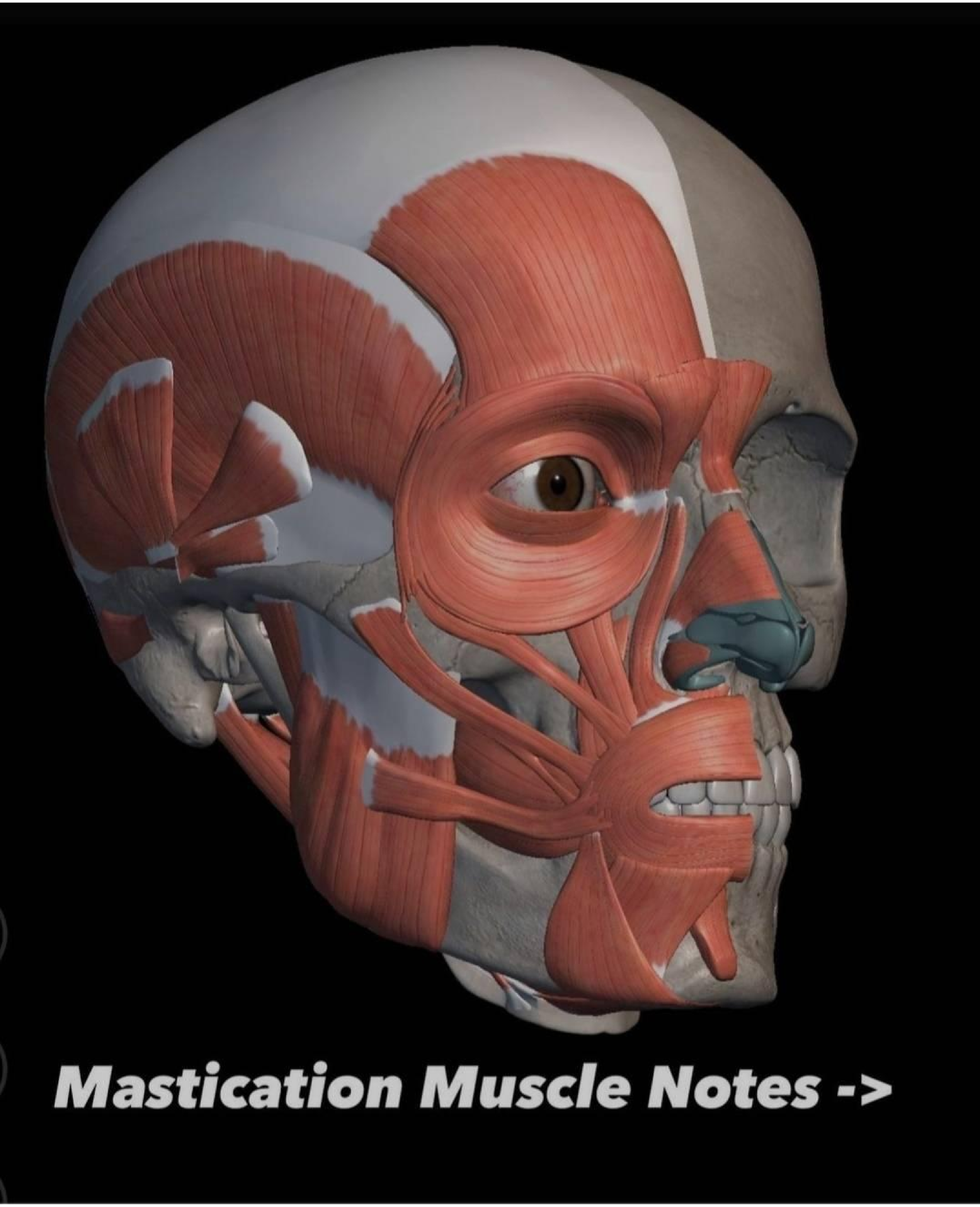
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* **Main trunk :-** meningeal branch
  nerve to medial pterygoid
* **Anterior trunk :-** - sensory branch :- Buccal nerve
  - Motor branch :- masseteric
  deep temporal nerve
  nerve to lateral pterygoid
* **Posterior trunk :-**
  - Auriculotemporal nerve
  - Lingual
  - inferior alveolar nerve

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Muscle of Mastication ...

Masseter :- Quadrilateral

Origin - superficial - Anterior 1/3 zygomatic arch
- zygomatic process of maxilla

Deep - zygomatic arch

middle - 1/3 of zygomatic arch

insertion - superficial - lower lateral surface of mandible

Deep - ramus of mandible

middle - central part of ramus of mandible

nerve - masseter nerve of mandibular nerve

Action - Elevation
Retraction

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Temporalis :- fan-shaped

Origin - Temporal fossa Excluding zygomatic bone

insertion - Deep surface of coronoid process
- Anterior border of ramus of mandible

nerve - Deep temporal branch of MN

Action - Elevation
Retraction

Lateral Pterygoid :-

Origin - upper head - infra-temporal surface, infra-temporal crest } greater wing of Sphenoid Bone

lower head - lateral Pterygoid plate

insertion - Pterygoid fovea on anterior surface of neck of mandible

- TMJ

nerve - anterior division of mandibular nerve

Action - Depress
Protrude ©

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medial pterygoid – Quadrilateral

Origin :- superficial head - tuberosity of maxilla
:- Deep head - medial surface of - lateral pterygoid plate

insertion :- medial surface of angle of adjoining ramus of mandible

nerve :- nerve to medial pterygoid → mandibular nerve

Action :- Elevate
Protrude

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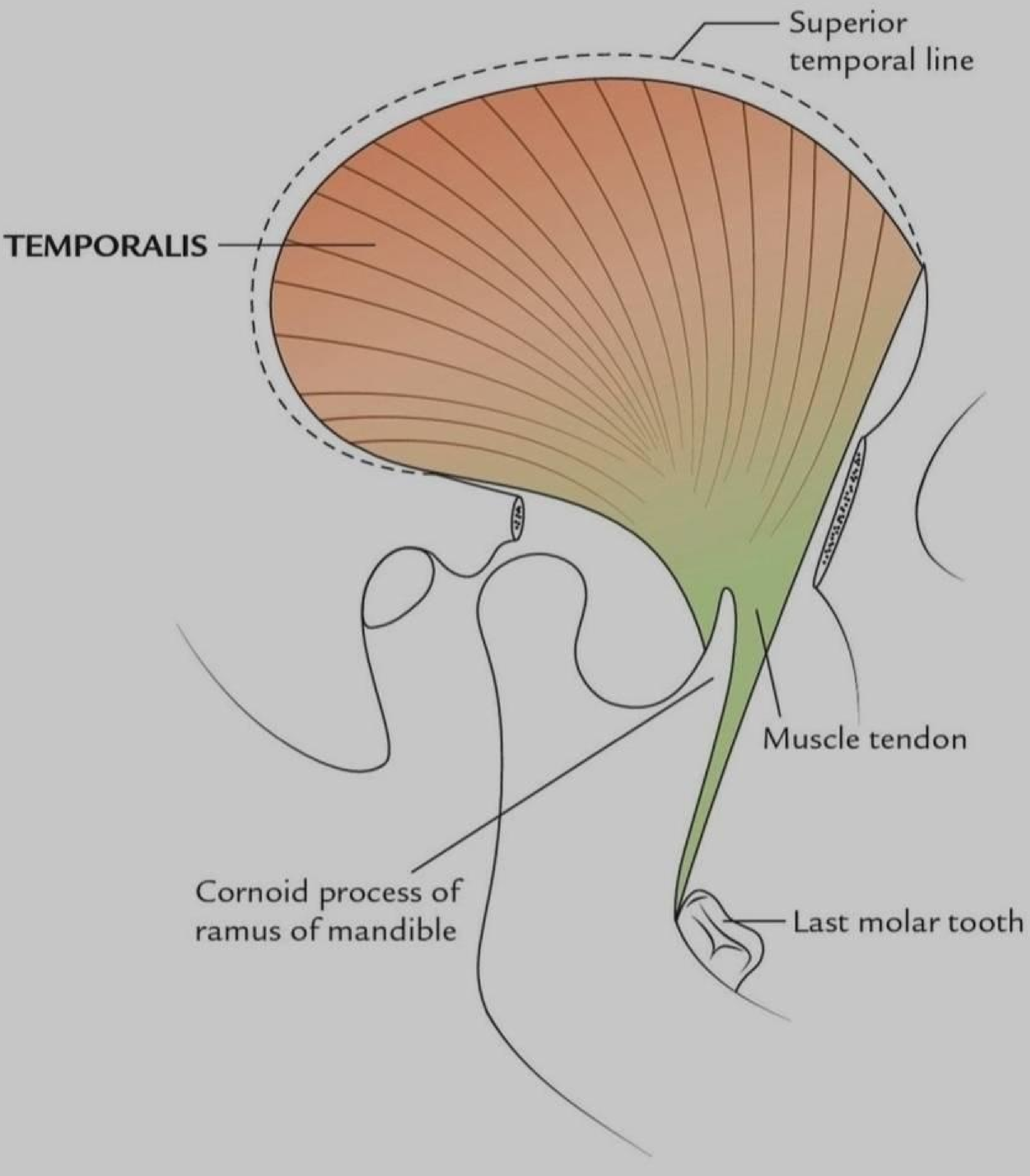


Fig. 10.21 Origin and insertion of the temporalis muscle.

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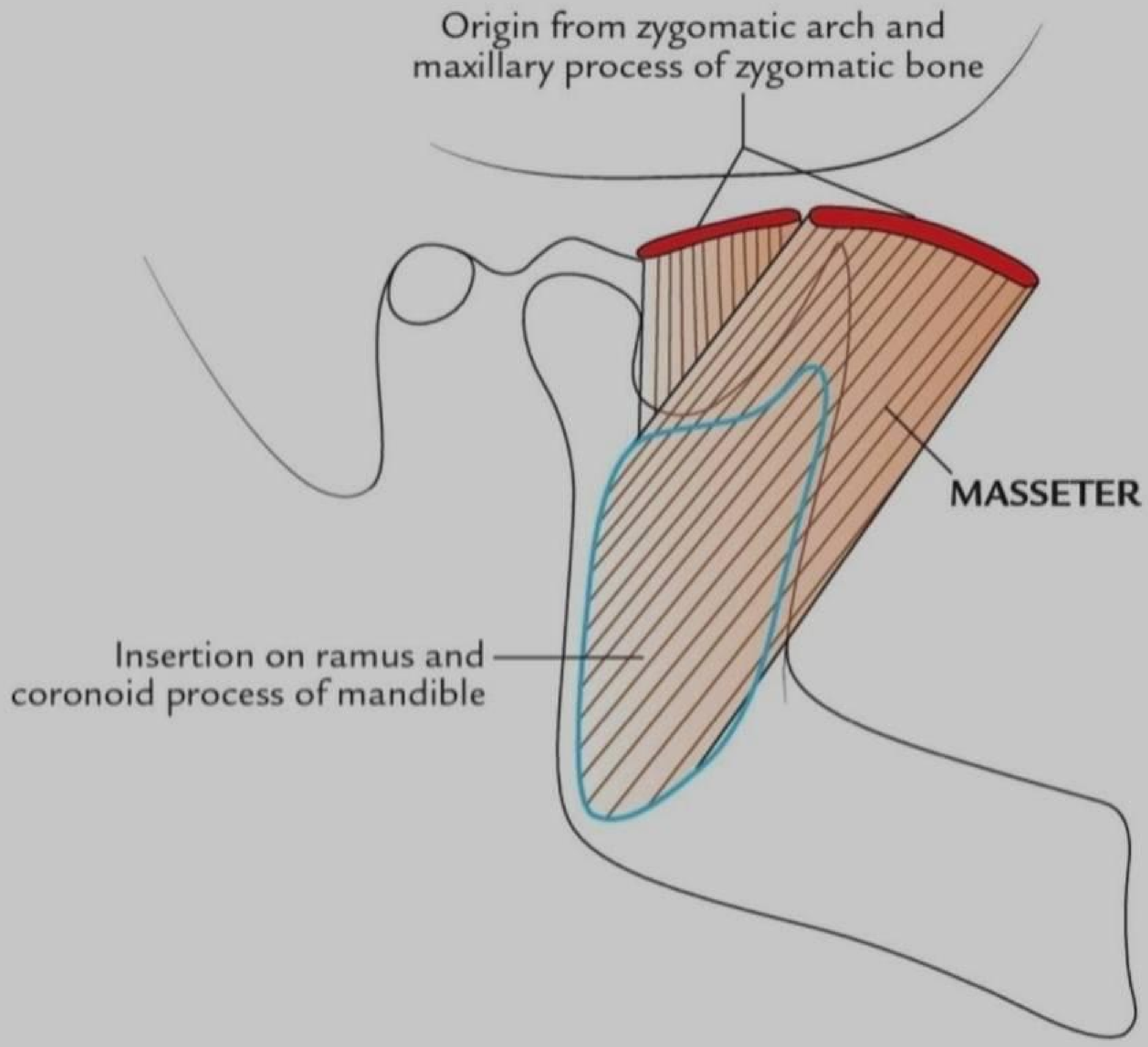


Fig. 10.22 Origin and insertion of the masseter muscle.

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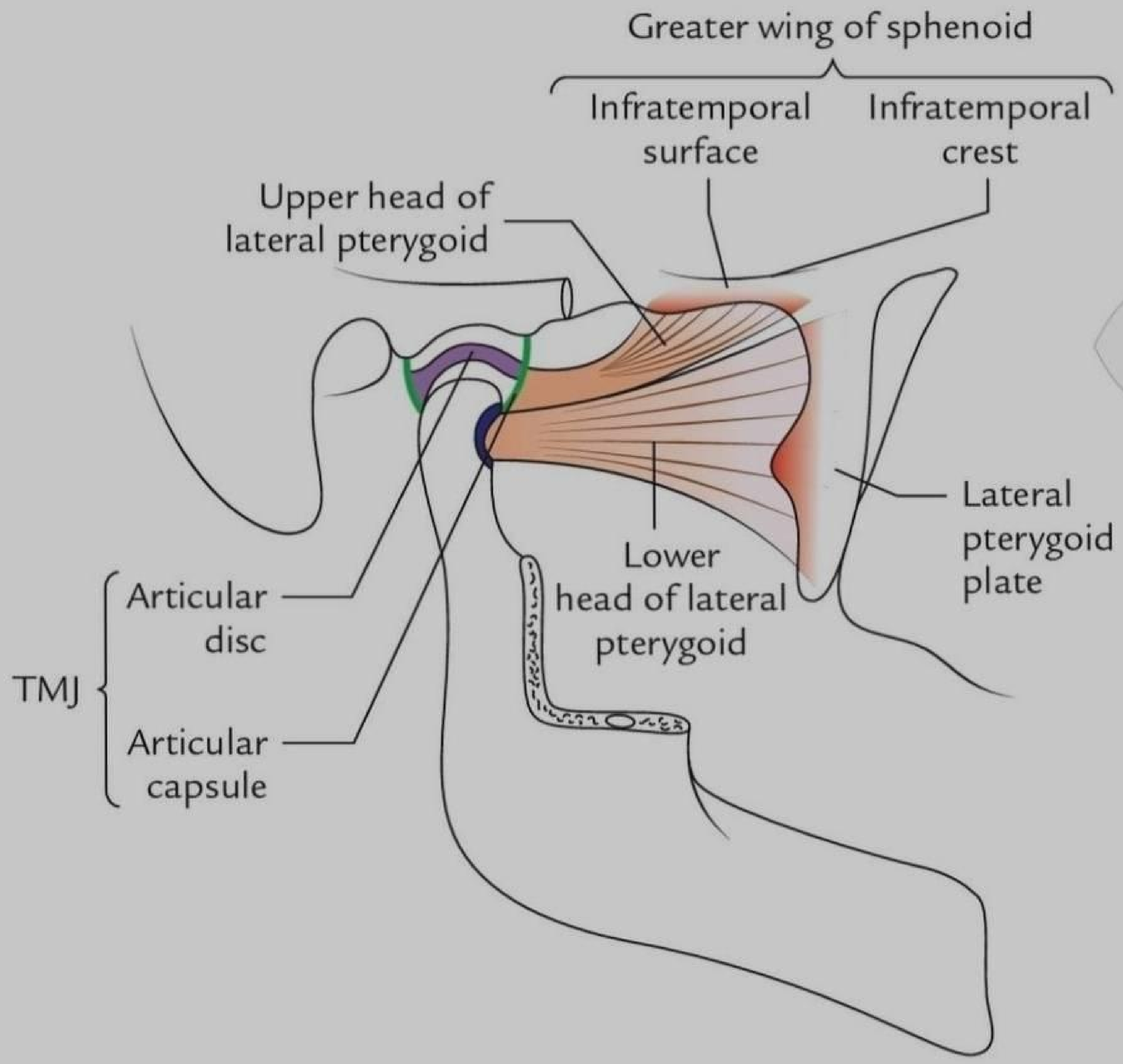


Fig. 10.4 Origin and insertion of the lateral pterygoid muscle.

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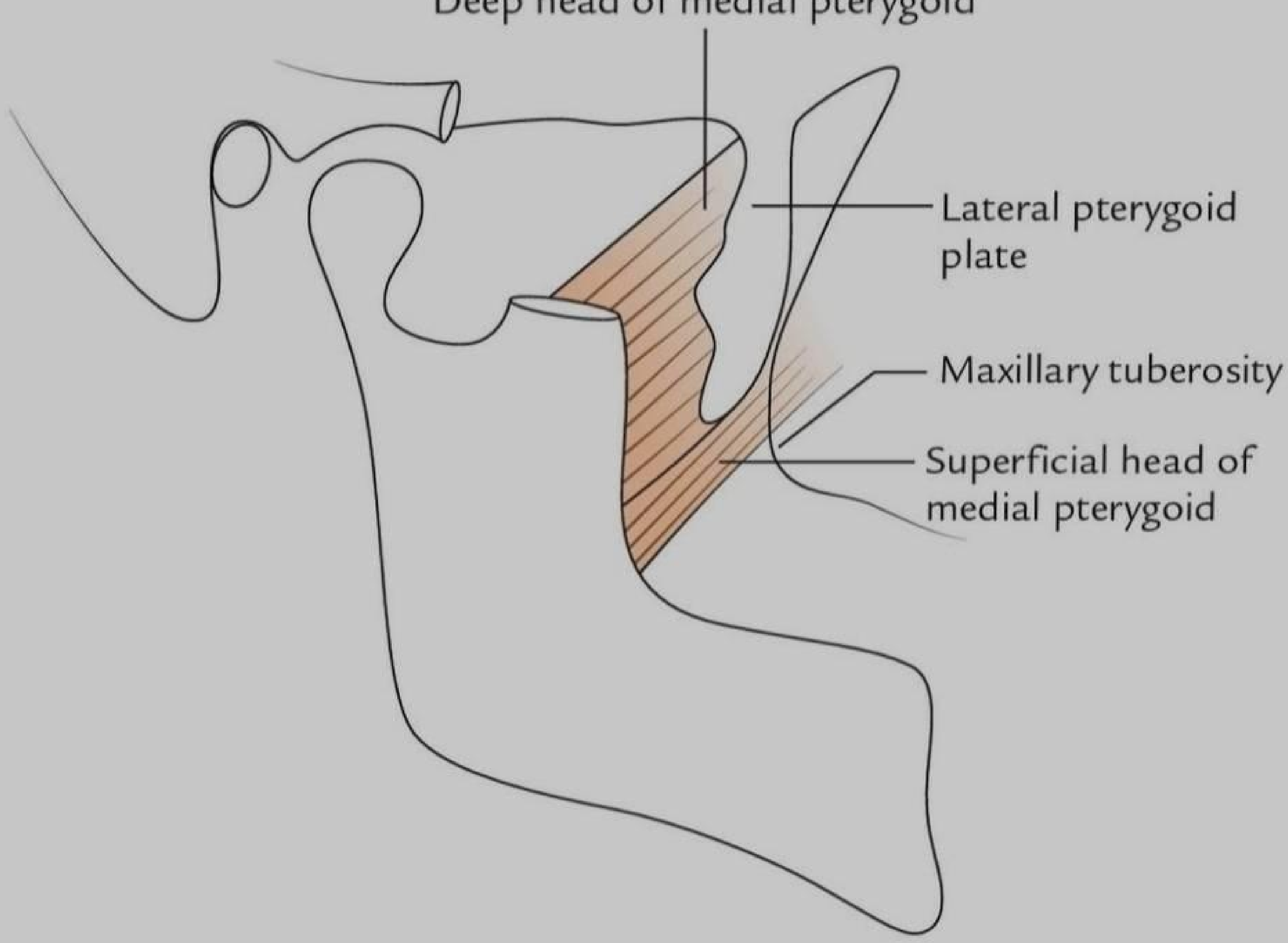


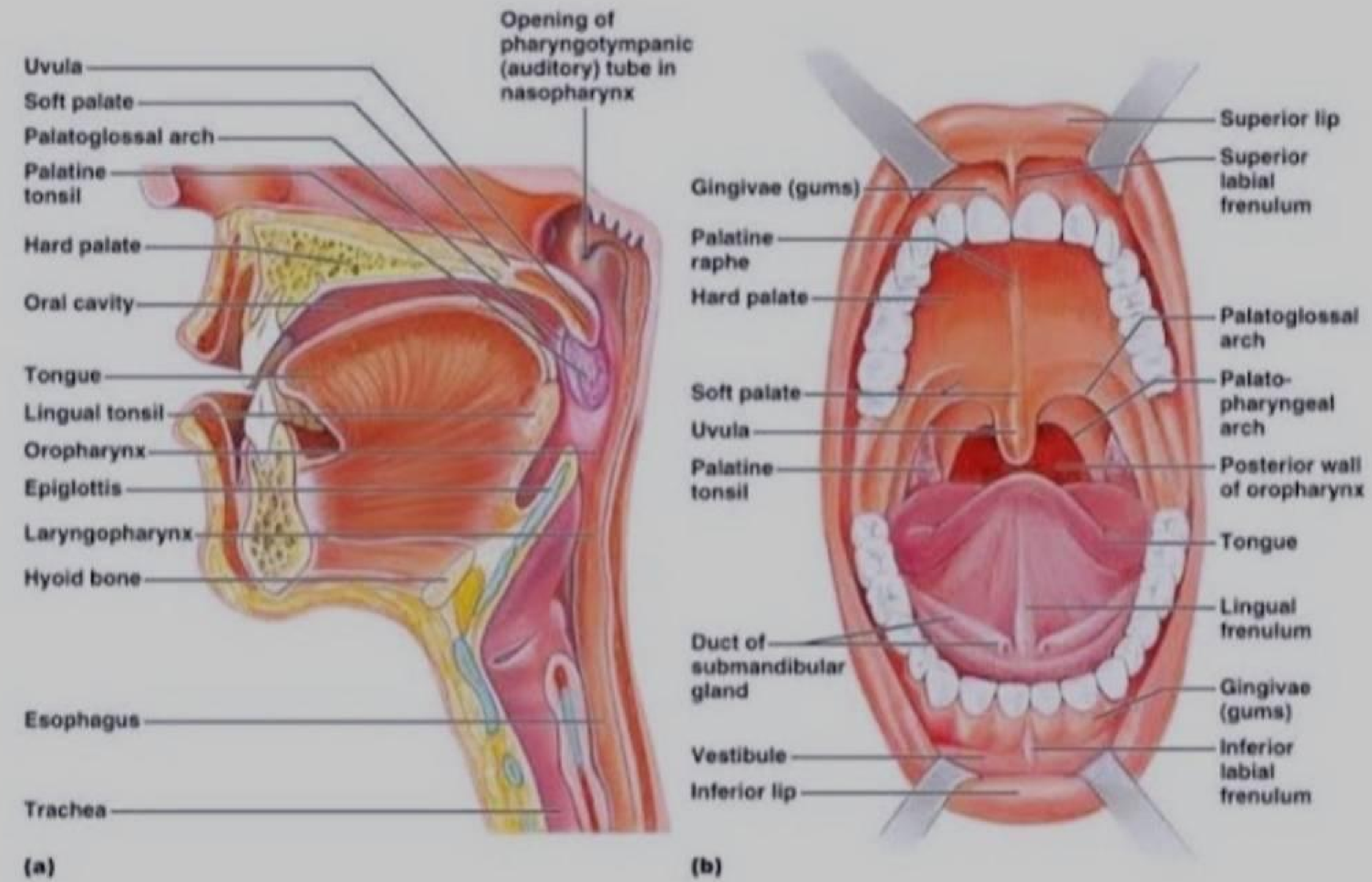
Fig. 10.6 Origin and insertion of the medial pterygoid muscle.

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Table 10.4 Origin, insertion, nerve supply, and actions of the muscles of the mastication

Muscles	Origin	Insertion	Nerve supply	Actions
Temporalis (fan shaped)	• Floor of temporal fossa • Temporal fascia	• Tip, anterior border, and medial surface of coronoid process • Anterior border of ramus of mandible	Mandibular division of trigeminal nerve	• Elevation of mandible by anterior and middle fibres • Retraction of mandible by posterior fibres
Masseter (quadrilateral)	Zygomatic arch adjoining part of zygomatic process of maxilla	• Lateral surface of ramus of mandible • Coronoid process	Mandibular division of trigeminal nerve	Elevation of mandible to occlude the teeth for forceful bite
Lateral pterygoid (Short, thick conical)	• Upper head from infratemporal surface and crest of greater wing of sphenoid • Lower head from lateral surface of lateral pterygoid plate	• Pterygoid fovea on anterior surface of neck of mandible • Articular disc and capsule of TMJ	Mandibular division of trigeminal nerve	• Depression of mandible by pulling the neck of mandible forward • Protraction
Medial pterygoid (quadrilateral)	• Superficial head from tuberosity of maxilla • Deep head from medial surface of lateral pterygoid plate	Medial surface of angle adjoining ramus of mandible	Mandibular division of trigeminal nerve	• Elevation of mandible • Protraction

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SOFT PALATE ANATOMY NOTES →

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Soft PALATE ...

- movable, muscular fold, suspended from - Posterior border of hard Palate.
- 2x surface → Anterior - concave
  Posterior - convex - floor of nasal cavity
- 2x Border → superior border attached → Posterior border of hard Palate.
  - inferior border - free
    - conical projection - "uvula"
      - from each side → base of uvula, two curved fold extend – laterally downward
        ↳ Anterior fold - Palatoglossal arch
        Posterior fold - Palatopharyneal arch

Structure :- Soft Palate is a fold of mucous membrane containing :-

- Palatine aponeurosis - flattened tendon of tensor veli palatini
  - near median plane – aponeurosis splits ↓ enclose musculus uvulae.
  - superior surface contain → levator veli palatini, Palatopharyngeus
  - inferior surface → Palatoglossus

- numerous mucous gland

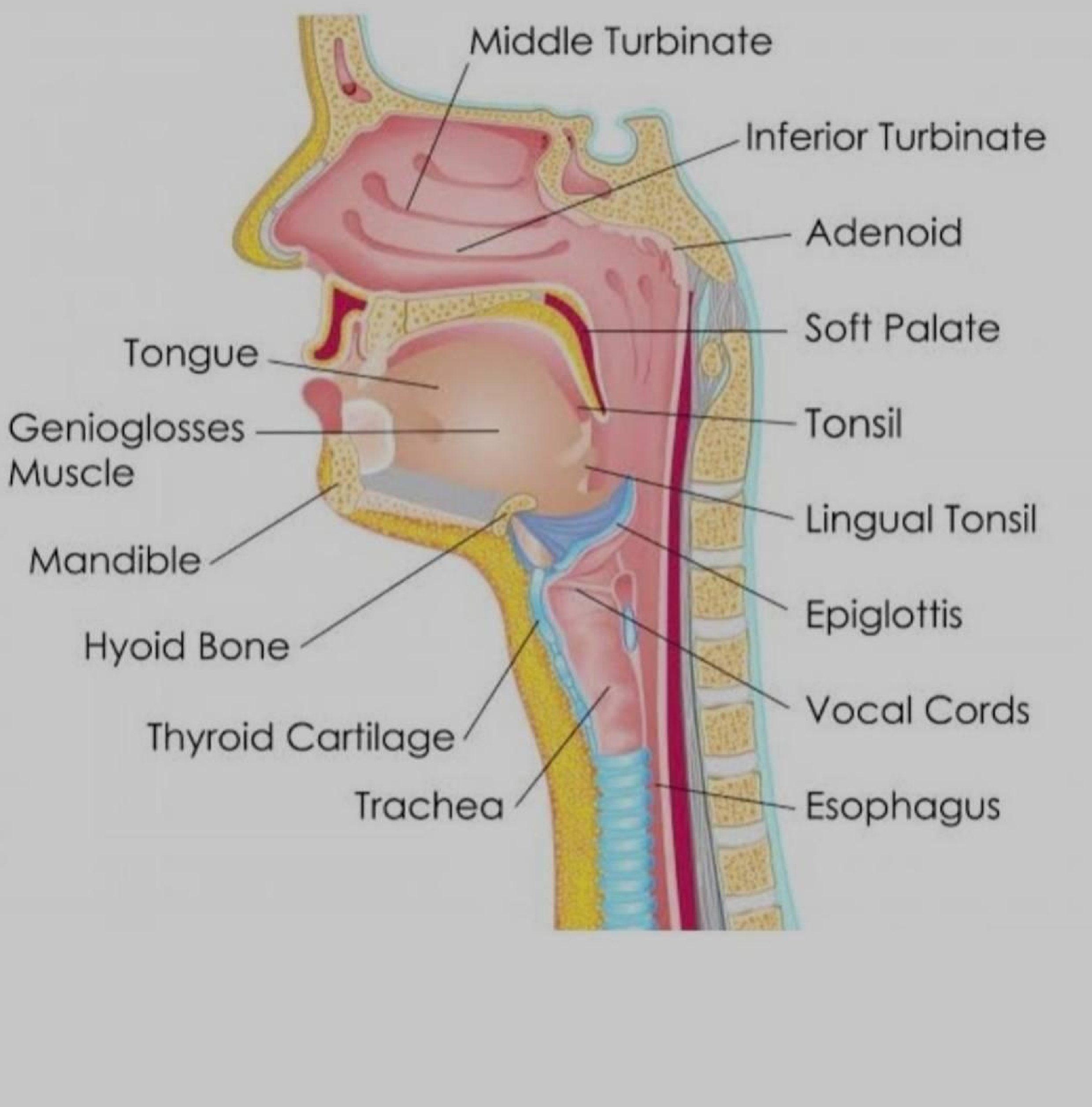
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- Muscle of soft Palate 5x
  - Tensor Palati
  - Levator palati
  - muscular uvulae
  - Palatoglossus
  - Palatopharyngeus

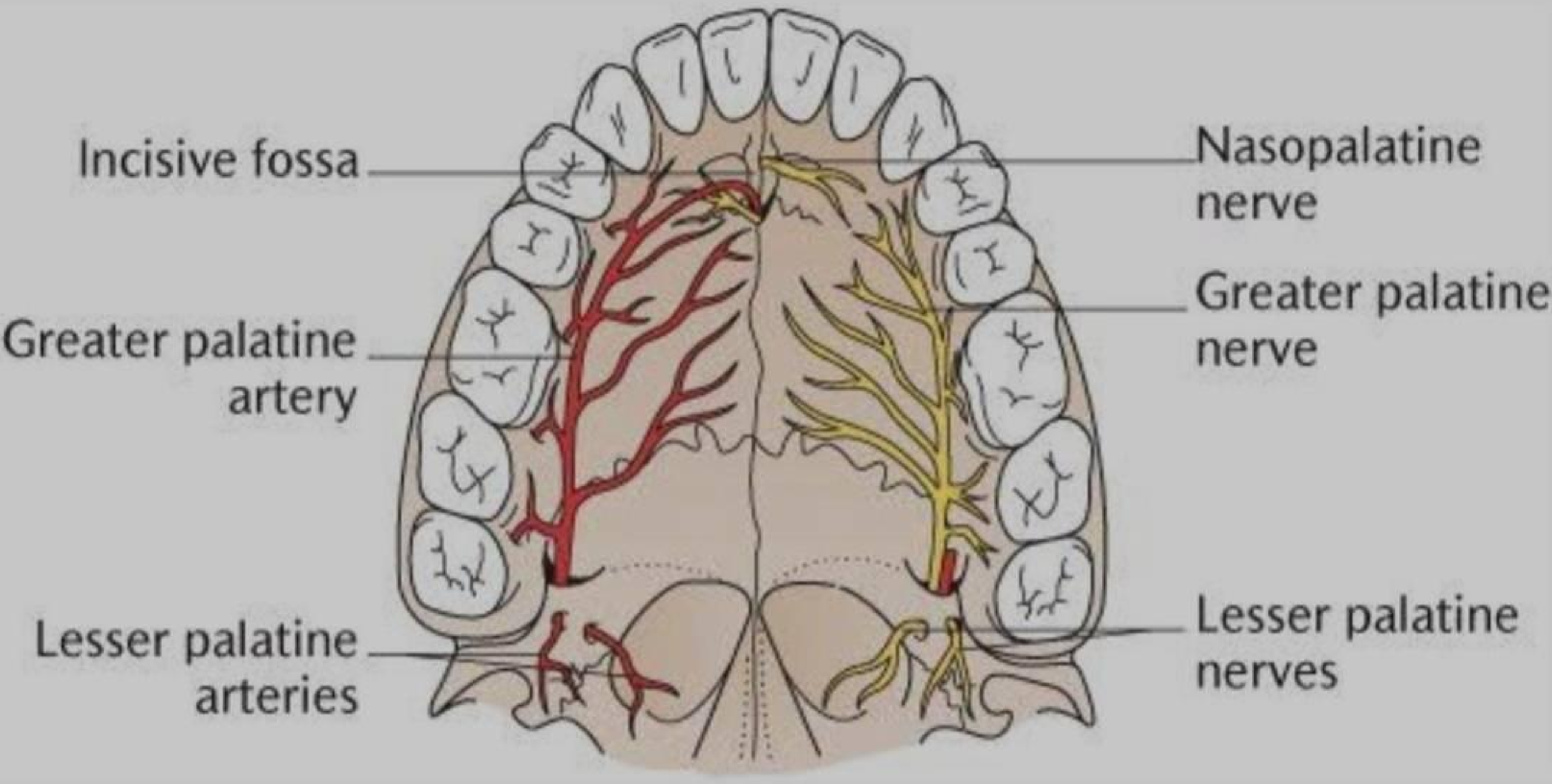
Blood supply :- Arteries – Greater Palatine branch → maxillary artery
– Ascending Palatine → facial artery
– Palatine branch of ascending Pharyngeal

Vein - tonsillar plexuses of vein

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Nerve supply of Palate ->

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Nerve Supply :- OF Soft PALATE !

* Motor nerve :- All muscle by - Pharyngeal plexus
  Except - tensor veli palatini - mandibular nerve

* Special sensory
  - Glasbatary nerve → carry tast sensation → from oral cavity
  - Tast sensation from soft palate → lesser palatine nerve
    ↓
    Anterior palatine nerve
    ↓
    Pterygo palatine ganglion (no relay)
    ↓
    greater petrosal nerve
    ↓
    Geniculate ganglion
    ↓
    nervus intermedius
    ↓
    nucleus of tractus solitarius.

LPGN

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Secretomotor nerve ..

Post ganglionic

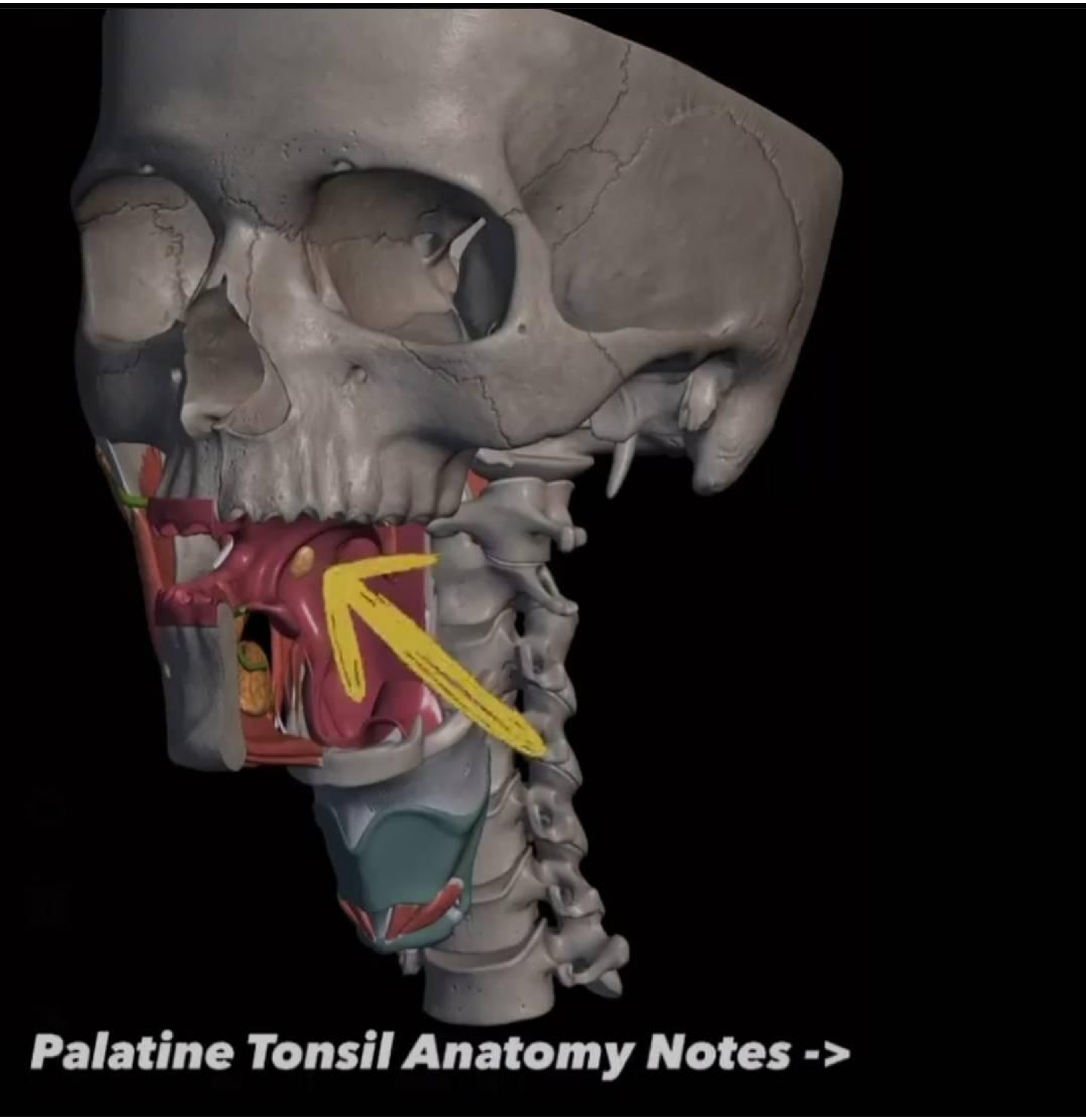
Soft palate glands
↓
lesser palatine nerve
↓
Anterior Palatine nerve
↓
Pterygopalatine ganglion (relay
↓
Deep petrosal nerve
↓
greater Petrosal nerve
↓
geniculate ganglion
↓
nervus intermedius
↓
Superior salivatory nucleus

Pre-ganglionic

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Maxillary

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PALATINE TONSIL

- secondary lymphoid tissue
- occupies - tonsillar fossa b/w palatoglossal and palatopharyngeal arches
- almond-shaped — 2 x surface - medial
  lateral

  2 x border - Anterior
  Posterior

  2 x Poles - Upper and lower

- Surface - medial surface → covered by stratified squamous epithelium
  - surface has - 12-15 crypts
  - largest of these → intratonsillar cleft

- lateral surface - covered by sheet of fascia which form hemicapsule of tonsile
  - capsule is an extension of → pharyngobasilar fascia
  - Tonsillar artery enter tonsil by piercing ↓ superior constrictor muscle
  - Paratonsillar vein → descend from palate ↓ in loose areolar tissue on lateral surface of capsule

- Border → Anterior border - Palatoglossal arch
  Posterior border - Palatopharyngeal arch

- Pole → upper pole - soft palate
  lower pole - Tongue

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- Bed of tonsil → is formed from within outwards by :-
  - Pharyngobasilar fascia
  - superior constrictor and palatopharyngeus muscle
  - buccopharyngeal fascia
  - glossopharyngeal nerve
  - lower part of → styloglossus

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Blood supply :-

- Main - Tonsillar branch of facial artery (FA)
- Ascending Palatine - FA
- Dorsal lingual branch - FA
- Ascending pharyngeal brannch - ECA
- greater Palatine - maxillary artery

- vein :- 2-3 vein → leave lower part of tonsil → join → Palatine, pharyngeal, facial → vein

- nerve :- glossopharyngeal nerve
  lesser palatine nerve

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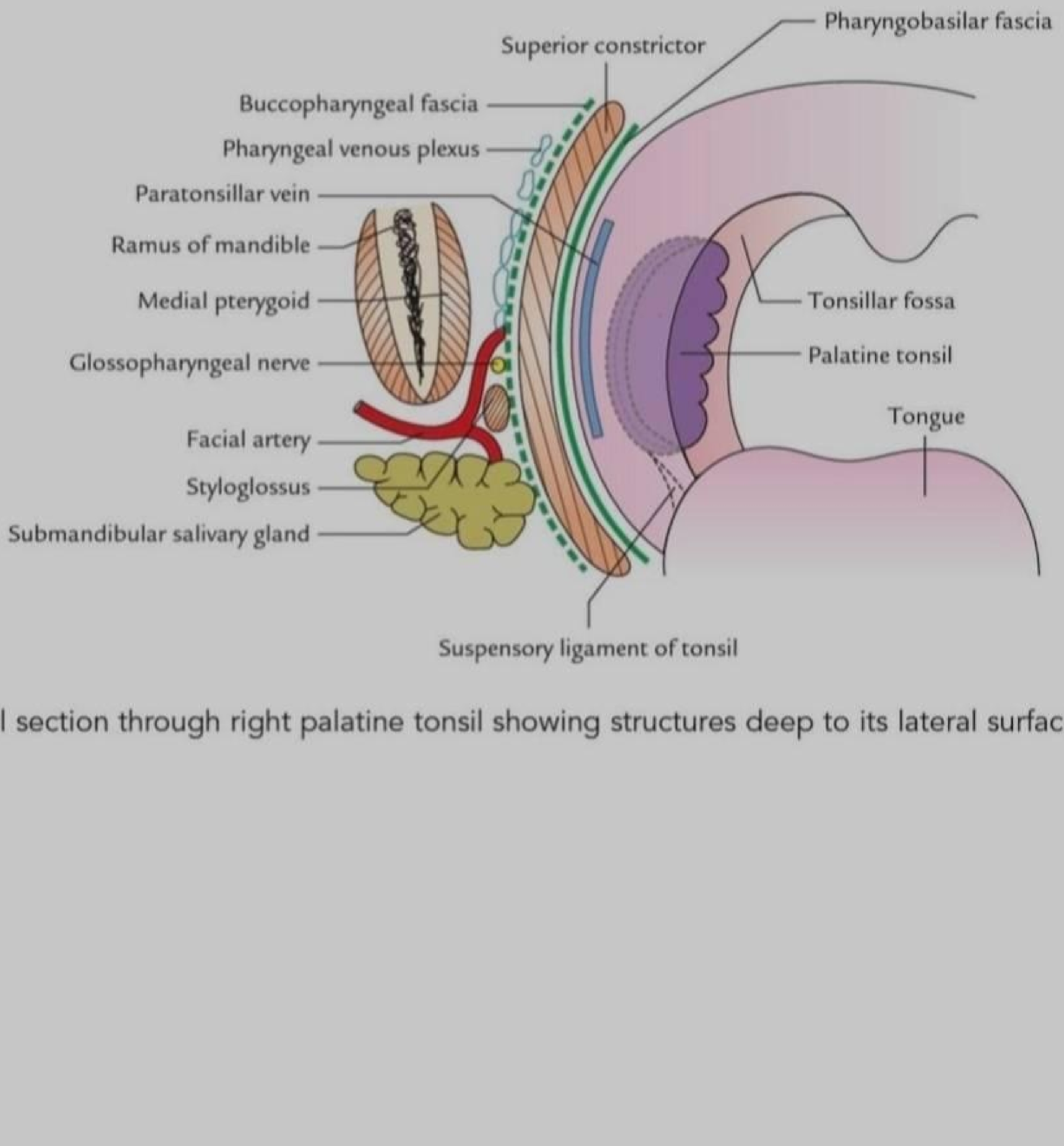


Fig. 14.16 Horizontal section through right palatine tonsil showing structures deep to its lateral surface.

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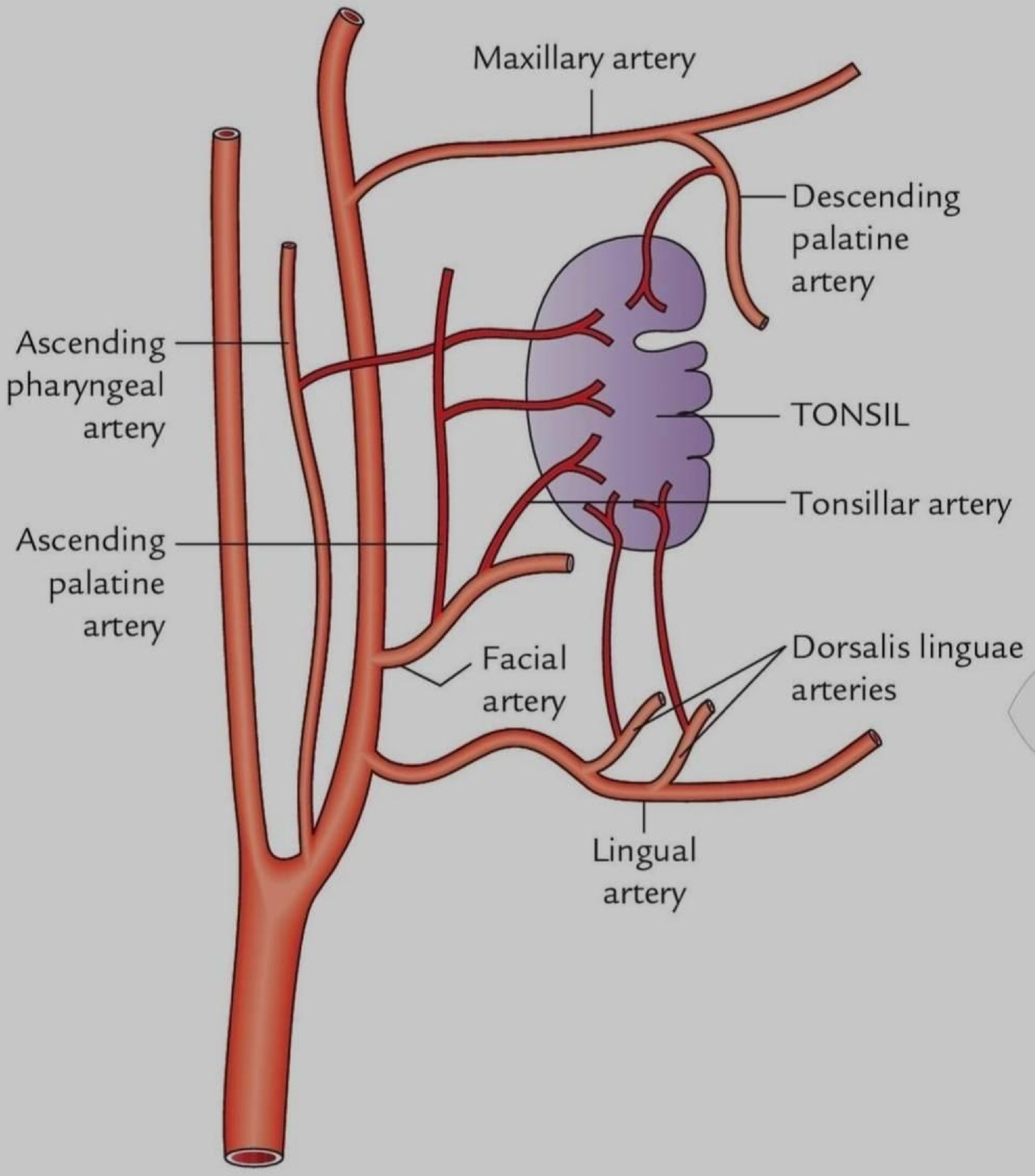
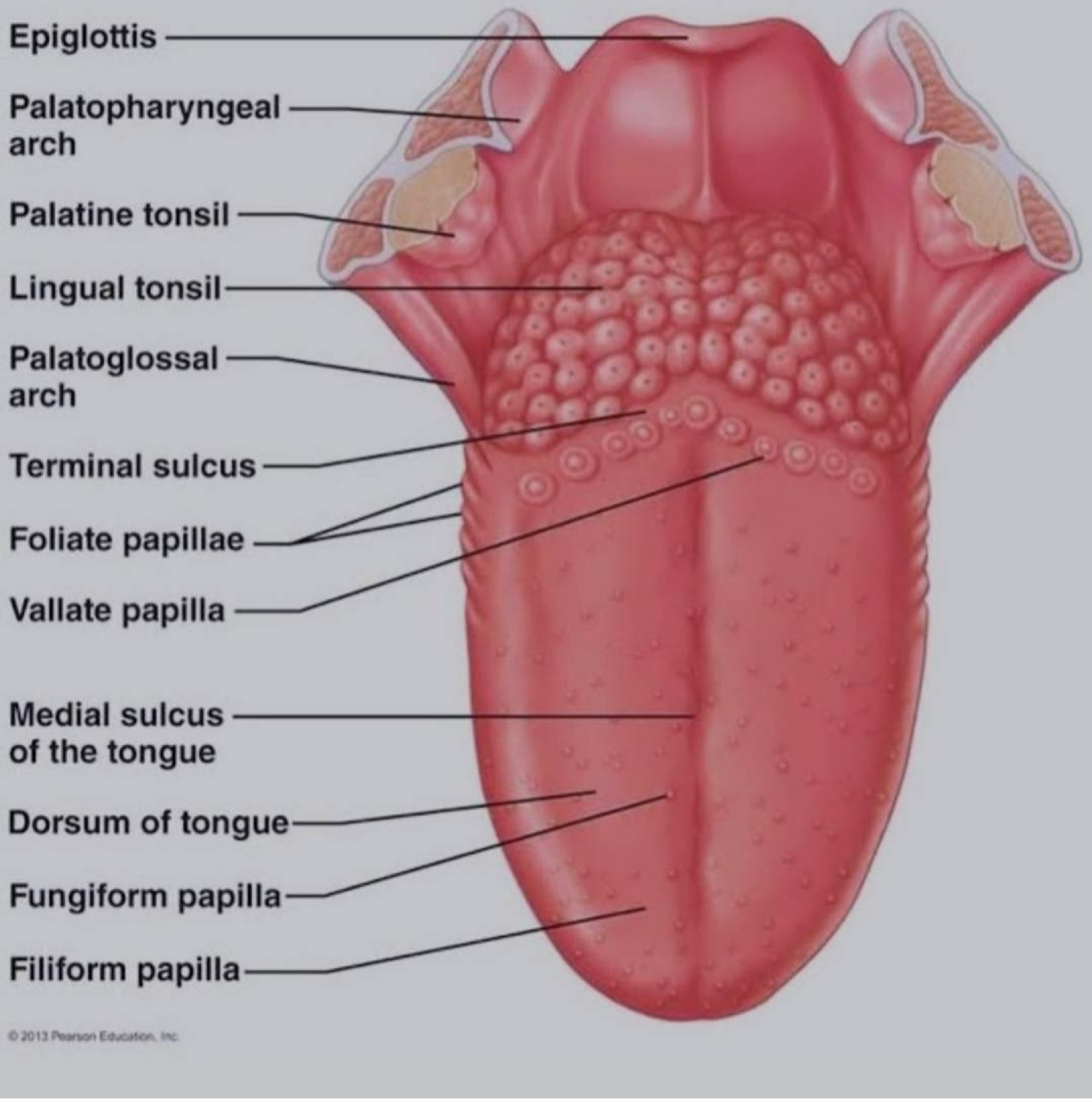


Fig. 14.17 Arteries supplying the tonsil.

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Tongue Anatomy Notes ->



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Tongue...

- muscular organ – situated → floor of mouth
- function- taste
  speech
  chewing and deglutition

External feature :- – Root
tip
Body → upper or Dorsum → oral or papillary part
→ Pharyngeal part
→ Posterior most part
→ inferior surface

- Root - attached → styloid process
  soft palate
  ] → above
  hyoid bone → below

- Tip - anterior free end → Behind upper incisor teeth

- Body - Dorsum - convex in all direction,
  - Oral part - floor of mouth
    - free margin – in contact with gums and teeth
    - superior surface → covered with papillae
    - inferior surface → covered with smooth membrane
      - median fold – frenulum
      - lateral fold - Plica frenulum

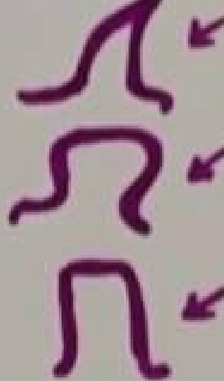
- Pharyngeal part - lie behind palatoglossal arches
  - Base of tongue
  - mucous membrane - no papillae
    - but lymphoid follicle

- Posterior most part - connected to epiglottis
  - 3x folds of mucous membrane
    - 2x lateral glossoepiglottic fold
    - median glossoepiglottic fold

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Papillae of tongue :-



- Circumvallate papillae → front of sulcus terminalis
  - 8 - 12 no.
  - largest size
- fungiform papillae → tip and margin of tongue
- filiform papillae → cover dorsum of tongue
  - → smallest
- foliate papillae → at lateral border → just infront of circumvallate Papillae

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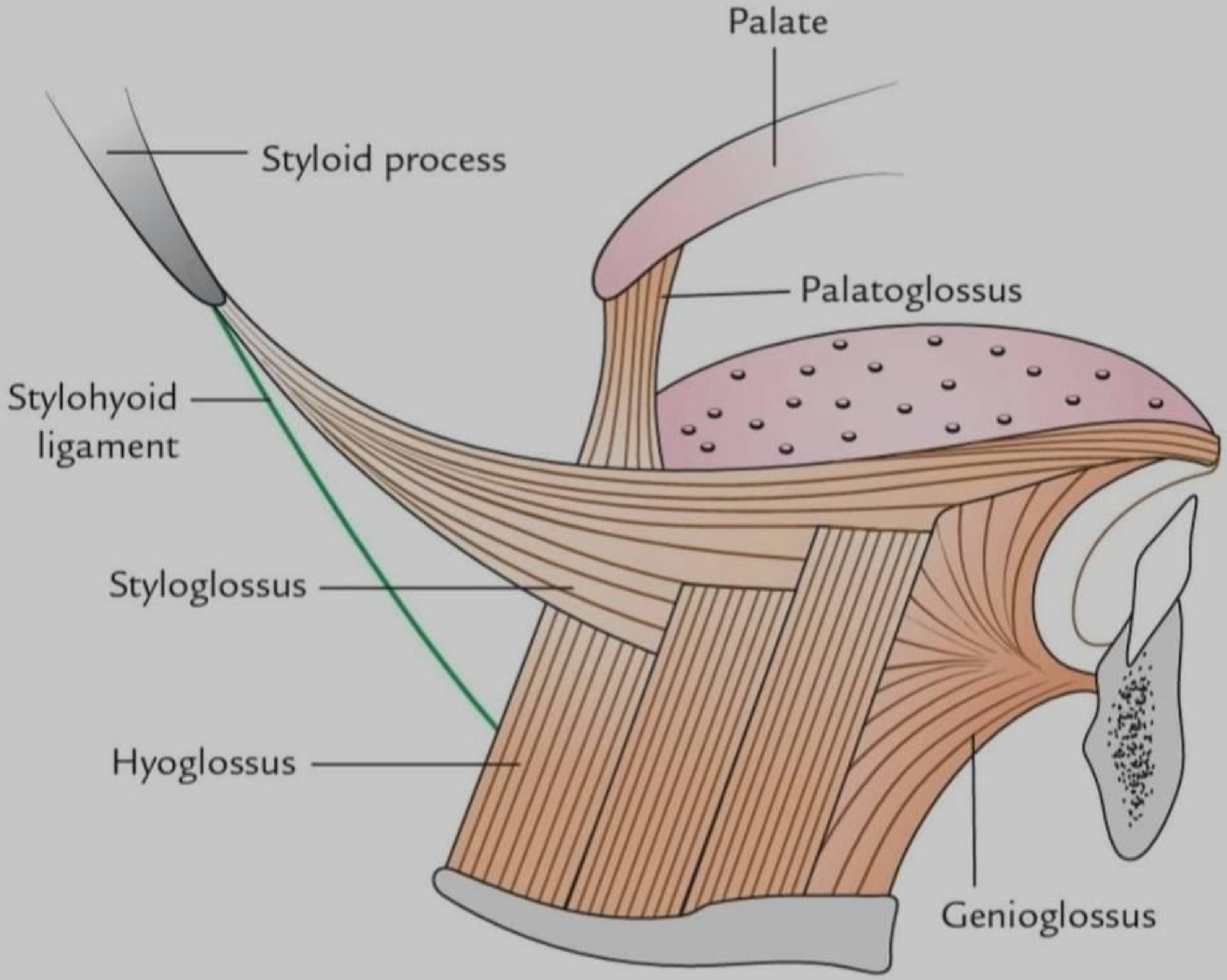


Fig. 13.15 Extrinsic muscles of the tongue.

Tongue Muscles Notes ->

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MUSCLE OF TONGUE...

- middle fibrous septum → divides tongue into → right halves / left halves
- Each half contain → 4x intrinsic muscle
  4x extrinsic muscle

Intrinsic muscles :-

- occupy upper part of tongue
- attached → submucous fibrous layer

- Superior longitudinal - arises from fibrous tissue → deep to mucous membrane
  - on dorsum of tongue
  - action - short tongue
    - make dorsum concave

- Inferior longitudinal - arises from fibrous tissue → beneath mucous membrane
  - on tip of tongue
  - action - short tongue
    - make dorsum convex

- Transverse - lie → on either side of mid line
  - origin - fibrous medial septum to submucous fibrous tissue
  - make - make tongue narrow

- Vertical - found at border of anterior part of tongue
  - action - flatted

Extrinsic Muscles :-

	Action
- Genioglossus →	Protrudes
- Hyoglossus →	Depresses
- Styloglossus →	Retracts
- Palatoglossus →	Elevates

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Palatoglossus → Origin - Oral sufface of Palatine aponeurosis
insertion - side of tongue
nerve - ~~Pala~~ Pharyngeal plexus

Hyoglossus → Origin - lateral part of hyoid bone
insertion - side of tongue
nerve - rest all 3x muscle → Hypoglossal nerve

Styloglossus → Origin - tip of styloid process
insertion - side of tongue

Genioglossus → Origin - upper genital tubercle of mandible
insertion - upper fibres into tip of tongue

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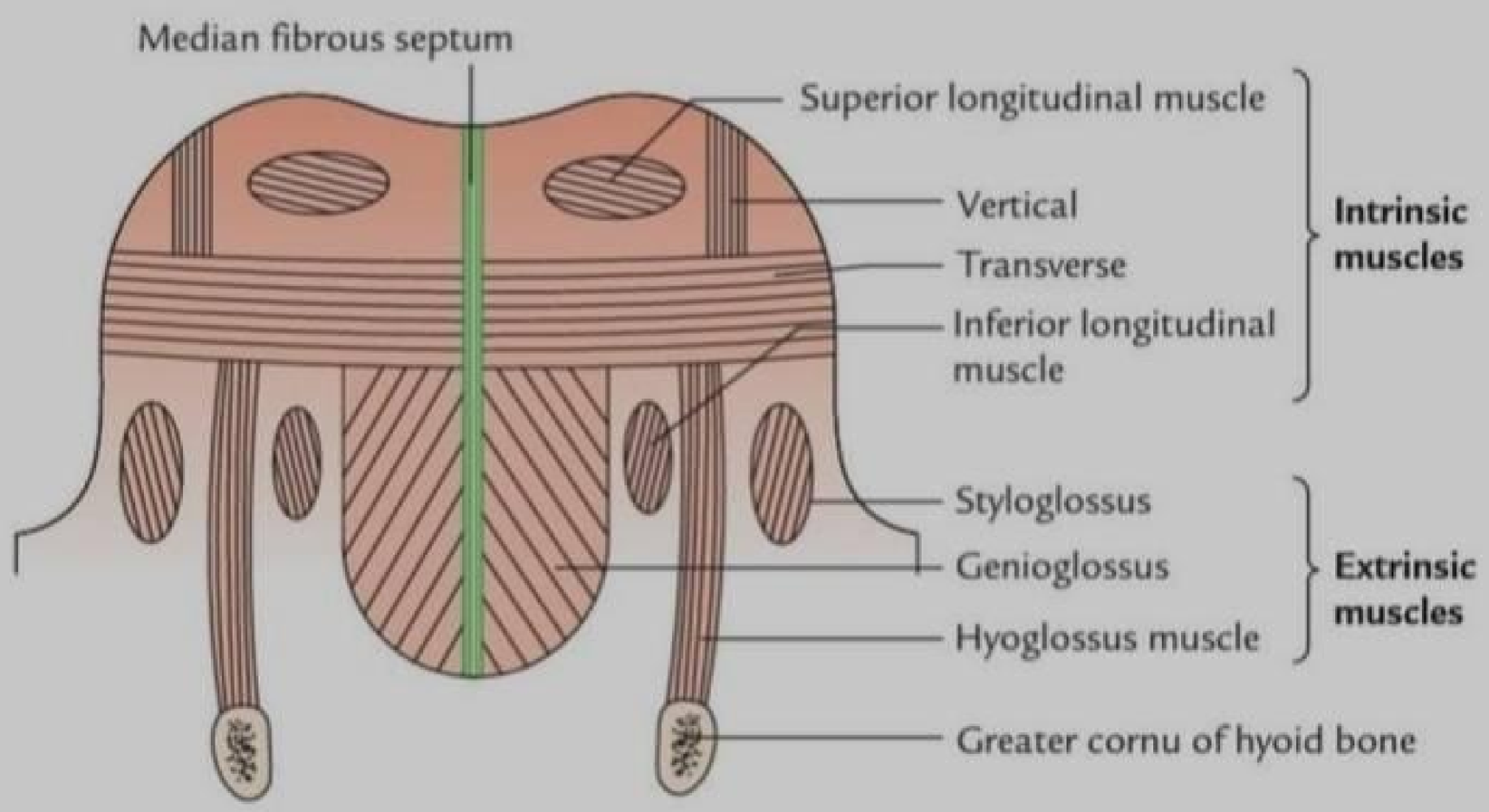


Fig. 13.14 Coronal section of the tongue showing arrangement of intrinsic and extrinsic muscles of the tongue.

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Tongue Nerve Supply Notes->

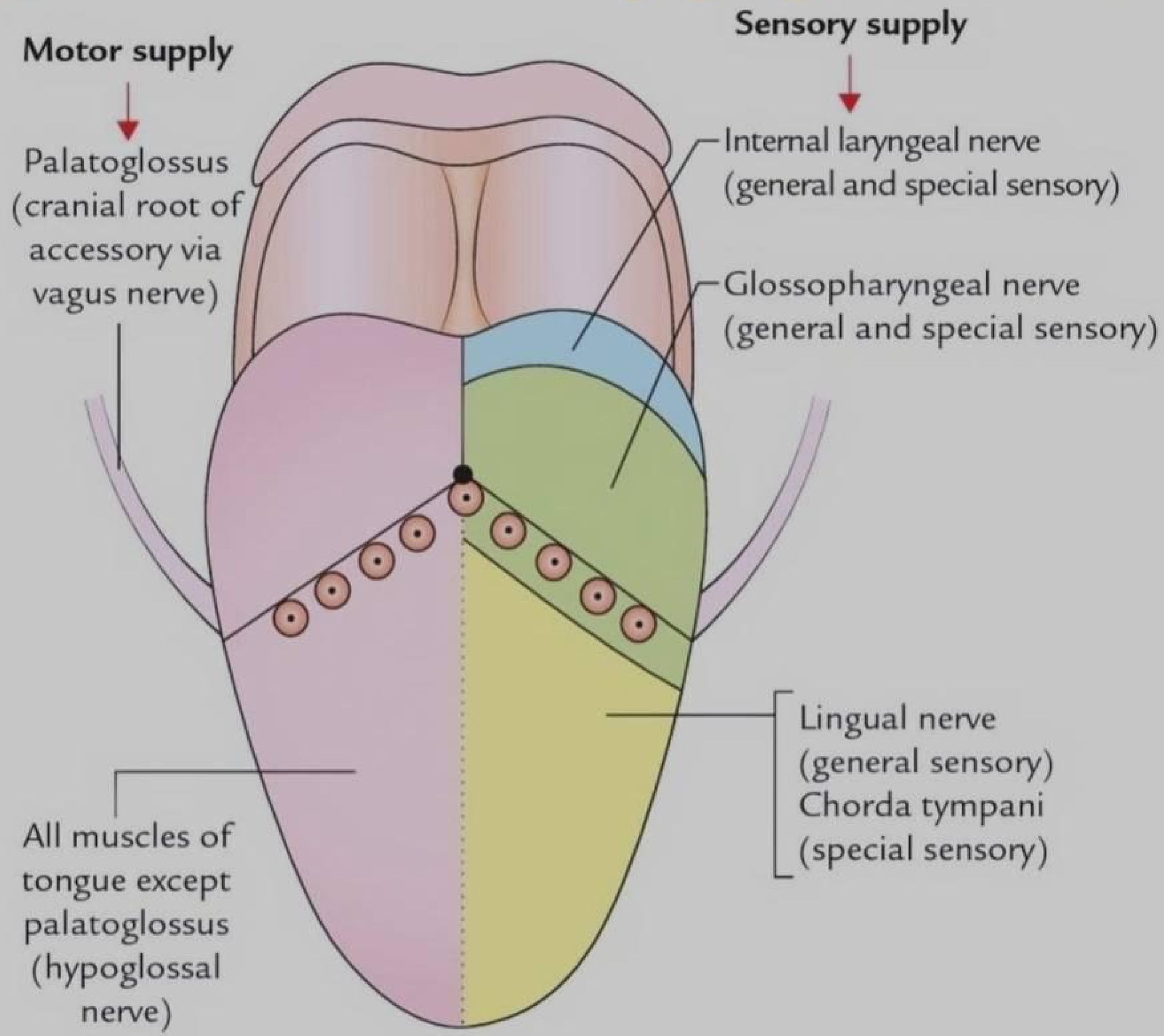


Fig. 13.20 Nerve supply of the tongue. Right half of the figure shows motor supply and left half shows sensory supply.

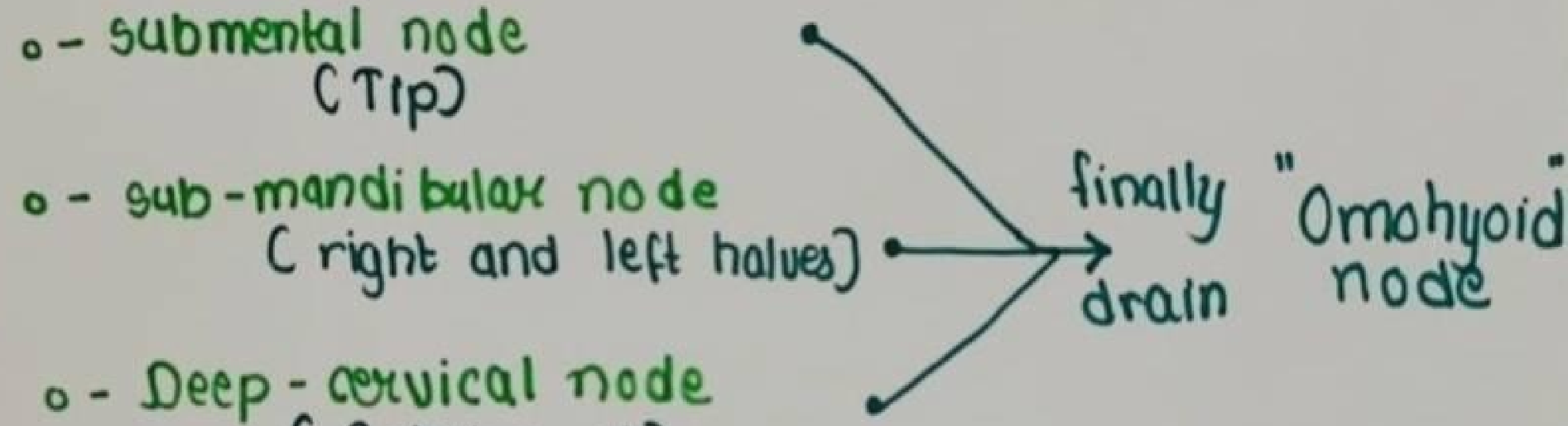
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Blood & nerve Supply of Tongue ...

- Artery → Derived from → Lingual artery ⟶ branch ECA
  - tip also supply by - tonsillar artery → FA

- venous drainage - Deep lingual vein - largest and principal vein
  - inferior surface of tongue
  - end in internal jugular vein through lingual vein

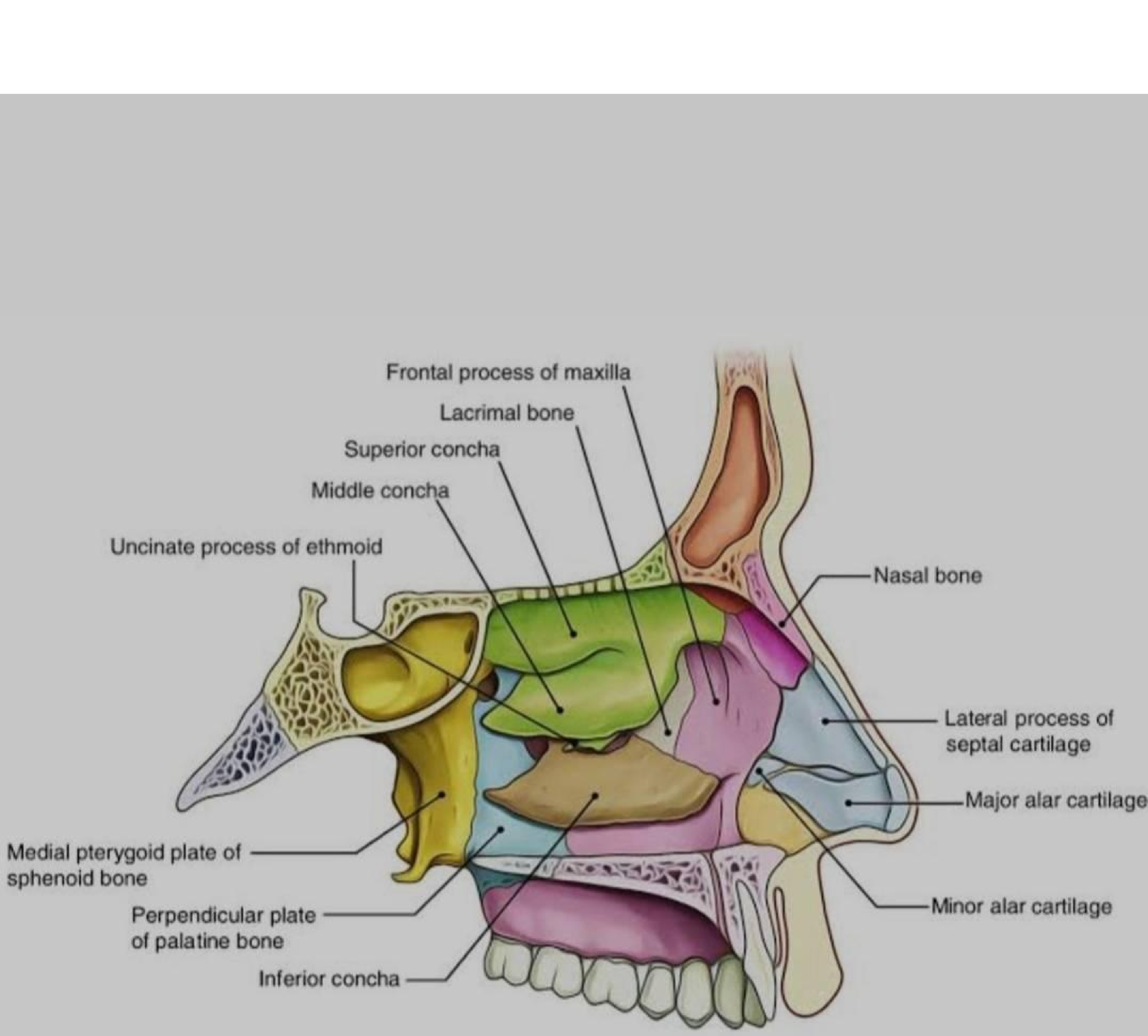
- lymphatic drainage



- nerve supply
  - Motor - all muscle - hypoglossal nerve
    except - Palatoglossus - Pharyngeal plexus
  - sensory - Anterior 2/3 - Lingual nerve
    chorda-tympani nerve
    - Posterior 1/3 - glossopharyngeal nerve

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Lateral wall of Nose Notes ->

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LATERAL WALL OF NOSE :-

- lateral wall of nose → irregular owing → presence of 3x shelf like bone Projection ↓ called conchae
  - conchae → increase surface area of nose
    - for effective air-conditioning of inspired air
- separate the nose - from orbit above
  - from maxillary sinus below
  - from lacrimal sac and nasolacrimal duct front
- sub-divided into 3x part - anterior - vestibule
  middle - atrium of middle meatus
  Posterior - meatuses
- skeleton of lateral wall is partly bony, partly cartilaginous and Partly made up of soft tissue
  - Bony part → Nasal
    frontal process of maxilla
    Lacrimal
    Labyrinth of ethmoid
    inferior nasal concha
    Perpendicular plate of palatine bone
    medial pterygoid plate
  - cartilaginous part → superior nasal cartilage
    inferior nasal cartilage
    3-4 small cartilages of ala
  - cuticular lower part → formed by fibrofatty tissue → covered with skin

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Conchae And Meatuses...

- nasal conchae → curved bone projection
  - directed → downward and medially
  - inferior concha – independent bone
  - middle concha – medial surface of ethmoidal labyrinth
  - superior concha – medial surface of ethmoidal labyrinth

- meatus of nose - Passage beneath the overhanging conchae

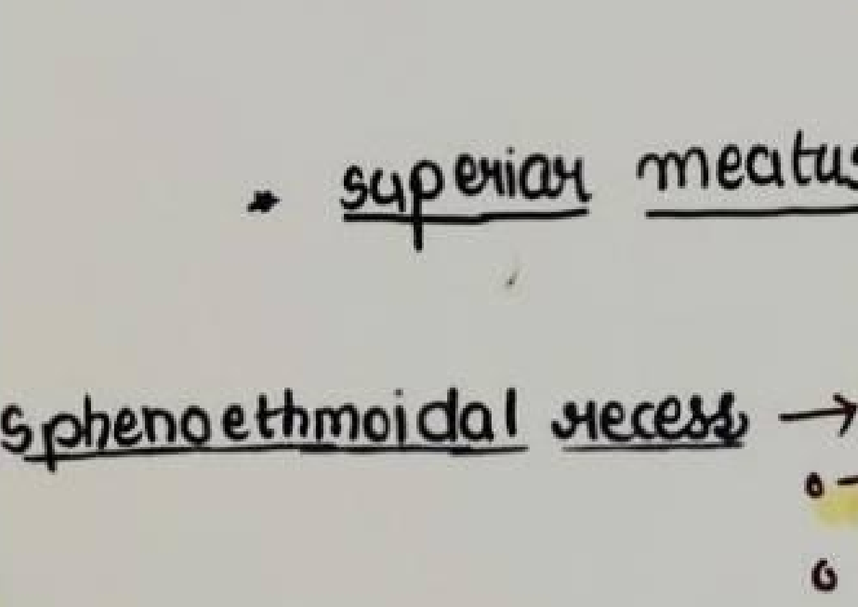
  * inferior meatus - largest of three meatuses
    - nasolacrimal duct open → guarded by lacrimal fold or 'Hasner's valve'
      ↓ into junction
      anterior 1/3 | Posterior 2/3

  * middle meatus - ethmoidal bulla → round elevation
    hiatus semilunaris → deep semicircular sulcus
    - opening of frontal air sinus → Anterior part of HS
    - opening of ethmoidal air sinus → Behind F.A
    - opening of maxillary air sinus → Posterior part of HS

  * superior meatus - shortest and shallowest of meatuses
    - opening of posterior ethmoidal air sinuses

Sphenoethmoidal recess → Δ fossa
- just above – superior concha
- receives – opening of sphenoidal air sinus

Arterial Supply :-



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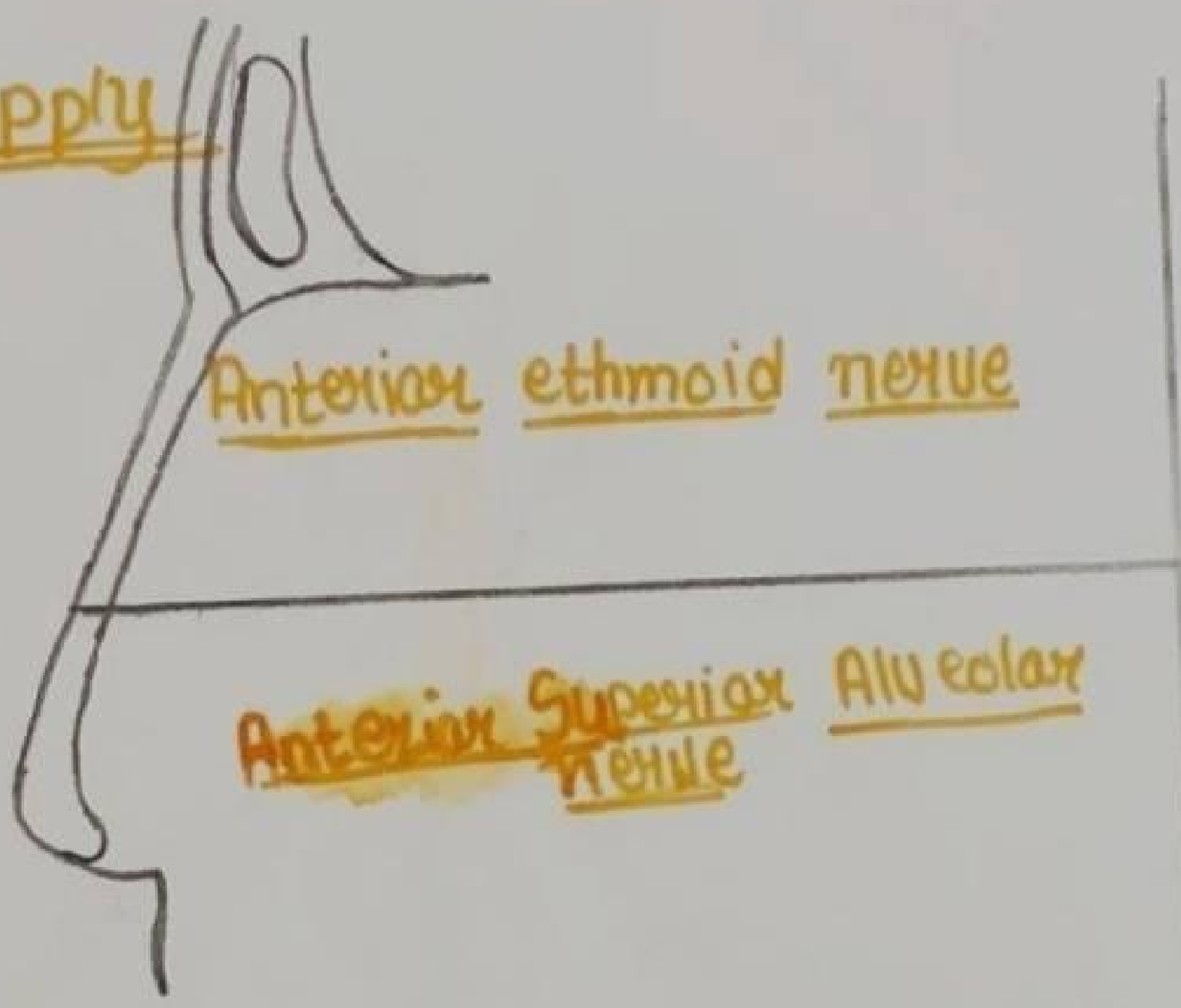
Anterior Posterior ethmoidal artery (Antero-superior quadrant)	Sphenopalatine artery (Posterosuperior quadrant)
Facial artery (Antero-inferior quadrant)	Greater palatine artery (Postero inferior quadrant)

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Venous Drainage :- → vein from a plexus which drains -

- Anteriorly - facial vein
- middle - Pterygoid plexus vein
- Posteriorly - Pterygoid plexus vein

nerve supply



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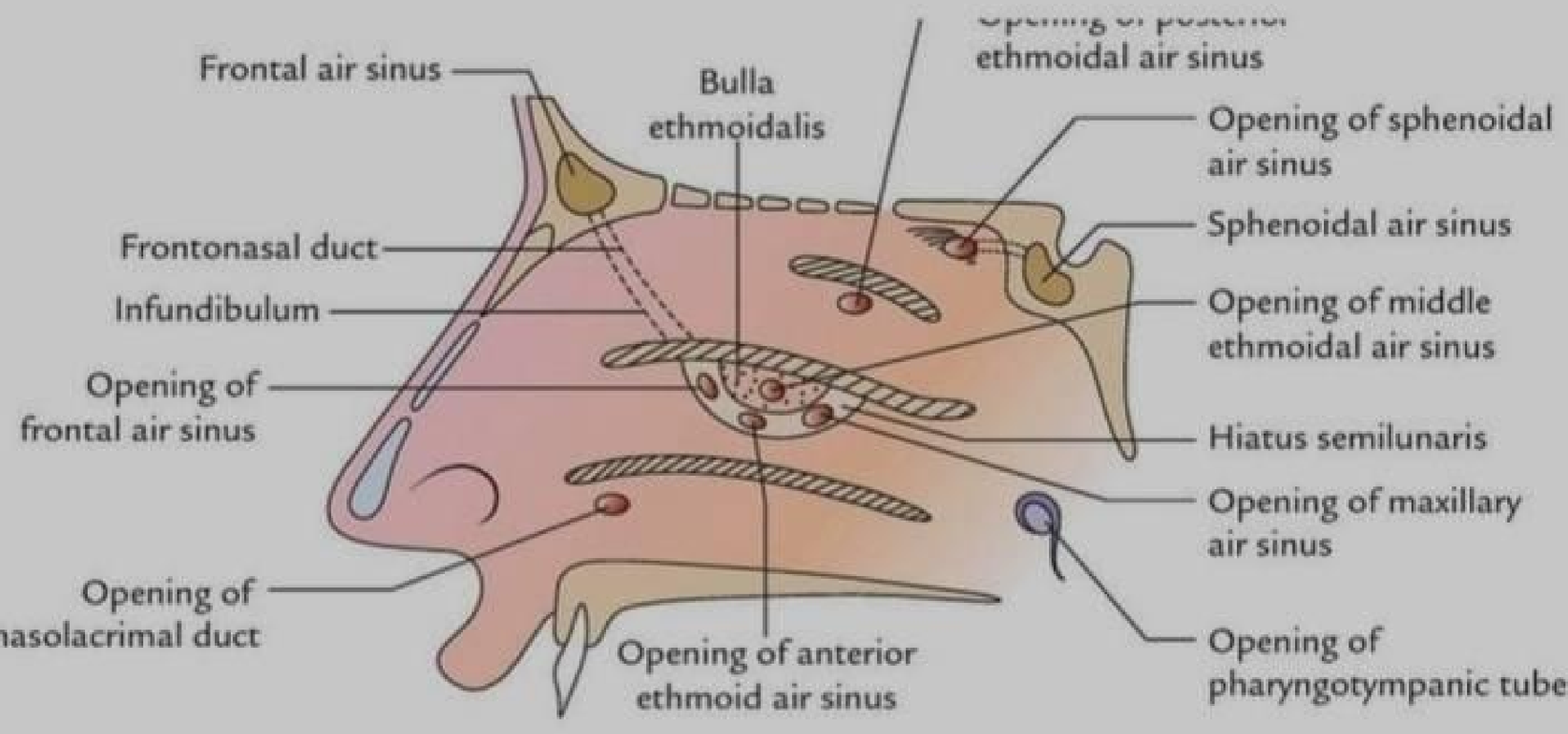
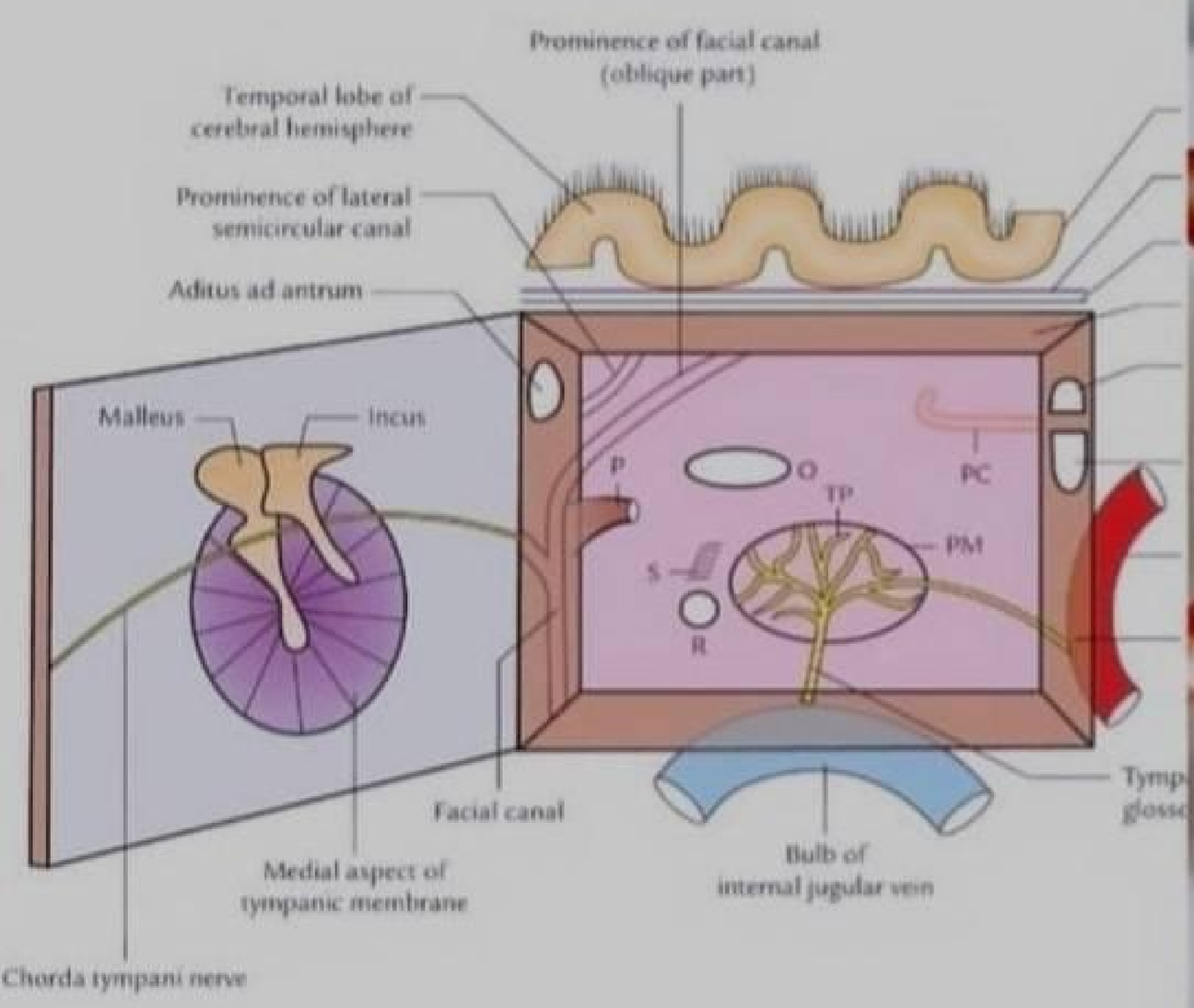
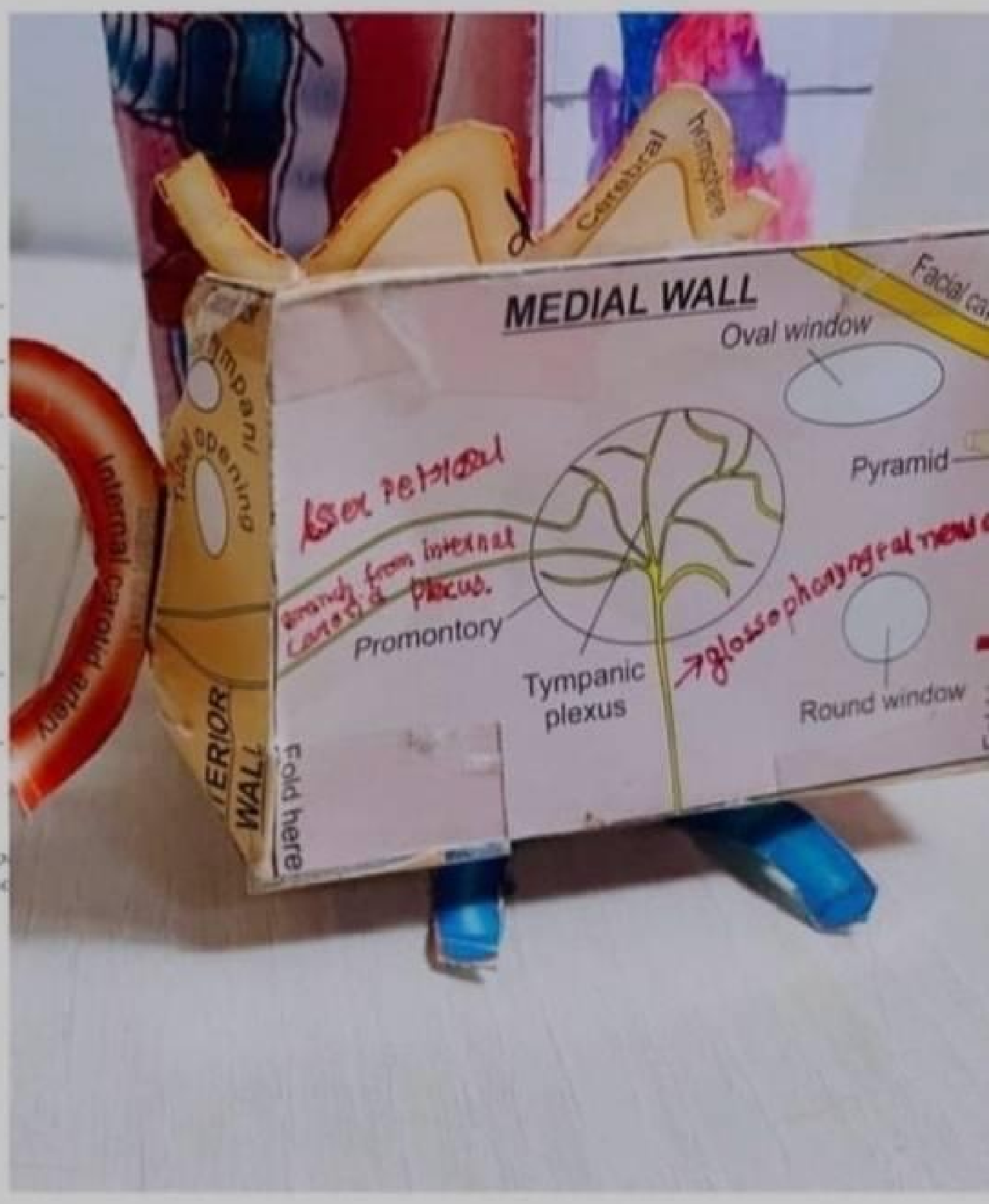


Fig. 17.7 Lateral wall of the nose with conchae removed showing openings of various sinuses and nasolacrimal duct.

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Middle Ear

3D Anatomy Model notes ->

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Middle EAR ...

- narrow air filled space $\xrightarrow{\text{situated in}}$ Petrous part of temporal bone $\xrightarrow{\text{between}}$ external ear and internal ear
- middle ear $\xrightarrow{\text{shaped}}$ like cube → lateral and medial wall - large
  other wall — narrow
- sub-divided → tympanic cavity proper - opposite the tympanic membrane
  epitympanic recess - lie above tympanic membrane
- communication → Anteriorly - nasopharynx $\xrightarrow{\text{through}}$ auditory tube
  Posteriorly - mastoid antrum → aditus

Boundaries :-

- Roof → thin plate bone → tegmen tympani
- floor → thin plate bone of temporal → separate superior bulb of - internal jugular vein
  - glossopharyngeal nerve

- Anterior wall → upper most - Tensor tympani
  middle most - auditory tube
  inner most - thin plate of bone ↓ which form posterior wall of carotid cana
  - Plate separated - internal carotid artery

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- Posterior wall → superiorly - aditus $\xrightarrow{\text{through which}}$ epitymanic recess ↓ communicate tympanic antrum
  - conical projection called pyramid
    lie - near junction of Posterior & medial wall
  - opening → for passage of tendon of "stapedius" muscle

- → Posterior edge of tympanic membrane ↓ is Posterior canaliculus $\xrightarrow{\text{for}}$ chorda tympani nerve

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- Lateral wall - separate middle ear from external acoustic meatus
  - Tympanic membrane —along→ tympanic ring and sulcus
  - Partly squamous temporal bone —in→ epitympanic recess

- Medial wall - separate middle ear from internal ear
  - Promontory - first turn of cochlear
    - grooved by - Tympanic plexus
  - fenestra vestibuli - oval opening
    - Postero superior → Promontory.
  - fenestra cochleae - round opening
    - Postero inferior → Promontory

- semicircular canal → above facial canal
- facial canal → above fenestra vestibuli

Contents :-

- 3x small bone → malleus, incus, stape
- Ligament of ear ossicle
- 2x muscle → tensor tympani, stapedius
- vessels supply and draining the middle ear
- nerve — chorda tympani, tympanic plexus

Arterial Supply :-

- Anterior tympanic branch — Maxillary artery
- Posterior tympanic branch — Posterior auricular artery
- Petrosal and superior tympanic branch — middle meningeal artery
- Branches of ascending pharyngeal artery
- Tympanic branches — ICA

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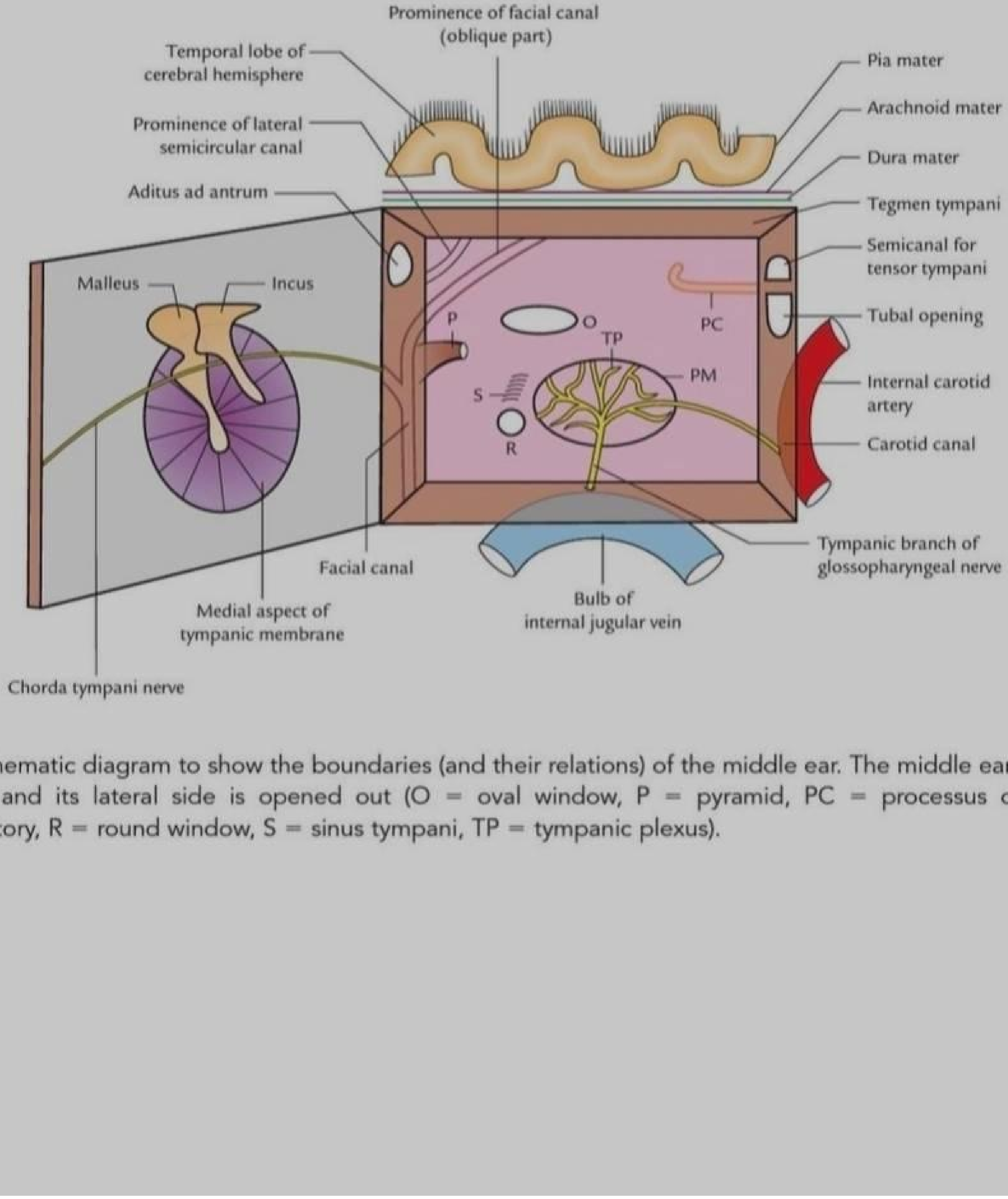
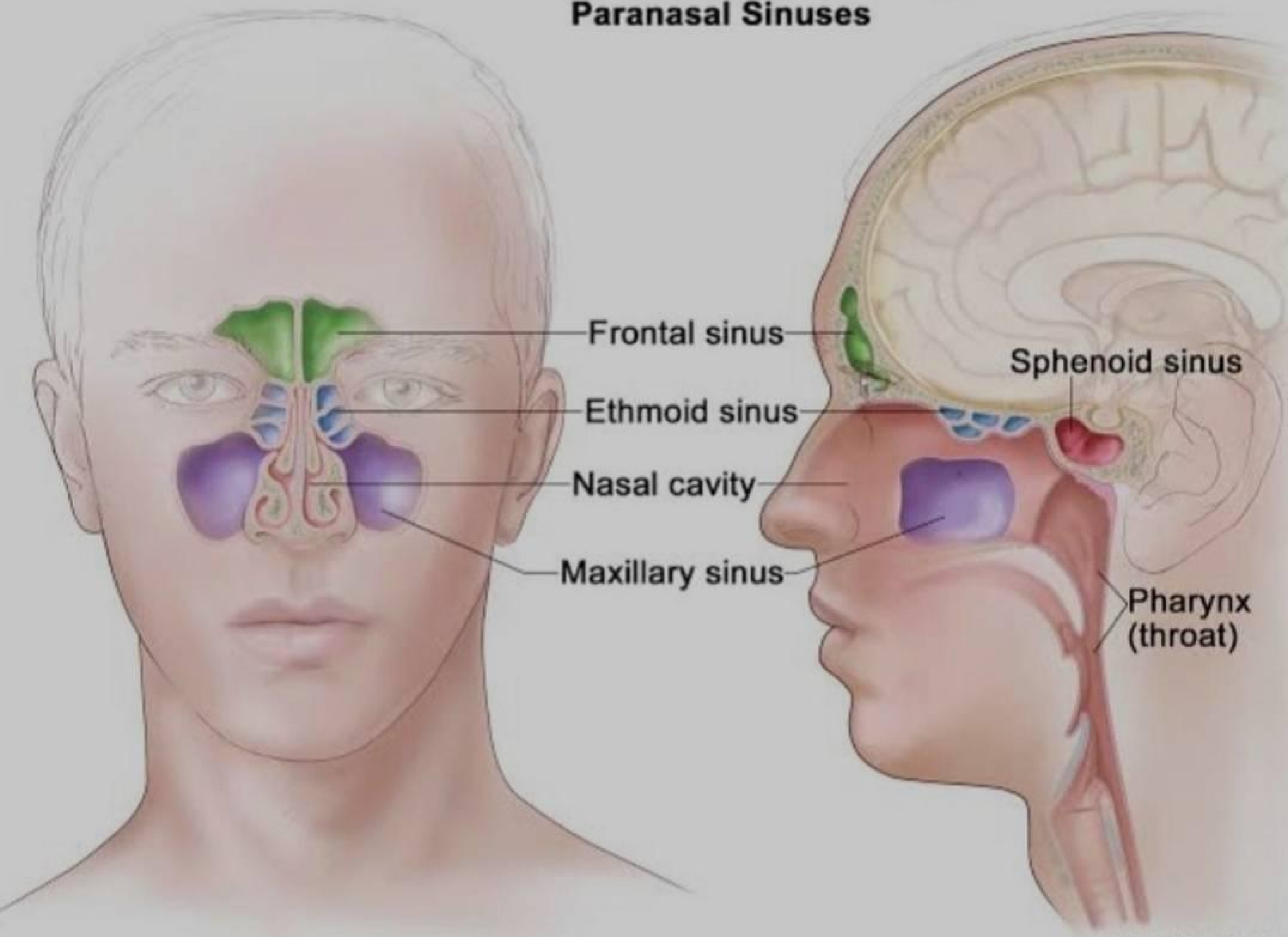


Fig. 18.10 Schematic diagram to show the boundaries (and their relations) of the middle ear. The middle ear is likened to a six-sided box and its lateral side is opened out (O = oval window, P = pyramid, PC = processus cochleariformis, PM = promontory, R = round window, S = sinus tympani, TP = tympanic plexus).

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Paranasal Sinuses Anatomy Notes ->



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PARANASAL Sinuses

- Air-filled space within some bone - frontal, maxilla, sphenoid, Ethmoid
- function - make skull lighter, warm up, humidify the inspired air
- Sinuses are rudimentary → absent at birth
  - enlarge during age 6-7 year
- from birth to adult life - growth of sinuses - due to enlargement of bone.

frontal sinus :-

- lie in → frontal bone deep to superciliary arch
- extend upwards, medial end of eyebrow
  backwards into medial part of roof of orbit
- open into → middle meatus of nose at → anterior end of hiatus semilunaris
- Right and left unequal in size

* Arterial supply → supra-orbital artery
* Vein → supra-orbital vein
  ophthalmic vein
* nerve → supra-orbital nerve
* Lymphatic drainage - sub-mandibular node

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Maxillary sinus :-

- lie in body of maxilla
- largest and first paranasal sinus to develop
- open into → middle meatus of nose → lower part of hiatus semilunaris
- isolated maxilla, → opening is large
- Intact skull, size of opening reduced to 3-4 mm overlapped by following: - above - uncinate process of ethmoid
  below - inferior nasal concha
  Behind - Perpendicular plate of palatine bone

* Arterial supply → facial
  infra-orbital
  greater palatine artery
* Vein - facial vein
  Pterygoid plexus
* nerve - Posterior superior alveolar branch - Maxillary nerve
  Anterior / middle superior alveolar branch - infraorbital nerve
* Lymphatic - submandibular node

→ GF
→ fruit Princess
→ PBM super attitude

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Sphenoidal Sinus :-

- right and left sphenoidal sinus lie within body of sphenoid bone ↓ separated by septum
- open into spheno-ethmoidal recess
- Relation → superior - hypophysis cerebri, optic chiasma
  laterally - internal carotid artery, cavernous sinus

* Arterial supply → Posterior ethmoidal, internal carotid arteries ic Pe
* vein → Pterygoid venous plexus, cavernous sinus
* nerve → Posterior ethmoidal nerve
* lymphatic drainage → Retropharyngeal nodes

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Ethmoidal sinuses :-

- lie - labyrinth of ethmoid bone
- divided into anterior, posterior, middle group

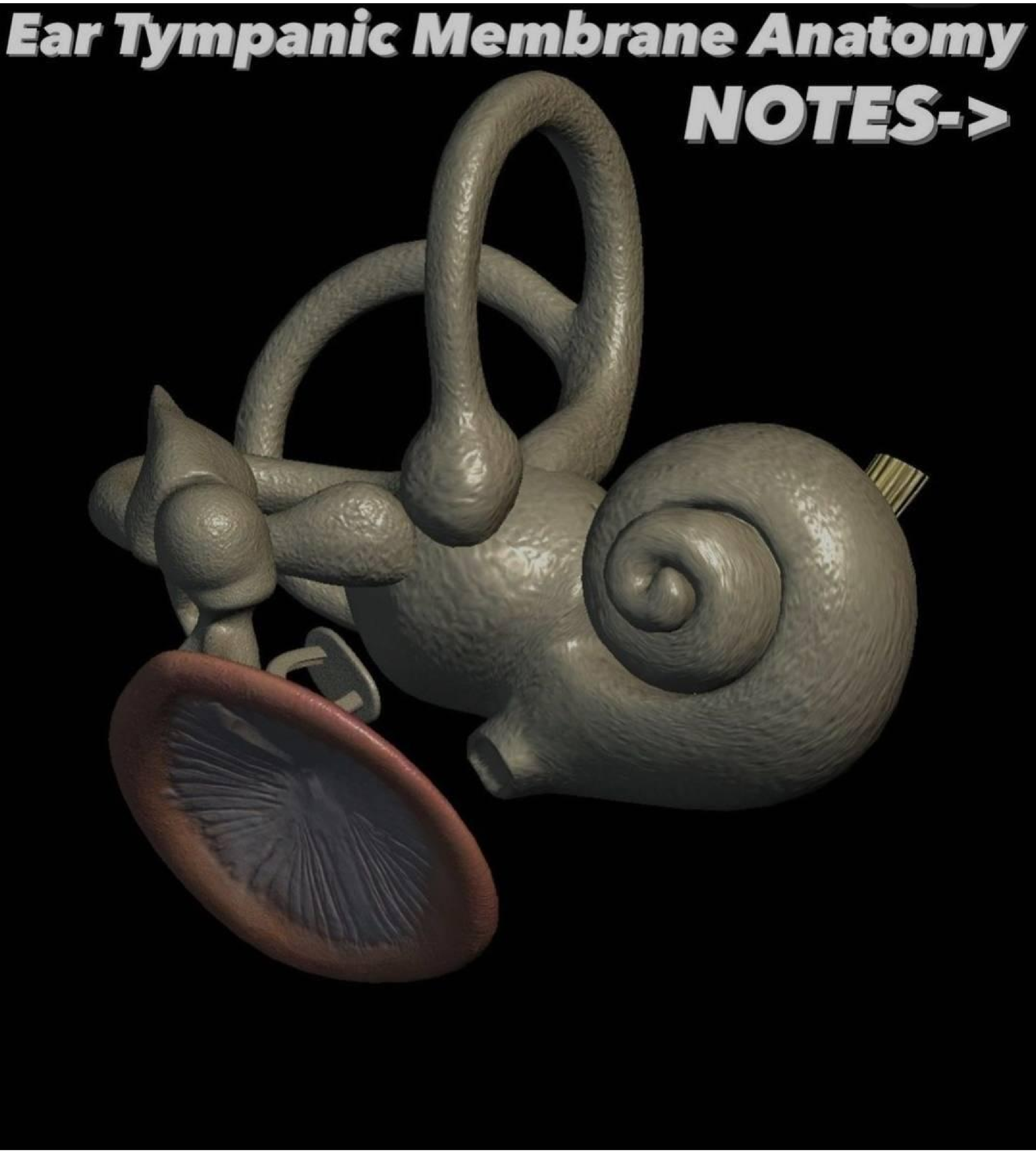
+ Anterior ethmoidal sinus
  - open - anterior part of hiatus semilunaris
  - supplied by - Anterior Ethmoidal nerve and vessel
  - lymphatic drain → sub-mandibular node

* middle ethmoidal sinus
  - open into middle meatus of nose
  - supplied by - Anterior Ethmoidal nerve and vessel
  - lymphatic drain → submandibular node

* Posterior ethmoidal sinus
  - open into superior meatus of nose
  - supplied by Posterior ethmoidal nerve and vessel
  - lymphatic drain - Retropharyngeal node

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Ear Tympanic Membrane Anatomy NOTES->



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TYMPANIC MEMBRANE...

- thin, translucent partition between external acoustic meatus and middle ear
- oval in shape
- placed - obliquely at an angle of 55° with floor of meatus
- face - downward, forward and laterally
- 2x surface → outer surface
  - lined by thin skin
  - concave
- ↳ inner surface
  - provide attachment to handle of malleus
  - inner convex
  - point of max. convexity lie → at tip of handle of malleus called → "umbo"

Bo

- membrane thickened at circumference which fixed tympanic sulcus of temporal plate
- superior sulcus deficient → here membrane attached to → tympanic notch
- from end of notch → 2x band → anterior / Posterior Malleolar fold prolonged
- greater part of TM is → tightly stretched → Pars tensa
- Part between 2x malleolar fold is loose → Pars flaccida

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Structure:-

- tympanic membrane is composed of following 3x layer:-
  - outer cuticular layer → skin
  - middle fibrous layer → superficial radiating fibres
    - deep circular fibres
  - inner mucous layer lined by → low ciliated columnar epithelium

Blood → Deep auricular artery

Anterior tympanic branch - maxillary artery
Posterior tympanic branch - Posterior auricular artery

vein → external jugular vein

Transverse sinus

nerve → auriculotemporal nerve

tympanic branch → glossopharyngeal nerve

Outer

inner

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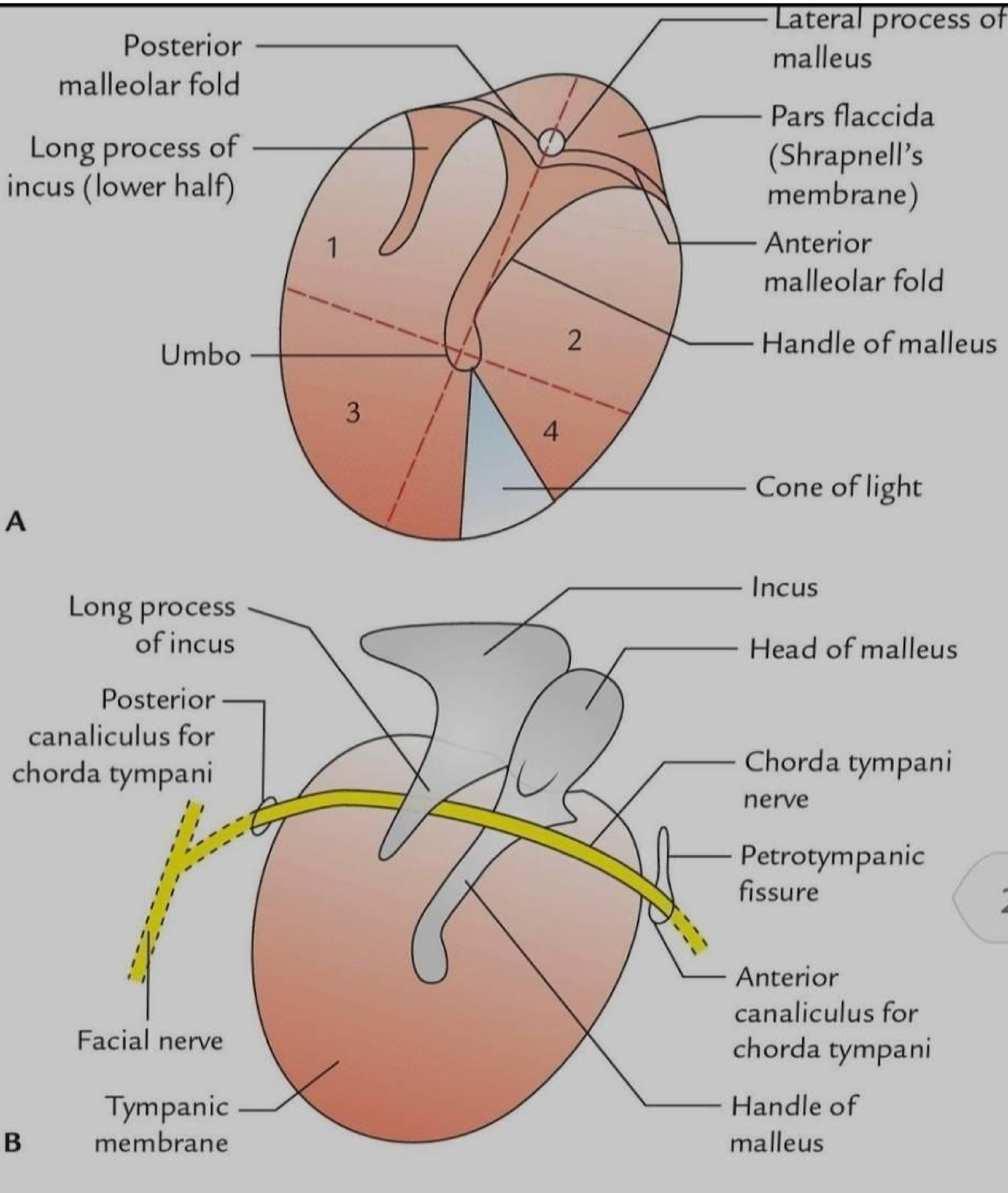


Fig. 18.5 Tympanic membrane (schematic diagram): A, external surface of tympanic membrane as seen through otoscope: 1 = posterosuperior quadrant, 2 = anterosuperior quadrant, 3 = posteroinferior quadrant, 4 = anteroinferior quadrant; B, medial surface of tympanic membrane showing

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Styloid Apparatus Notes->

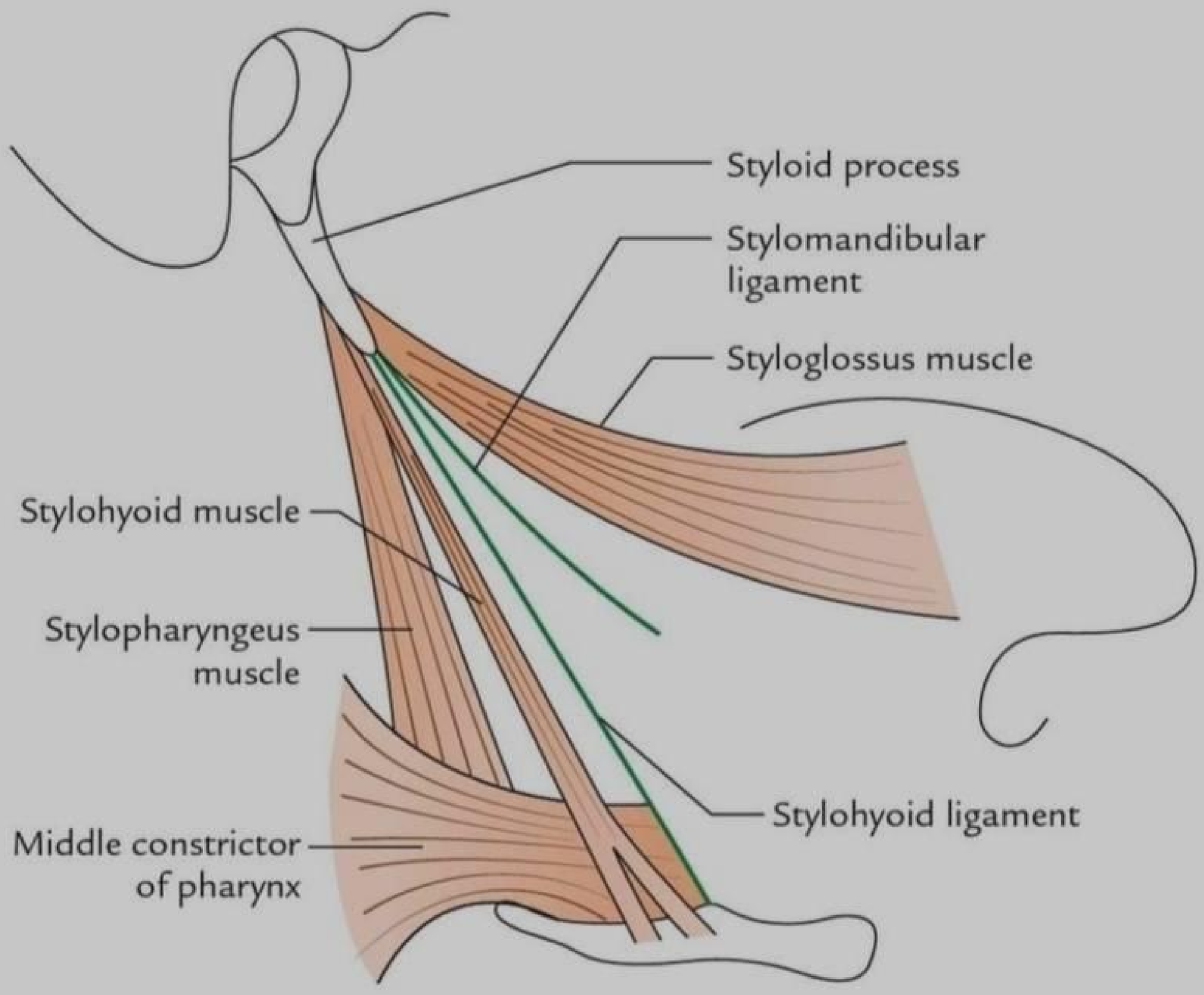


Fig. 16.19 Styloid apparatus.

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Styloid apparatus...

- Styloid process + attached structure → styloid apparatus

- 3x muscle
  - Styloglossus
  - Stylohyoid
  - Stylopharyngeus
- 2x ligaments
  - Stylohyoid
  - Stylomandibular

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- apparatus is of diverse origin
  - Stylohyoid muscle and ligament ⇒ 2nd brachial arch
  - Stylopharyngeus ⇒ 3rd branchial arch
  - Styloglossus ⇒ occipital myotomes
  - Stylomandibular ligament ⇒ Deep fascia of neck

Styloid process:

- long pointed bony process of temporal bone
- downward, forward, medially
- descends b/w external and internal carotid artery ↓ to reach side of Pharynx

Styloglossus ∘ arises → anterior surface of styloid process
insert → side of tongue

Stylopharyngeus arises → medial surface of base of styloid process
insert → Posterior border of thyroid cartilage

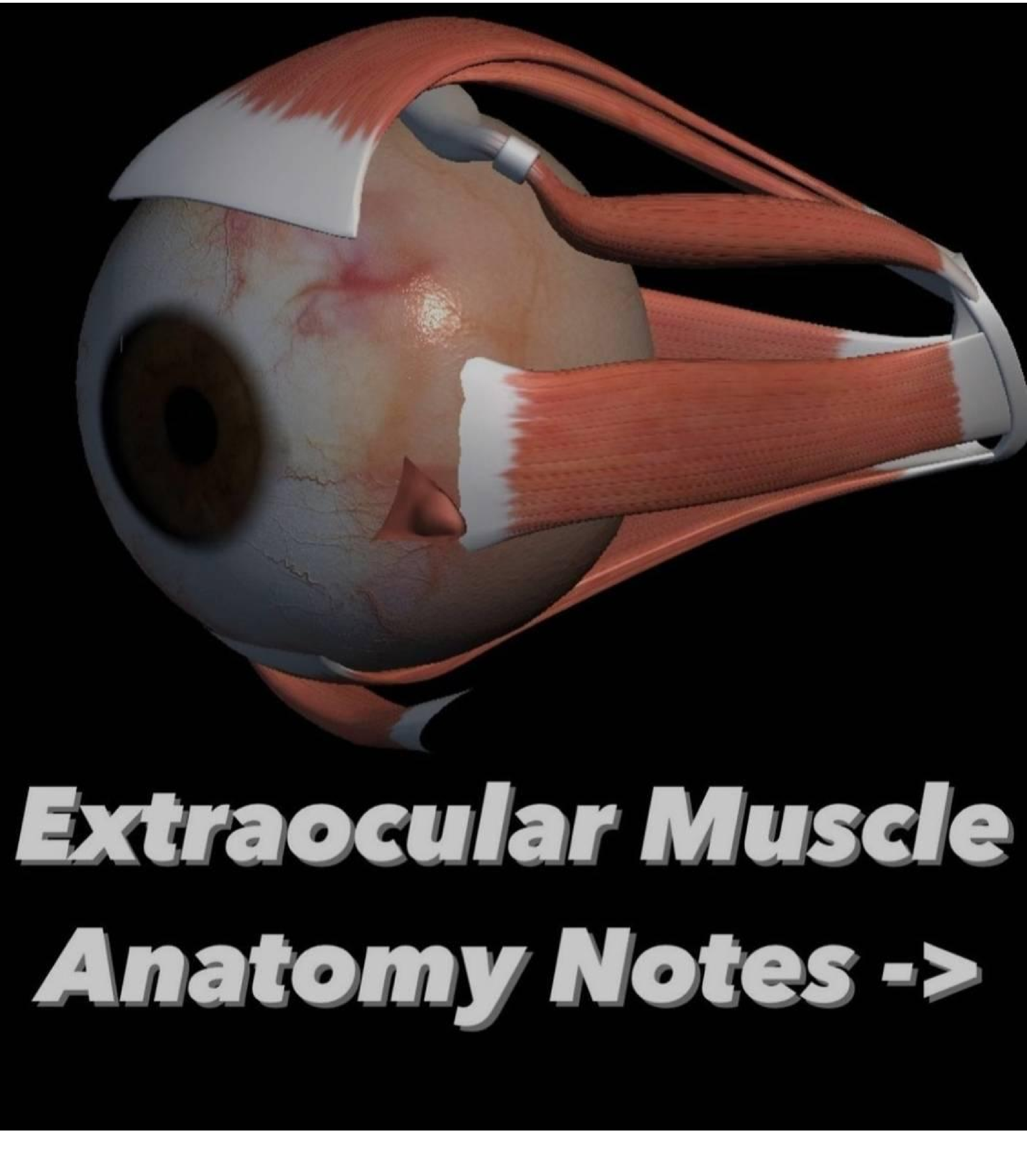
Stylohyoid ∘ arises → Posterior surface of styloid process
insert → hyoid bone

Stylomandibular ligament attached → laterally to styloid process - above
insert → angle of mandible - below

Stylohyoid ligament extend from → tip of styloid proces to lesser cornua of hyoid bone

* **Feature:**
  - External carotid artery → tip of styloid process
  - Facial nerve crosses → base of styloid process

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Extraocular Muscle ...

- Voluntary muscle :-

4x recti ◦- superior rectus
inferior rectus
medial rectus
Lateral rectus

2x oblique ◦- superior oblique
inferior oblique

levator palpebrae superioris

◦- Involuntary muscle :-

◦- superior tarsal muscle ◦- deeper portion of LPS superior
◦- inserted - upper margin of tarsus
◦- elevate - upper eyelid

◦- inferior tarsal muscle ◦- extend - fascial sheath of inferior rectus
◦- depresses the lower eyelid

◦- Orbitalis bridge → the inferior orbital fissure

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* voluntary muscles :-

Origin :-

◦- 4x recti ◦- common annular tendon —ring attached→ superior orbital fissure

◦- lateral rectus —additional origin→ greater wing of sphenoid bone

◦- superior oblique ◦ —arises→ lesser wing of sphenoid bone
◦- inferior oblique —arises→ orbital surface of maxilla
◦- levator palpebrae superior → lesser wing of sphenoid bone

insertion :- ◦- Recti —inserted→ sclera (Posterior to limbus)
◦- superior oblique —Passes through→ fibro-cartilage pully
↓ attached to trochlear fossa of frontal bone

insertion → sclera behind equator of eyeball

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inferior oblique muscle inserted close to SO.

Nerve :-

- Rectus x 3 = CN=3 / occulomotor nerve
- superior oblique = CN=4 / trochlear nerve
- lateral rectus = CN=6 / abducent nerve

All | 3
SO | 4
LR | 6

Action :-

- upward / elevation - superior rectus
  inferior oblique



- Downard / Depressor - inferior rectus
  superior oblique



- medial rotation / Adduction - medial rectus
  superior rectus
  inferior rectus



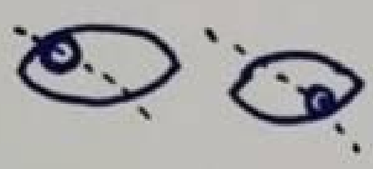
- lateral rotation / Abduction - lateral rectus
  superior, inferior - oblique

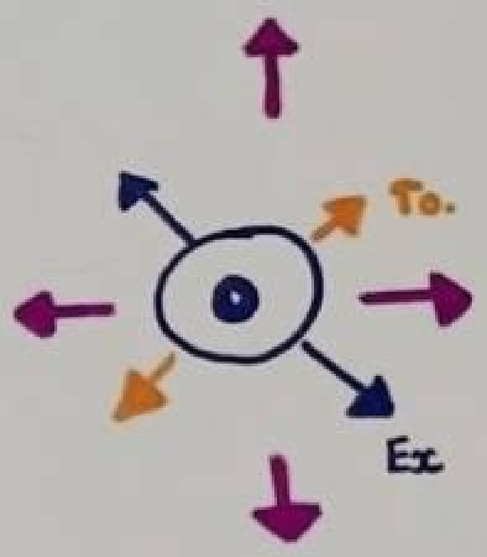


- Torsion - superior oblique
  superior rectus



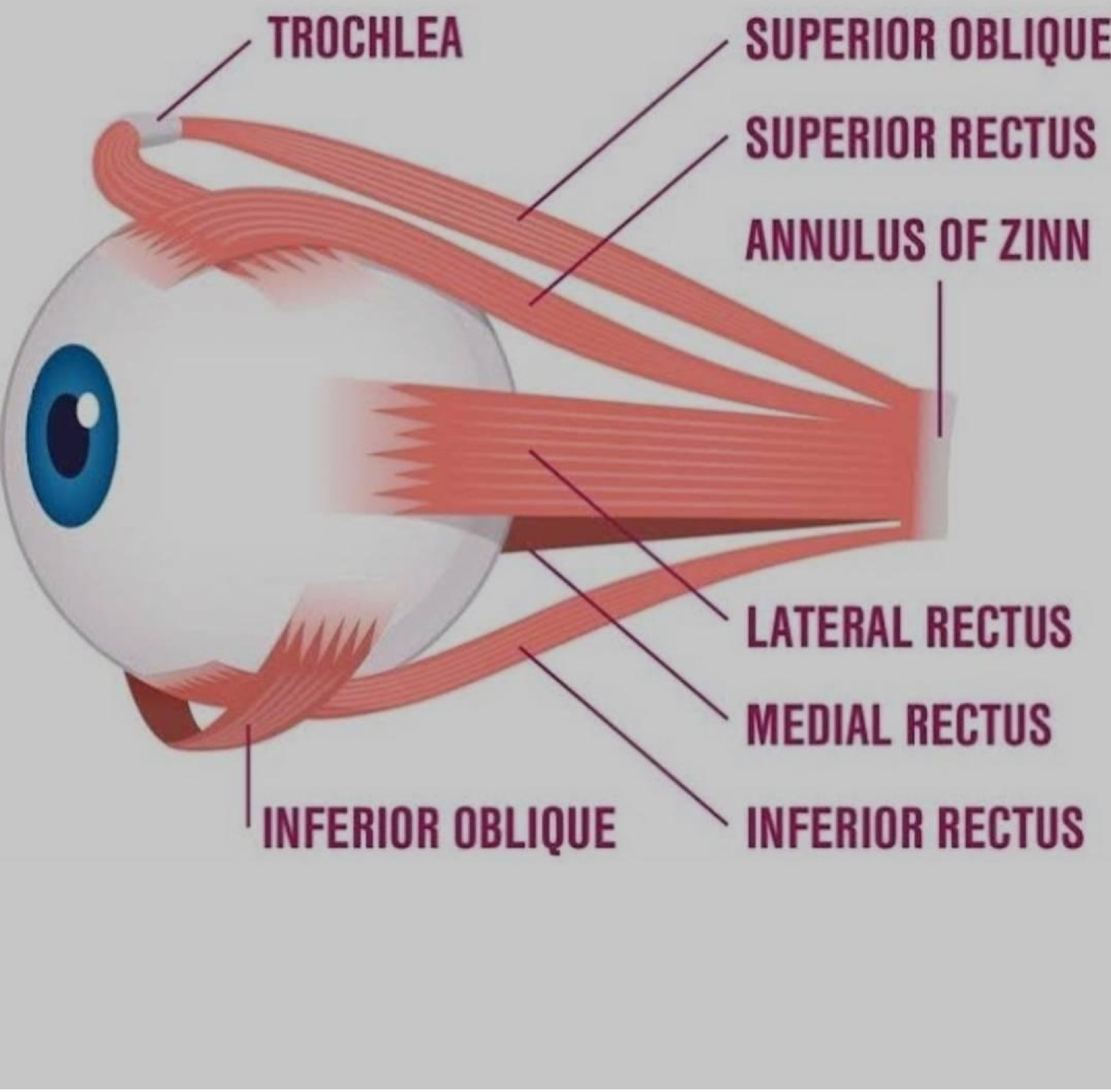
- Extorsion - inferior oblique
  inferior rectus



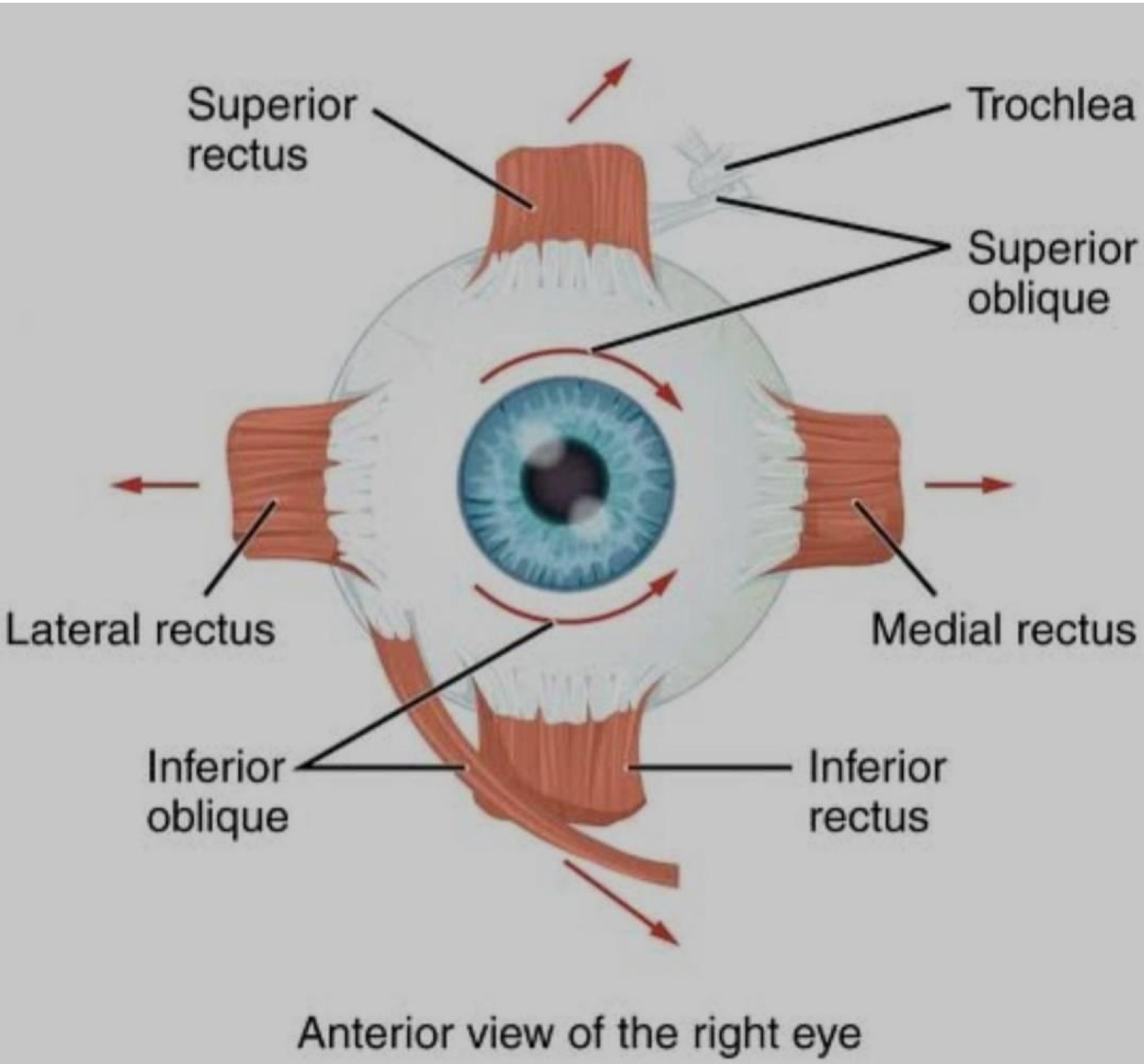


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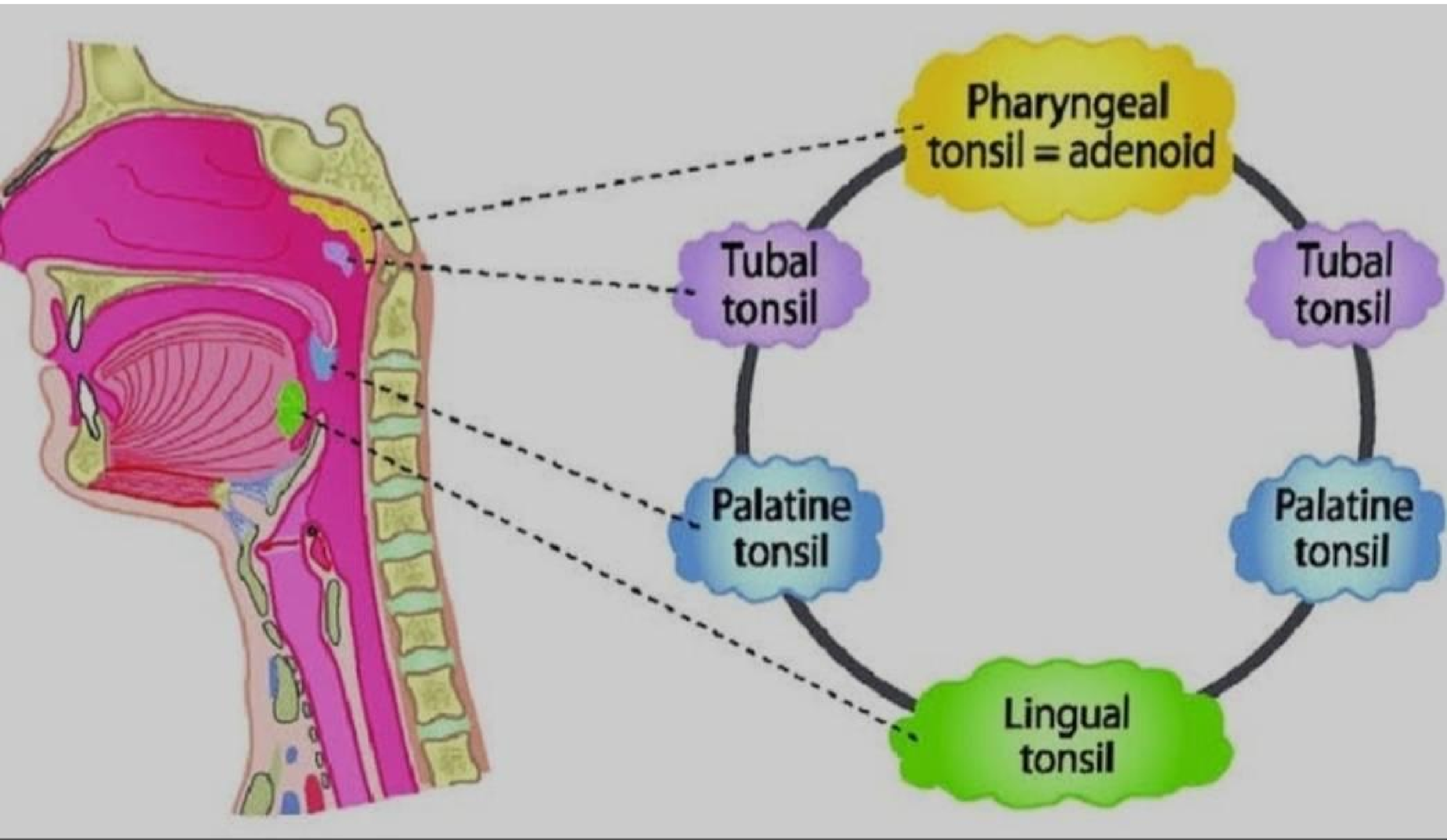
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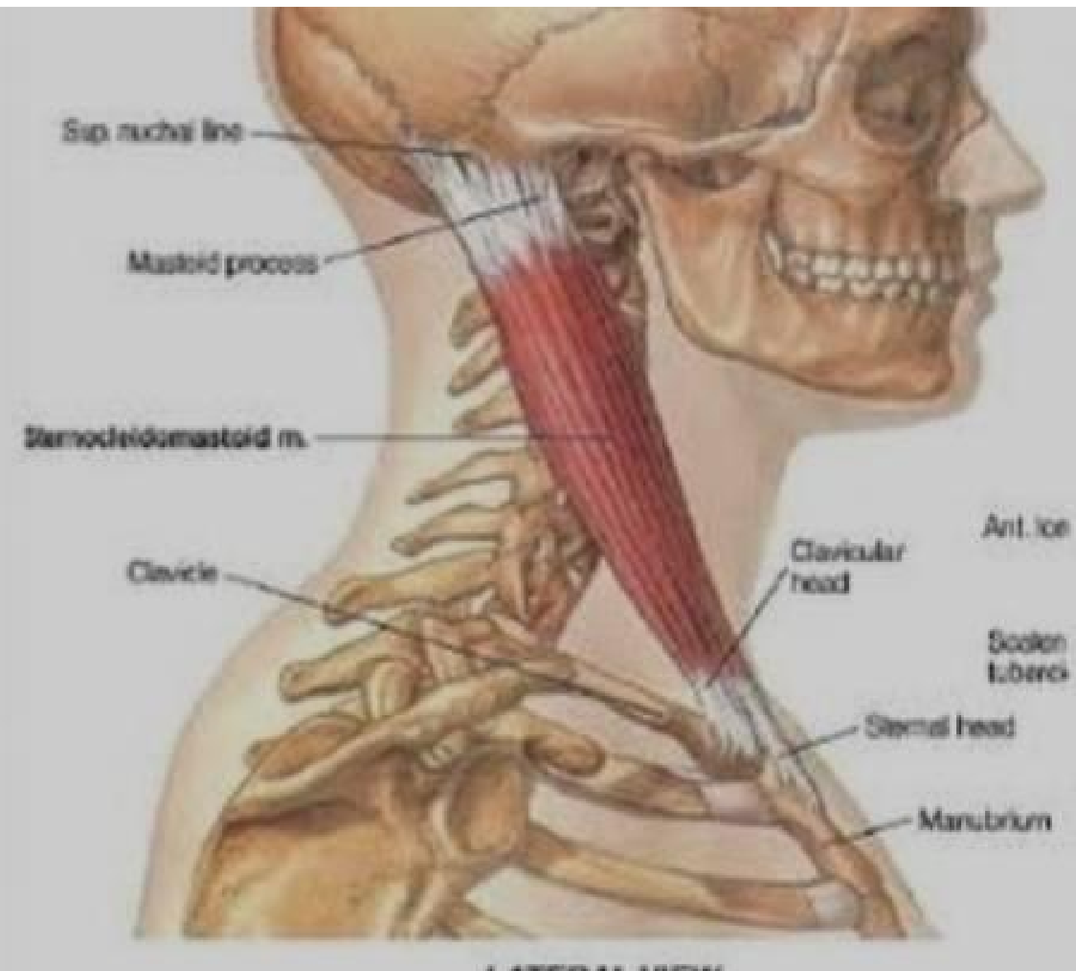


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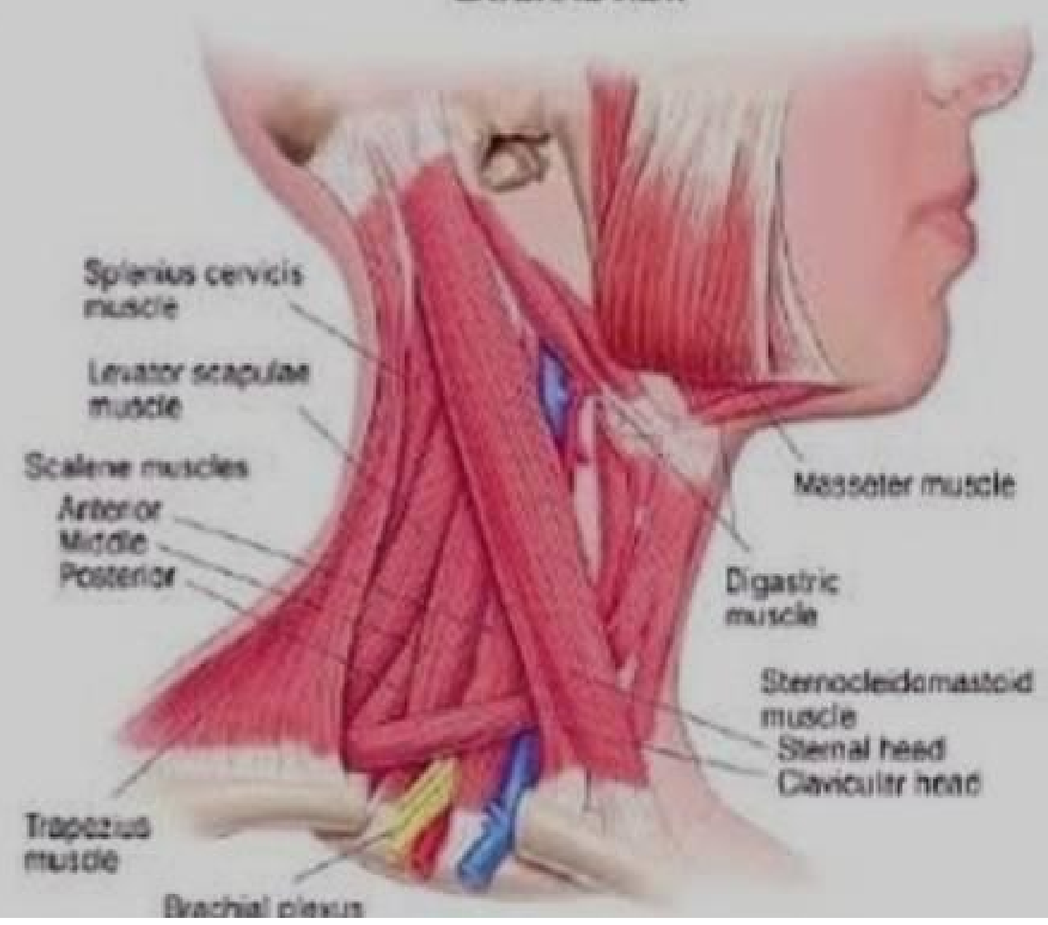
Sterno-cleidomastoid muscle

- **Origin: upper border of manubrium sterni and medial third of upper surface of clavicle.**

- **Insertion: mastoid process and lateral third of superior nuchal line.**
- **Nerve supply: Spinal accessory nerve (motor) and C2,3 (proprioceptive).**



- **Action:**
- **Both muscles extend the atlanto-occipital joint and flex the other cervical intervertebral joints.**
- **One muscle turns the head to make the face looks upward and to the opposite side.**
- **On fixation of the insertion, the muscle act as accessory muscle of inspiration.**



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DIGASTRIC

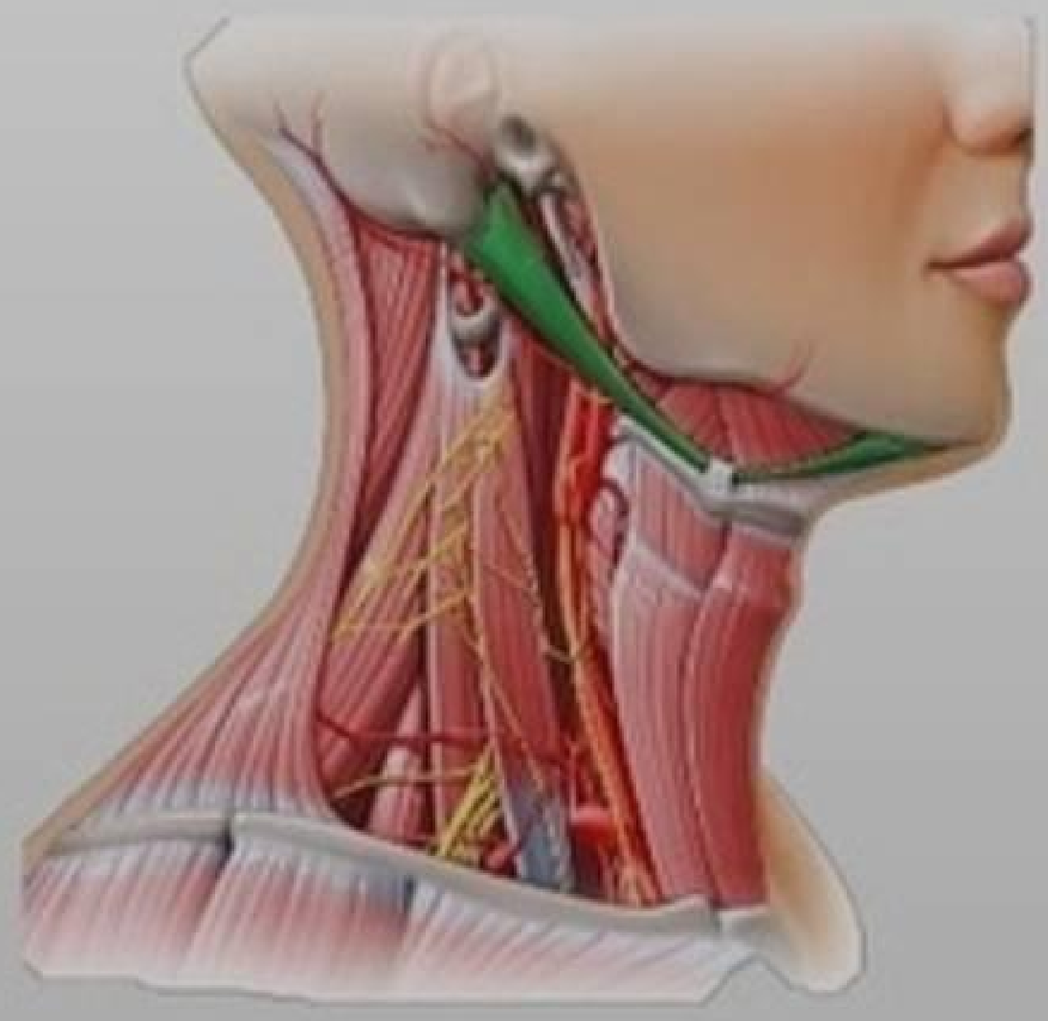
IT HAS TWO BELLIES UNITED BY AN INTERMEDIATE TENDON.

ORIGIN:

- ANTERIOR BELLY: ORIGINATES FROM THE DIGASTRIC FOSSA OF MANDIBLE.
- POSTERIOR BELLY: ORIGINATES FROM MASTOID NOTCH OF TEMPORAL BONE.

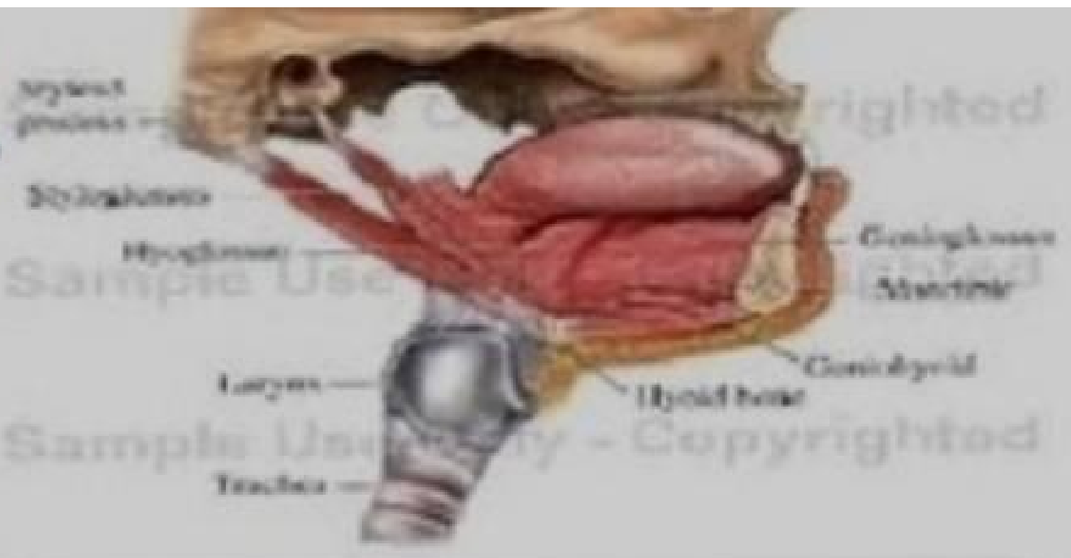
INSERTION:

- BOTH THESE BELLIES MEET AT THE INTERMEDIATE TENDON WHICH PERFORATES THE STYLOHYOID AND IS HELD BY A FIBROUS PULLEY TO THE HYOID BONE.



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DIGASTRIC MUSCL



NERVE SUPPLY :

ANTERIOR BELLY:

- **Mylohyoid nerve, a branch of the mandibular division of trigeminal nerve**

POSTERIOR BELLY:

- **Nerve from posterior auricular branch of facial nerve**

ACTIONS :

- **Depress and retract the mandible, so assisting the lateral pterygoid muscle in opening the mouth**
- **Elevation of the hyoid bone, utilized during swallowing and speech**

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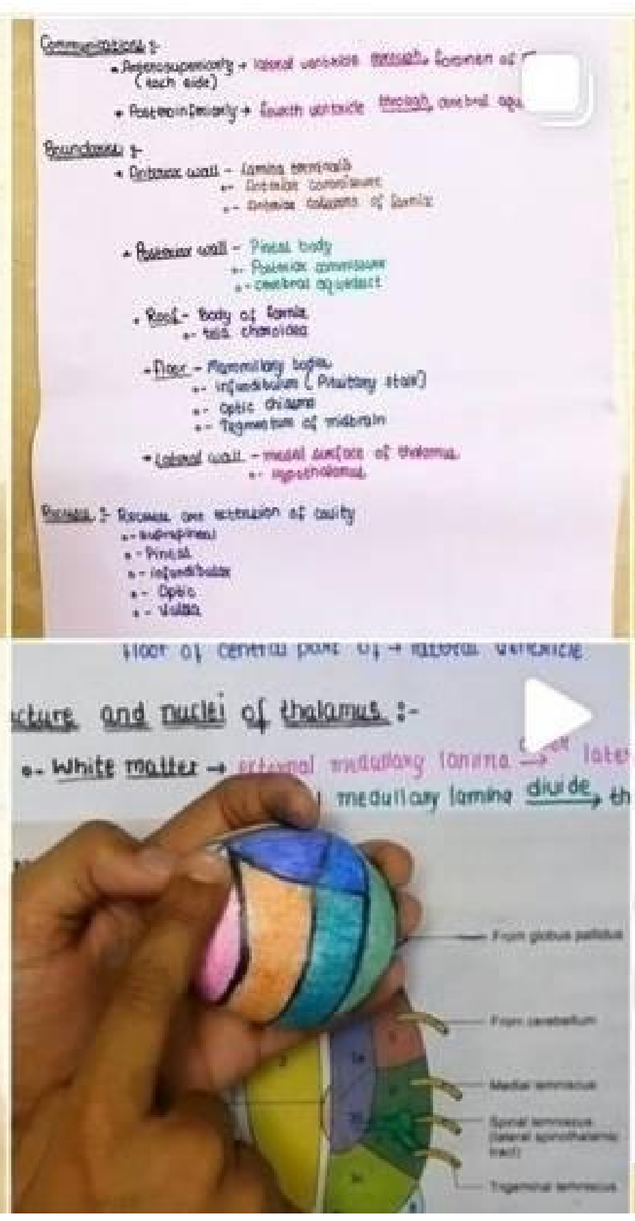
Histology

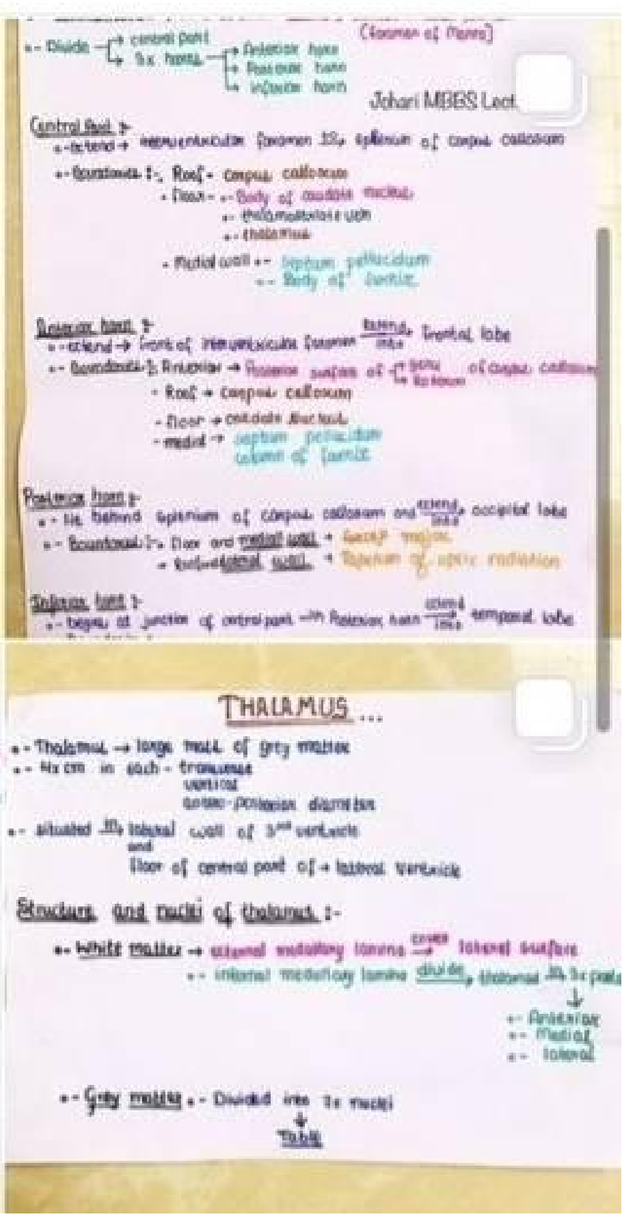


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Histology 34w

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mucus salivary glands

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